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Stochastic eco-evolutionary dynamics of multivariate traits: A framework for modeling population processes illustrated by the study of drifting **G**-matrices

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ABSTRACT

I derive a novel stochastic equation for the evolution of the additive genetic variance-covariance matrix **G** in response to mutation, selection, drift, and fluctuating population size. Common wisdom holds that the effect of drift on **G** is simply to reduce each of its entries by a common proportional amount while preserving its orientation. In contrast, I find that drift causes significant and directional shifts in the orientation of **G** by driving genetic correlations to their extremes. Biologically, this is a consequence of linkage build-up introduced by drift. I compare these theoretical results to empirical observations based on experiments conducted by Phillips et al. (2001). Additionally, to derive the model of **G**-matrix evolution, I developed a novel synthetic framework for modelling ecological and evolutionary dynamics of populations carrying multivariate traits. This framework is optimized for deriving new models across a wide range of topics in population biology. Foundations of the framework are formalized by the theory of measure-valued processes, but application of the framework only requires multivariate calculus, and heuristics are presented in the main text for making additional calculations involving stochastic processes. Collectively, this work establishes a powerful framework enabling efficient formal analysis of integrated population processes across evolution and ecology, and its potential for making new discoveries is illustrated by novel findings on fundamental aspects of **G**-matrix evolution.

1. Introduction

A fundamental principle of evolutionary biology is that random genetic drift erodes heritable variation at a rate inversely proportional to effective population size. At the level of allele frequencies, models for the distribution of genetic variation responding to drift are well-known (Kimura, 1964, 1968; Ewens, 2004; Hill and Robertson, 1966). In contrast, at the level of quantitative characters, models of additive genetic variation responding to drift focus on the average outcome using deterministic models (Latter, 1970; Bulmer, 1972; Lande, 1976, 1980; Chakraborty and Nei, 1982; Turelli and Barton, 1994; Lynch and Hill, 1986; Bürger, 2000; Barton and Turelli, 2004; Débarre and Otto, 2016; Walsh and Lynch, 2018). Empirical work has supported theoretical predictions for the average response of additive genetic variation to drift (Phillips et al., 2001; McGuigan et al., 2005; Whitlock et al., 2002), but has also emphasized the need to predict the distribution of outcomes (Phillips et al., 2001; Whitlock, 1995) especially for multivariate traits and genetic covariances (Phillips and McGuigan, 2006; Mallard et al., 2023a).

In the setting of multivariate traits, a common summary statistic for genetic architecture is the **G**-matrix. This matrix has the additive genetic variance of each trait on the associated diagonal entry. Off-diagonal entries quantify genetic covariances between traits which may be maintained by pleiotropic loci and linkage between loci affecting different traits. The standard view is that drift produces a proportional decrease in **G** (Phillips and McGuigan, 2006; Cano et al., 2004; McGuigan, 2006; Chapuis et al., 2008; Dugand et al., 2021; Mallard et al., 2023a), and thus independent populations that have diverged due to drift should have proportional **G**-matrices (Roff, 2000; Steppan et al., 2002; Aguirre et al., 2013). However, this insight rests on results from a deterministic model for the response of **G**-matrices to drift (Lande, 1979) and a model that assumes recombination happens sufficiently fast to break-up linkage produced by selection (Lande, 1980). Hence, there is a need to develop theoretical predictions for the stochastic evolution of **G**-matrices driven solely by drift, and especially for the effect of drift on genetic covariances (Mallard et al., 2023a,b).

The reason for this gap in evolutionary theory stems, in part, from the lack of formal approaches to place stochastic models of **G**-matrix

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evolution on a concrete mathematical foundation. In this paper, I aim to fill this gap by leveraging the powerful theory of measure-valued processes, while also keeping the presentation as accessible as possible. Furthermore, taking this approach to formally derive **G**-matrix dynamics revealed a much broader framework for modelling a wide-range of population processes. In particular, given the growing appreciation for the interplay between ecological and evolutionary processes (Reznick, 2015; Hendry, 2017; Kuosmanen et al., 2022), this framework makes an important contribution by enabling the formal interfacing of **G**-matrix evolution with models of eco-evolutionary feedbacks (e.g., Patel et al., 2018). I therefore focus this paper on presenting the framework in its full generality, and return to the study of **G**-matrix evolution as an example to illustrate its utility.

In its full generality, this framework offers tools to model the integrated ecological and evolutionary dynamics of populations with multivariate traits that respond to mutation, selection (including frequency and abundance dependence), demographic stochasticity, and consequential random genetic drift. In particular, this framework can be used to obtain generalizations of many classical models in evolution and ecology, such as Lotka-Volterra dynamics (Cattiaux and Méléard, 2009; Akjouj et al., 2024), coevolution (Gilman et al., 2012; Débarre et al., 2014), and evolutionary rescue (Klausmeier et al., 2020; Xu et al., 2023). Further details on how to apply this framework to generalize known models are given in the discussion section.

As in classical quantitative genetics, the framework here assumes a linear genotype-phenotype map with fixed mutational covariance. Under this modelling choice, development enters implicitly through the genotype-phenotype map rather than via explicit mechanistic developmental feedbacks (see Milocco and Salazar-Ciudad, 2022; González-Forero, 2022, 2024b,a). This paper therefore focuses on mutation-selection-drift dynamics of multivariate traits, not on the evolution of developmental architecture.

To apply this framework, the most important biological details relevant to a modeler are the mechanisms mediating fitness. In contrast, details involving the genetic architecture of traits are abstracted in a way that captures basic biological principles while optimizing analytical tractability. For instance, asexual reproduction is assumed and mutation is modeled following approaches similar to Kimura (1965) and Débarre and Otto (2016) by assuming offspring traits are distributed around their parental traits. This approach to modelling inheritance is similar to the infinitesimal model in which traits are determined by many loci of small effect, which implies offspring are normally distributed around their parental values (Barton et al., 2017). However, an important distinction is that the infinitesimal model restricts the response of allele frequencies to selection such that changes in genetic variance are only temporary and due to build-up of genetic linkage (Bulmer, 1971). Instead, I begin by assuming offspring traits are normally distributed around their parental traits (referred to as the Gaussian descendants model by Turelli, 2017), in which case the response of population variance to selection can be permanent. Abstract approaches similar to this have been successful for obtaining valuable analytical insights into genetic variation maintained by mutation-selection balance (Kimura, 1965; Lande, 1975; Turelli, 1984, 1986) and by mutation-drift balance (Lande, 1976, 1979; Barton, 1989; Débarre and Otto, 2016). Hence, by optimizing a trade-off between genetic detail and analytical tractability, this framework provides an accessible approach for obtaining and communicating a wide array of novel theoretical insights.

To establish this framework, I build on the work of Week et al. (2021), which presented a stochastic differential equation framework focused on modelling the simultaneous dynamics of abundances, 1-dimensional mean traits, and 1-dimensional trait variances responding to mutation, selection, demographic stochasticity, and random genetic drift. This 1-dimensional framework was based on the development of heuristics (i.e., practical rules that allow exact calculations without requiring familiarity with the underlying martingale theory or its formal derivation) for working with stochastic partial differential equations (spde). However, for multivariate traits, the spde approach breaks down (Dawson, 1993; Etheridge, 2000; Perkins, 2002). To overcome this challenge and establish a rigorous analytical framework for deriving population processes, I take an approach based on so-called *martingale problems* (Ethier and Kurtz, 1986; Dawson, 1993; Stroock and Varadhan, 1997; Rogers and Williams, 2000). Mathematical aspects of this approach are provided in Supplementary Material, Section 4. In the main text, I focus on the resulting dynamical equations and heuristics for performing calculations with minimal technical background.

1.1. Overview

I begin by outlining the derivation of the deterministic version of the framework without making any assumptions on the shape of trait distributions, which is summarized by a system of ordinary differential equations. This leads to expressions of selection in terms of covariances with fitness, which I refer to collectively as the Deterministic Covariance version (or *DC* for short). By assuming traits follow multivariate normal distributions, covariances with fitness are replaced by multivariate gradients of fitness with respect to mean traits and trait variances, and I refer to the resulting system of differential equations as the Deterministic Gradient version (or *DG* for short). To simplify presentation of the deterministic version of the framework, I assume traits are perfectly heritable. However, because this work is motivated by understanding the consequences of drift for **G**-matrix evolution, I briefly describe an approach to model imperfect heritability after introducing *DC* and *DG*. This model of imperfect heritability is adopted while introducing the stochastic extensions of the framework.

The stochastic extensions of the framework include the effects of demographic stochasticity (i.e., random reproductive output) and random genetic drift (which occurs here as a consequence of demographic stochasticity). I introduce two stochastic extensions. Both build on *DG* by assuming multivariate normal trait distributions and by expressing selection in terms of fitness gradients. The first form expresses dynamics in terms of Brownian motions as drivers of stochasticity (referred to as the Brownian Motion Gradient version, or *BG* for short), which is particularly useful for numerical analysis. The second form expresses dynamics in terms of a more general underlying martingale process (referred to as the Martingale Gradient version, or *MG* for short), and I use this form to introduce heuristics for deriving analytical models. To demonstrate the heuristics (summarized in Table 1), I derive a stochastic equation for the evolution of additive genetic correlations between trait values. To bring this paper full-circle, I then discuss how this exercise provides novel insights into the evolutionary response of **G**-matrices to random genetic drift and compare these theoretical results with observations obtained from experiments (Phillips et al., 2001; McGuigan et al., 2005; Whitlock et al., 2002).

The Supplementary Material provides details to support conclusions and arrive at expressions presented throughout the main text. Supplementary Material, Section 1 summarizes numerical implementations of the framework. Detailed derivation of the deterministic versions of the framework are given in Supplementary Material, Section 2. section 3 of the Supplement formally interfaces the continuous-time framework presented here with discrete time models of classical quantitative genetics. The mathematical foundation of this framework is communicated in Supplementary Material, Section 4. Using this foundation, Section 5 of the Supplement presents detailed calculations for the derivation of the stochastic versions of the framework presented in the main text. In Supplementary Material, Section 6, I derive the dynamics of genetic correlations responding to drift. An outline for one approach to derive this framework from a diffusion-limit is given in Section 7 of the Supplement. Finally, Supplementary Material, Section 8 formalizes a connection with Gillespie's work on the evolutionary consequences of offspring number variation (Gillespie, 1974, 1975, 1977).

2. The framework

The framework tracks the dynamics of the population density across trait space for asexually reproducing populations. To model d-dimensional traits, I assume trait space is the entire Euclidean space $\mathbb{R}^d$. Given the d-dimensional trait $\mathbf{z} = (z_1, \dots, z_d)^\top$ (with $^\top$ denoting matrix transposition so $\mathbf{z}$ is a column vector), I write $\nu(\mathbf{z})$ for the population density at $\mathbf{z}$. Then the total abundance of the population is given by $n = \int_{\mathbb{R}^d} \nu(\mathbf{z})\, d\mathbf{z}$. The distribution of the trait $\mathbf{z}$ is then $p(\mathbf{z}) = \nu(\mathbf{z})/n$, and this is also referred to as the relative abundance of $\mathbf{z}$. Using $p(\mathbf{z})$, the mean trait vector is given by $\bar{\mathbf{z}} = \int_{\mathbb{R}^d} \mathbf{z}\, p(\mathbf{z})\, d\mathbf{z}$, and the trait covariance matrix is $\mathbf{P} = \int_{\mathbb{R}^d} (\mathbf{z} - \bar{\mathbf{z}})(\mathbf{z} - \bar{\mathbf{z}})^\top p(\mathbf{z})\, d\mathbf{z}$, with P_{ij} being the covariance between z_i and z_j. The chosen notation facilitates comparison with classical multivariate quantitative genetic models (Lande, 1980; Lande and Arnold, 1983; Jones et al., 2003; Arnold et al., 2008).

In the following Section 2.1 I describe how to obtain the deterministic version of the framework using multivariate calculus. I do this in two parts. The first part (*DC*) makes no assumption on the shape of $p(\mathbf{z})$. The second part (*DG*) assumes $p(\mathbf{z})$ is the density of a multivariate normal distribution. I continue to make this assumption in Section 2.2 where I introduce the stochastic extension of this framework.

2.1. Deterministic dynamics

2.1.1. The deterministic covariance version (DC)

To establish a flexible, but tractable framework to model the dynamics of n, $\bar{\mathbf{z}}$, and $\mathbf{P}$, I generalize the deterministic model used by Week et al. (2021), referred to there as the Deterministic Asexual Gaussian Allelic model (DAGA), to the multivariate setting. Specifically, DAGA focuses on dynamics due to just mutation and selection for asexually reproducing populations. Mutation is modeled as the variance $\mathbf{M}$ of a normal distribution determining offspring traits centered on their parental traits, and selection results from the covariance of fitness and phenotype.

In the multivariate trait setting, mutation is modeled as a $d \times d$ covariance matrix $\mathbf{M}$, with the off-diagonal entries contributing to trait covariances. Here, I assume $\mathbf{M}$ does not depend on $\mathbf{z}$. This setup fits within the conceptual framework that has been popular for theoretical and empirical studies of $\mathbf{G}$-matrix evolution (Jones et al., 2007; Arnold et al., 2008; Mallard et al., 2023a). Fitness is quantified by a rate $m(\nu, \mathbf{z})$, which is the growth rate for the sub-population of individuals with trait value $\mathbf{z}$ in a population summarized by ν. The dependency of m on ν and $\mathbf{z}$ allows for the modelling of interwoven ecological and evolutionary dynamics. I will also refer to $m(\nu, \mathbf{z})$ as a *fitness function*, and this function may also depend on environmental parameters such as the trait values of individuals in interacting species. However, I omit notation accounting for such possibilities to simplify the frameworks presentation. Putting these components together, the multivariate generalization of DAGA is given by the partial differential equation

$$\dot{\nu}(\mathbf{z}) = m(\nu, \mathbf{z})\, \nu(\mathbf{z}) + \tfrac{1}{2} \nabla^\top \mathbf{M} \nabla\, \nu(\mathbf{z}), \tag{1}$$

where $\dot{\nu}(\mathbf{z})$ is the instantaneous rate of change of $\nu(\mathbf{z})$ over time, the symbol $\nabla := (\partial/\partial z_1, \dots, \partial/\partial z_d)^\top$ is the gradient operator with respect to the d-dimensional trait $\mathbf{z}$, and

$$\tfrac{1}{2} \nabla^\top \mathbf{M} \nabla := \tfrac{1}{2} \sum_{i,j=1}^{d} M_{ij} \frac{\partial}{\partial z_i} \frac{\partial}{\partial z_j} \tag{2}$$

is the mutation operator. Eq. (1) can be used to obtain a multivariate generalization of Kimura's continuum-of-alleles model (Kimura, 1965) that assumes diffusion instead of convolution with an arbitrary mutation kernel. Justification for multivariate DAGA Eq. (1) as a deterministic approximation of the full stochastic framework is given in Supplementary Material, Section 7.

If $M_{ij} = 0$ when $i \neq j$ and $M_{ii} = M_{jj}$ is constant across all ij, then the above mutation operator becomes proportional to the Laplacian operator on $\mathbb{R}^d$, which causes symmetric diffusion of the population density. Hence, unequal entries imply that mutation results in asymmetric diffusion across trait space so that mutation affects some traits more than others. Additionally, if $M_{ij} \neq 0$ when $i \neq j$, mutation contributes to covariance between traits z_i and z_j. This model of mutation can be obtained from a diffusion-limit of an individual-based model that assumes independence of reproduction and mutation in which the phenotypic effect size of mutation is small (see Supplementary Material, Section 7). A model where mutation and reproduction interact has been studied by Wickman et al. (2023).

Mathematically, assuming n is finite allows us to write $\dot{n} = \int_{\mathbb{R}^d} \dot{\nu}(\mathbf{z})\, d\mathbf{z}$. We can then apply integration-by-parts to obtain

$$\dot{n} = \bar{m}\, n, \tag{3}$$

where $\bar{m}(\nu) = \int_{\mathbb{R}^d} m(\nu, \mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$ is mean fitness. Biologically, n should be very large since drift is being ignored. However, this approach can still be useful for gaining insights when n is small, which occurs for example with Lotka-Volterra models (Akjouj et al., 2024). A version of Eq. (3) based on a finite number of genotypes (in place of continuous trait variation) is given in Crow and Kimura (1970) on page 10 above their equation 1.2.7.

Dynamics of the mean trait vector are obtained by applying the quotient rule and integration-by-parts to $\dot{\bar{\mathbf{z}}} = \int_{\mathbb{R}^d} \mathbf{z}\, \dot{p}(\mathbf{z})\, d\mathbf{z}$, which provides

$$\dot{\bar{\mathbf{z}}} = \mathrm{Cov}(m, \mathbf{z}), \tag{4}$$

where $\mathrm{Cov}(m, \mathbf{z})$ is a d-dimensional vector with ith entry given by

$$\mathrm{Cov}(m, z_i) = \int_{\mathbb{R}^d} \left(m(\nu, \mathbf{z}) - \bar{m}(\nu) \right) (z_i - \bar{z}_i)\, p(\mathbf{z})\, d\mathbf{z}, \tag{5}$$

where $\bar{z}_i$ is the ith entry of $\bar{\mathbf{z}}$. Eq. (4) is similar to the Price equation without transmission bias in that evolutionary change is expressed as a covariance between fitness and trait value (Price, 1970, 1972) . However, unlike the Price equation (which is a general identity) the present result is derived from an explicit population dynamical model and therefore depends on specific biological and analytical assumptions.

The same techniques can also be applied to obtain $\dot{\mathbf{P}}$ as

$$\dot{\mathbf{P}} = \mathbf{M} + \mathrm{Cov}(m, (\mathbf{z} - \bar{\mathbf{z}})(\mathbf{z} - \bar{\mathbf{z}})^\top), \tag{6}$$

where $\mathrm{Cov}(m, (\mathbf{z} - \bar{\mathbf{z}})(\mathbf{z} - \bar{\mathbf{z}})^\top)$ is a $d \times d$ matrix with ijth entry given by $\mathrm{Cov}(m, (z_i - \bar{z}_i)(z_j - \bar{z}_j))$. The response of the trait covariance matrix to selection as a covariance of fitness with $(\mathbf{z} - \bar{\mathbf{z}})(\mathbf{z} - \bar{\mathbf{z}})^\top$ was previously found by Lande and Arnold (1983), without an explicit mutation term. An expression including $\mathbf{M}$ was later presented by Phillips and McGuigan (2006), but not derived from an underlying dynamical model. The steps required to arrive at Eqs. (3)–(6) are detailed in the Supplementary Material, Section 2.

Eqs. (3), (4), and (6) follow from the underlying trait-structured dynamical model and make no assumptions on the form of the trait distribution. Consequently, the model can be used to study populations with non-trivial higher moments, such as skewed or otherwise non-Gaussian distributions. However, this flexibility of accommodating general trait distributions comes at the cost of a limited range of fitness functions that are amenable to study. This is due to i) the challenge of calculating covariances between arbitrary fitness functions and phenotypic moments, and ii) moment-closing issues that often arise during these calculations (Barton and Turelli, 1987; Gilpin and Feldman, 2019; Guerand et al., 2023). In spite of this limitation, a family of biologically important fitness functions are tractable to study using *DC*. These function take the form

$$m(\nu, \mathbf{z}) = r + \mathbf{b}^\top \mathbf{z} - \tfrac{1}{2} (\boldsymbol{\theta} - \mathbf{z})^\top \boldsymbol{\Psi} (\boldsymbol{\theta} - \mathbf{z}) - c\, n, \tag{7}$$

where $\mathbf{b} = (b_1, \dots, b_d)^\top$ (which does not depend on $\mathbf{z}$) confers directional selection (positive or negative for trait z_i depending on the sign of b_i), and $\boldsymbol{\Psi}$ is a symmetric matrix that mediates stabilizing or disruptive selection around the vector $\boldsymbol{\theta}$. Classically, $\boldsymbol{\Psi}$ is denoted $\boldsymbol{\omega}^{-1}$ in quantitative genetic theory (e.g., Lande, 1979; Jones et al., 2012). The absence of stabilizing selection under the classical notation occurs when all eigenvalues of $\boldsymbol{\omega}$ diverge towards $+\infty$. To avoid this technicality, I use $\boldsymbol{\Psi}$ so that

the absence of stabilizing selection is associated with the zero matrix $\boldsymbol{\Psi} = \mathbf{0}$. Additionally, the parameter $c \geq 0$ is the strength of competition, which here acts globally between individuals regardless of trait, and r is the growth rate in the absence of selection and competition (i.e., when $\mathbf{b}, \boldsymbol{\Psi}$, and c are all zero). It is possible to generalize this family so that competition is non-global, but this requires additional technical details as the fitness function becomes operator-valued (which is treated, for example, by Volpert, 2014; and by Etheridge et al., 2024). Supplementary Material, 3 connects these fitness functions to those known from classical discrete time quantitative genetic models.

Combining Eqs. (3), (4), and (6) with Eq. (7) provides (derivation given in Supplementary Material, Section 2.3)

$$\dot{n} = \left(r + \mathbf{b}^\top \bar{\mathbf{z}} - \tfrac{1}{2}(\boldsymbol{\theta} - \bar{\mathbf{z}})^\top \boldsymbol{\Psi} (\boldsymbol{\theta} - \bar{\mathbf{z}}) - \tfrac{1}{2}\mathrm{tr}(\boldsymbol{\Psi}\mathbf{P}) - c\, n \right) n, \tag{8a}$$

$$\dot{\bar{\mathbf{z}}} = \mathbf{P}\,\mathbf{b} + \mathbf{P}\,\boldsymbol{\Psi}\,(\boldsymbol{\theta} - \bar{\mathbf{z}}) - \tfrac{1}{2}\mathbf{S} : \boldsymbol{\Psi}, \tag{8b}$$

$$\dot{\mathbf{P}} = \mathbf{M} + \tfrac{1}{2}\mathrm{tr}(\boldsymbol{\Psi}\mathbf{P})\,\mathbf{P} + \mathbf{S} \cdot (\boldsymbol{\Psi}(\boldsymbol{\theta} - \bar{\mathbf{z}})) + \mathbf{S} \cdot \mathbf{b} - \tfrac{1}{2}\,\mathbf{K} : \boldsymbol{\Psi}, \tag{8c}$$

where $\mathbf{S}$ is the third-order skew tensor of $\mathbf{z}$ defined by $S_{ijk} = \int_{\mathbb{R}^d} (z_i - \bar{z}_i)(z_j - \bar{z}_j)(z_k - \bar{z}_k) p(\mathbf{z}) d\mathbf{z}$. Products with $\mathbf{S}$ are given by $(\mathbf{S} : \boldsymbol{\Psi})_i = \sum_{jk} S_{jki} \Psi_{jk}$, $(\mathbf{S} \cdot \boldsymbol{\Psi})_{ij} = \sum_k S_{ijk} \Psi_{kj}$, and $(\mathbf{S} \cdot \mathbf{b})_{ij} = \sum_k S_{ijk} b_k$. In addition, $\mathbf{K}$ is the fourth-order kurtosis tensor of $\mathbf{z}$ defined by $K_{ijkl} = \int_{\mathbb{R}^d} (z_i - \bar{z}_i)(z_j - \bar{z}_j)(z_k - \bar{z}_k)(z_l - \bar{z}_l) p(\mathbf{z}) d\mathbf{z}$. The product with $\mathbf{K}$ is given by $(\mathbf{K} : \boldsymbol{\Psi})_{ij} = \sum_{kl} K_{ijkl} \Psi_{kl}$. Eq. (8) collects the abundance, mean, and covariance dynamics (including skewness and kurtosis contributions) into a single system for clarity. While related formulations exist, this particular consolidation does not appear to be standard in the literature.

Eq. (8) demonstrate that, for fitness functions taking the form of (7), the dynamics of abundance, mean trait, and trait covariance matrix depend on higher phenotypic moments for non-normal trait distributions. In fact, a complete description requires an infinite number of equations. However, by combining (7) with multivariate DAGA (i.e., with (1)), I demonstrate in Supplementary Material, Section 2.5 that, when $\boldsymbol{\Psi}$ is positive definite, $\nu(\mathbf{z})$ has a globally asymptotically stable equilibrium proportional to the density of a multivariate normal distribution with covariance matrix

$$\hat{\mathbf{P}} = \boldsymbol{\Psi}^{-1/2}(\boldsymbol{\Psi}^{1/2}\mathbf{M}\boldsymbol{\Psi}^{1/2})^{1/2}\boldsymbol{\Psi}^{-1/2} \tag{9}$$

(with matrix square roots defined using eigenvalue decomposition: $\mathbf{Q} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{-1}$ implies $\mathbf{Q}^{1/2} = \mathbf{U}\boldsymbol{\Lambda}^{1/2}\mathbf{U}^{-1}$) and equilibrium mean vector

$$\hat{\bar{\mathbf{z}}} = \boldsymbol{\Psi}^{-1}\mathbf{b} + \boldsymbol{\theta}, \tag{10}$$

and with a total abundance

$$\hat{n} = \frac{1}{c}\left(r + \mathbf{b}^\top\boldsymbol{\theta} + \tfrac{1}{2}\mathbf{b}^\top\boldsymbol{\Psi}^{-1}\mathbf{b} - \tfrac{1}{2}\mathrm{tr}\left((\boldsymbol{\Psi}^{1/2}\mathbf{M}\boldsymbol{\Psi}^{1/2})^{1/2}\right)\right). \tag{11}$$

Derivations for these equilibrium expressions are given in Supplementary Material, Section 2.5.1.

The mutation-selection balance of phenotypic variance $\hat{\mathbf{P}}$ is a multivariate generalization of the univariate classical result obtained from multi-locus models (e.g., Bulmer, 1972) and continuum-of-alleles models (e.g., Bürger, 1986). Lande (1980) found a similar result using a multilocus approach that included recombination. In particular, $\hat{\mathbf{P}}$ has the same matrix-square-root structure as the equilibrium found by Lande (1980), and the two coincide when linkage is negligible and when $\mathbf{M}$ corresponds to the sum of Lande's locus-specific mutation matrices. Further work is needed to extend these results for more general fitness functions and higher phenotypic moments, which are already known to have important evolutionary consequences (Débarre et al., 2015). A new approach to study the dynamics of higher moments for univariate traits was recently introduced by Gilpin and Feldman (2019).

2.1.2. The deterministic gradient version (DG)

As noted above, deriving population dynamics from arbitrary fitness functions can be challenging in the more general framework based on covariances between fitness and phenotype. To overcome this we can make the useful simplifying assumption that traits follow a multivariate normal distribution. Under this assumption, all phenotypic moments (such as skew and kurtosis) can be expressed in terms of the mean and variance which greatly simplifies expressions such as Eq. (8). While deviations from normality can have consequences for ecological and evolutionary processes (Turelli, 1988; Débarre et al., 2015), normality has been an important initial assumption for studying a wide range of topics such as reaction-norm evolution (Lande, 2014), coevolving mutualistic networks (Nuismer et al., 2018), and niche construction (Fogarty and Wade, 2022). Furthermore, normality is a well-established approximation that holds under many genetic and selective scenarios (Turelli and Barton, 1994). Then, as a first pass, I assume traits are multivariate normally distributed for the rest of this paper.

In the context of this framework, the assumption of multivariate normality is particularly useful because it allows us to rewrite covariances between fitness and phenotype as gradients of fitness functions with respect to moments of the trait distribution. Such gradients can be analytically calculated for a broad range of fitness functions. The calculations to obtain these expressions begin with the definitions of covariances between fitness and phenotype, and then apply properties of the multivariate Gaussian function and integration-by-parts (see Supplementary Material, Section 2.3). As a result, the deterministic *DG* version of the framework is given by

$$\dot{n} = \bar{m}\, n, \tag{12a}$$

$$\dot{\bar{\mathbf{z}}} = \mathbf{P}\,(\nabla_{\bar{\mathbf{z}}}\,\bar{m} - \overline{\nabla_{\bar{\mathbf{z}}}\, m}), \tag{12b}$$

$$\dot{\mathbf{P}} = \mathbf{M} + 2\,\mathbf{P}\,(\nabla_{\mathbf{P}}\,\bar{m} - \overline{\nabla_{\mathbf{P}}\, m})\,\mathbf{P}, \tag{12c}$$

where $\nabla_{\bar{\mathbf{z}}}\,\bar{m}$ and $\overline{\nabla_{\bar{\mathbf{z}}}\, m}$ are d-dimensional vectors that respectively capture the effects of frequency independent and frequency dependent selection on mean trait evolution. More precisely, writing $\partial_i = \partial/\partial\bar{z}_i$ as the partial derivative operator with respect to the ith mean trait, the ith entry of $\nabla_{\bar{\mathbf{z}}}\,\bar{m}$ and $\overline{\nabla_{\bar{\mathbf{z}}}\, m}$ are respectively given by $\partial_i\,\bar{m}$ and $\overline{\partial_i\, m} = \int_{\mathbb{R}^d} p(\mathbf{z})\,[\,\partial_i\, m(\nu, \mathbf{z})\,]\, d\mathbf{z}$. Similarly, $\nabla_{\mathbf{P}}\,\bar{m}$ and $\overline{\nabla_{\mathbf{P}}\, m}$ are $d \times d$ matrices that respectively quantify the dependence of fitness on trait variances and covariances. Writing $\partial_{ij} = \partial/\partial P_{ij}$ as the partial derivative operator with respect to the covariance between trait components i and j, the ijth entries of $\nabla_{\mathbf{P}}\,\bar{m}$ and $\overline{\nabla_{\mathbf{P}}\, m}$ are respectively given by $\partial_{ij}\,\bar{m}$ and $\overline{\partial_{ij}\, m} = \int_{\mathbb{R}^d} p(\mathbf{z})\,[\,\partial_{ij}\, m(\nu, \mathbf{z})\,]\, d\mathbf{z}$. Although m is not written explicitly as a function of $\bar{\mathbf{z}}$ or $\mathbf{P}$, it depends on them through $\nu(\mathbf{z})$. The derivatives of m above are therefore understood as sensitivities of m to perturbations of the distribution that alter $\bar{\mathbf{z}}$ or $\mathbf{P}$, and are not zero in general.

In Supplementary Material, Section 2.3, I show how to apply integration-by-parts to obtain the alternative expressions $\dot{\bar{\mathbf{z}}} = \mathbf{P}\,\overline{\nabla m}$ and $\dot{\mathbf{P}} = \mathbf{M} + \mathbf{P}\,\overline{\nabla^2 m}\,\mathbf{P}$ where $(\nabla)_i = \partial/\partial z_i$ (so $\overline{\nabla m}$ is the trait averaged gradient of fitness with respect to phenotype $\mathbf{z}$) and $(\nabla^2)_{ij} = \partial^2/\partial z_i \partial z_j$ (so $\overline{\nabla^2 m}$ is the trait averaged Hessian of fitness with respect to phenotype $\mathbf{z}$). A discrete-time counterpart to $\overline{\nabla^2 m}$ is given by equation (14b) in Lande and Arnold (1983).

The Hessian form for the response of $\mathbf{P}$ to selection is similar to previously known discrete-time expressions (Lande and Arnold, 1983; Débarre et al., 2014; Phillips and Arnold, 1989; Mullon and Lehmann, 2019). An important difference is that the discrete-time expressions include an additional term equal to the outer product of the per-generation response of the mean trait to selection, commonly denoted $(\Delta\bar{\mathbf{z}})(\Delta\bar{\mathbf{z}})^\top$, which appears with a negative sign in the discrete-time update of $\mathbf{P}$. However, this extra term disappears under weak selection approximations often made in discrete-time quantitative genetic models (Lande, 1976, 1982; Turelli, 1984; Barton and Turelli, 1989; Turelli and Barton, 1990; Lynch and Walsh, 1998; Bürger, 2000; Walsh and Lynch, 2018). Furthermore, under a continuous-time scaling, the mean changes only $O(\Delta t)$ over a small interval, so its contribution to covariance is $O(\Delta t^2)$. Dividing by Δt to form a rate equation and letting $\Delta t \to 0$ removes this quadratic term, leaving only the curvature-driven (Hessian) part of selection in the continuous-time covariance dynamics, in agreement with Eq. (12c).

Mean trait and trait covariance dynamics can be expressed in index form as

$$\dot{\bar{z}}_i = \sum_{j=1}^{d} P_{ij}\,(\partial_j\,\bar{m} - \overline{\partial_j\,m}), \tag{13a}$$

$$\dot{P}_{ij} = M_{ij} + 2\sum_{k,l=1}^{d} P_{ik}\,(\partial_{kl}\,\bar{m} - \overline{\partial_{kl}\,m})\,P_{lj}. \tag{13b}$$

Inclusion of the terms $\overline{\nabla_{\bar{\mathbf{z}}}\,m}$ and $\overline{\nabla_{\mathbf{P}}\,m}$ in Eqs. (12b) and (12c) make it clear that this framework accounts for frequency-dependent selection. The decomposition of selection into components arising from frequency-dependent and frequency-independent sources has been made explicit in previous quantitative genetic treatments of mean trait dynamics (see eqn. (9) of Lande, 1976), but was only implicit in models of trait variance evolution due to their expressions in terms of Hessians (Lande and Arnold, 1983; Débarre et al., 2014; Phillips and Arnold, 1989; Mullon and Lehmann, 2019). A decomposition of the Hessian of fitness into frequency-dependent and frequency-independent components does not appear to be standard in existing treatments of trait-covariance evolution. Additionally, the approach outlined here can make a useful complement to adaptive dynamics and evolutionary game-theoretic treatments of frequency-dependent selection (Dieckmann and Law, 1996; Hofbauer and Sigmund, 1998; Traulsen et al., 2005).

An important caveat of both *DC* and *DG* versions of the framework taking the form presented above is that they rely on perfect heritability of trait values. However, traits are in general not perfectly heritable, and this is of fundamental importance in evolutionary biology. I therefore address this additional complexity in the following sub-subsection. I note here that extending *DC* and *DG* to include imperfect heritability results in nearly identical equations, the important difference is that average growth rates must additionally be averaged over a phenotypic residual term. I state this explicitly for the *DG* version below.

2.1.3. Imperfect inheritance

Following classical quantitative genetics, imperfect inheritance of trait values can be accounted for by assuming traits decompose into an additive genetic component and a residual component: $\mathbf{z} = \mathbf{g} + \mathbf{e}$ (Lynch and Walsh, 1998).

Biologically, this decomposition assumes that non-additive genetic effects and genotype-environment interactions are absent, so that $\mathbf{g}$ reflects only additive genetic effects, while $\mathbf{e}$ reflects the remaining non-heritable factors. This justifies interpreting $\mathbf{e}$ as the residual from fitting a linear statistical model of trait values to genetic predictors (Falconer and Mackay, 1996; Lynch and Walsh, 1998). The absence of non-additive genetic effects is especially important in this asexual framework, where the lack of recombination would otherwise allow multilocus allelic associations to render such effects heritable (Barton and Turelli, 1991). It is further assumed that the only trait-mediating factors that are heritable are additive genetic factors (i.e., cytoplasmic inheritance, persistent epigenetic marks, maternal effects, microbiome transmission, and cultural inheritance are all excluded). Under these assumptions, $\mathbf{e}$ is uncorrelated with $\mathbf{g}$ and is uncorrelated among individuals, consistent with interpreting $\mathbf{g}$ and $\mathbf{e}$ as the genetic predictor and residual in a linear model relating trait values to genetic variation.

In contrast to $\mathbf{e}$, the additive genetic component $\mathbf{g}$ follows the same Gaussian mutation model described above. In particular, given that $\mathbf{g}$ is the additive genetic component of the trait of a parent, an offspring will have an additive genetic component that is multivariate normally distributed with mean $\mathbf{g}$ and covariance matrix $\mathbf{M}$.

Assuming the residual $\mathbf{e}$ is independent and identically distributed for all individuals with mean zero and covariance matrix $\mathbf{E}$, and denoting $\gamma(\mathbf{g})$ the population density at genetic value $\mathbf{g}$, and $\varepsilon(\mathbf{e})$ the distribution of the residuals, the population density of trait values is given by $\nu(\mathbf{z}) = \int_{\mathbb{R}^d} \varepsilon(\mathbf{e})\gamma(\mathbf{z}-\mathbf{e})d\mathbf{e}$. This preserves total abundance so that $n = \int_{\mathbb{R}^d} \nu(\mathbf{z})d\mathbf{z} = \int_{\mathbb{R}^d} \gamma(\mathbf{g})d\mathbf{g}$. These assumptions also imply that the growth rate for the sub-population of individuals carrying genetic value $\mathbf{g}$ is given by $\tilde{m}(\gamma,\mathbf{g}) = \int_{\mathbb{R}^d} \varepsilon(\mathbf{e})m(\nu,\mathbf{g}+\mathbf{e})d\mathbf{e}$. Furthermore, the fitness function for trait values $m(\nu,\mathbf{z})$ and the fitness function for additive genetic values $\tilde{m}(\gamma,\mathbf{g})$ have the same mean value across the population

$$\begin{aligned}\bar{\tilde{m}} &= \frac{1}{n}\int_{\mathbb{R}^d} \tilde{m}(\gamma,\mathbf{g})\,\gamma(\mathbf{g})\,d\mathbf{g} \\ &= \frac{1}{n}\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} m(\nu,\mathbf{z})\,\varepsilon(\mathbf{e})\,\gamma(\mathbf{z}-\mathbf{e})\,d\mathbf{e}\,d\mathbf{z} = \frac{1}{n}\int_{\mathbb{R}^d} m(\nu,\mathbf{z})\,\nu(\mathbf{z})\,d\mathbf{z} = \bar{m}.\end{aligned} \tag{14}$$

This approach to deriving the fitness associated with an additive genetic value given the fitness associated with a trait value and the distribution of residuals was originally introduced by Kimura and Crow (1978). It also appears in Eq. (4) of Lande (1979). The dynamics of γ are given in analogy to $\dot{\nu}$ as

$$\dot{\gamma}(\mathbf{g}) = \tilde{m}(\gamma,\mathbf{g})\,\gamma(\mathbf{g}) + \tfrac{1}{2}\nabla^{\top}\mathbf{M}\nabla\,\gamma(\mathbf{g}). \tag{15}$$

The distribution of additive genetic values in the population is given by $\tilde{p}(\mathbf{g}) = \gamma(\mathbf{g})/n$. In combination with the model for phenotypic residuals, the mean trait vector is calculated as $\bar{\mathbf{z}} = \bar{\mathbf{g}}$ and the trait covariance matrix is $\mathbf{P} = \mathbf{G} + \mathbf{E}$. To simplify calculating fitness, and to accommodate the assumption of multivariate normal traits, I assume $\varepsilon(\mathbf{e})$ is the density of a multivariate normal distribution (with mean zero and covariance matrix $\mathbf{E}$).

Under these assumptions the expression for abundance dynamics does not change, but the mean trait dynamics can be calculated as

$$\dot{\bar{\mathbf{z}}} = \mathbf{G}\,(\nabla_{\bar{\mathbf{z}}}\,\bar{m} - \overline{\nabla_{\bar{\mathbf{z}}}\,m}), \tag{16}$$

and $\mathbf{G}$-matrix dynamics are given by

$$\dot{\mathbf{G}} = \mathbf{M} + 2\,\mathbf{G}\,(\nabla_{\mathbf{G}}\,\bar{m} - \overline{\nabla_{\mathbf{G}}\,m})\,\mathbf{G}. \tag{17}$$

I apply this model of imperfect inheritance while describing the stochastic extensions of the framework below. Further details about how imperfect inheritance is interfaced with the stochastic extension of this framework, and how to obtain Eqs. (16) and (17) are provided in the Supplementary Material, Section 5.

In summary, this section derives deterministic equations for abundance, mean traits, and trait covariances from the DAGA formulation (1). It presents both the non-closed moment system for general trait distributions (Eqs. (3)–(8)) and the closed gradient/Hessian form obtained under normality with imperfect inheritance (Eqs. (12a), (16), and (17)), which the stochastic extensions build on below.

2.2. Stochastic dynamics

In this section, I extend the framework to the case where demographic stochasticity induces random genetic drift. This extension introduces the stochastic version of the framework and builds on the Deterministic Gradient (*DG*) formulation developed above. Two closely related stochastic variants are presented.

The first variant, called the Brownian Motion Gradient version (*BG*), expresses stochastic dynamics with respect to Brownian motion processes, as is standard for stochastic differential equations (Øksendal, 2013; Evans, 2012). *BG* is particularly useful for numerical applications, and can be implemented using the Euler-Maruyama algorithm (Bayram et al., 2018). I illustrate this during a brief study of genetic correlations evolving in response to random genetic drift. Further information is provided in Supplementary Material, Section 1. Implementations of this approach using the DifferentialEquations.jl package in Julia (Rackauckas and Nie, 2017), and a manual implementation are provided at the Zenodo repository https://doi.org/10.5281/zenodo.19087096.

The second variant, called the Martingale Gradient version (*MG*), is useful for deriving the dynamics of more specific quantities. This use-case is illustrated below to formally obtain a stochastic differential equation tracking the evolution of genetic correlations in response to random genetic drift. To do so, I introduce a powerful set of novel heuristics

(i.e., practical rules that allow exact calculations without requiring familiarity with the underlying measure-valued martingale theory, see Table 1) that can be used to derive an array of models from this variant of the framework. These heuristics are obtained by examining a so-called martingale process underlying this entire framework, which is also the namesake of *MG*. Mathematical details are provided in Section 4 of the Supplement.

Both stochastic extensions of the framework include an additional variable v, which is associated with the variance of lifetime reproductive output of individuals. Supplementary Material, Section 7.1 shows how to derive v from an individual-based model. This parameter can be interpreted as the population average of what Gillespie (1974, 1975, 1977) calls the variance in offspring number. Supplementary Material, Section 8 formally demonstrates this interpretation. Importantly, v can also be intrepreted as either the rate of reproduction for semelparous populations (demonstrated in Supplementary Material, Section 7.1) or the sum of birth and death rates in iteroparous populations (discussed in Week et al., 2021). I refer to v as the *reproductive variance*.

Previous work has shown that a novel form of selection can emerge when v depends on trait value, referred to as *noise-induced selection*, and for which evolutionary responses behave radically different from classical deterministic selection (Gillespie, 1974, 1977; Constable et al., 2016; Parsons et al., 2010; Kuosmanen et al., 2022; Bhat and Guttal, 2025). However, accounting for noise-induced selection leads to significantly more complex expressions for the evolution of mean trait vector and trait covariance matrix (e.g., see Week et al., 2021, for the univariate case). While genotype-dependent differences in the variance of offspring number have been documented in laboratory asexual systems (Turner and Chao, 1999), there is no evidence that such covariance is widespread, and selection acting explicitly on this variance component has not been directly measured in empirical populations. Then, as a first pass, I keep the expressions relatively simple by assuming v is constant across additive genetic values.

2.2.1. The brownian motion gradient version (BG)

Accounting for the effects of demographic stochasticity, the abundance dynamics can now be expressed as the following stochastic differential equation

$$dn = \bar{m}\, n\, dt + \sqrt{v\, n}\, dB_n, \tag{18}$$

where the noise process B_n is a standard Brownian motion (i.e., $B_n(t)$ has variance equal to t and $B_n(0) = 0$). Assuming multivariate normality, and the above model of imperfect inheritance, the mean trait dynamics (derived in Supplementary Material, Section 5.3) are given by

$$d\bar{\mathbf{z}} = \mathbf{G}\,(\nabla_{\bar{\mathbf{z}}}\, \bar{m} - \overline{\nabla_{\bar{\mathbf{z}}}\, m})\, dt + \sqrt{\frac{v}{n}}\, \mathbf{G}\, d\mathbf{B}_{\bar{\mathbf{z}}}, \tag{19}$$

with $\mathbf{B}_{\bar{\mathbf{z}}}$ a d-dimensional vector of independent standard Brownian motions, and the matrix square root $\sqrt{\mathbf{G}}$ can be any $d \times d$ matrix $\mathbf{A}$ that satisfies $\mathbf{A}\mathbf{A}^{\top} = \mathbf{G}$ (for examples, see Kessy et al., 2018). Eq. (19) provides a continuous-time extension to the framework of multivariate evolution introduced by Lande (1979). Alternatively, the stochastic dynamics of mean traits can be expressed using index notation as

$$d\bar{z}_i = \sum_{j=1}^{d} G_{ij}\, (\partial_j\, \bar{m} + \overline{\partial_j\, m})\, dt + \sqrt{\frac{v}{n}\, G_{ii}}\, dB_{\bar{z}_i}, \tag{20}$$

where $\partial_i = \partial/\partial \bar{z}_i$ and, for each i, $B_{\bar{z}_i}$ is a standard Brownian motion and non-independence for each ij is encoded by the heuristic $dB_{\bar{z}_i}\, dB_{\bar{z}_j} = \rho_{ij}\, dt$, with $\rho_{ij} = G_{ij}/\sqrt{G_{ii}\, G_{jj}}$ the genetic correlation between traits i and j. This heuristic is particularly useful when applying Itô's formula (the stochastic analog of the chain rule, see Klebaner, 1998; Øksendal, 2013; or Evans, 2012) to derive dynamics for functions of mean trait values. This approach may be used, for example, to track the dynamics of interaction coefficients for coevolving species as a multivariate generalization of the approach taken by Week and Nuismer (2021).

In Eq. (19) the correlated effects of genetic drift on mean trait evolution are encoded by the product $(\sqrt{\mathbf{G}}\, d\mathbf{B}_{\bar{\mathbf{z}}})_i = \sum_j (\sqrt{\mathbf{G}})_{ij} (d\mathbf{B}_{\bar{\mathbf{z}}})_j$. In contrast, for Eq. (20), these correlated effects are encoded directly by the non-independence of the Brownian motions $B_{\bar{z}_1}, \ldots, B_{\bar{z}_d}$. In particular, because $\mathbf{B}_{\bar{\mathbf{z}}}$ is defined such that its entries are independent, the ith entry of $d\mathbf{B}_{\bar{\mathbf{z}}}$ cannot be equal to $dB_{\bar{z}_i}$ as $(d\mathbf{B}_{\bar{\mathbf{z}}})_i\, (d\mathbf{B}_{\bar{\mathbf{z}}})_j = \delta_{ij}\, dt$, where $\delta_{ij} = 1$ when $i = j$ and zero otherwise. These heuristics work in the absence of multivariate normality, but the deterministic component of $d\bar{\mathbf{z}}$ expressed above does depend on multivariate normality.

The expression of $d\bar{z}_i$ in Eq. (20) using non-independent Brownian motions is not standard. However, this form will be useful below to illustrate the relationship between the *BM* (Eqs. (18), (20) and (23)) and *MG* (Eqs. (26)) versions of the framework.

Supplementary Material, Section 5.4 shows that the application of the multivariate normal approximation to the stochastic dynamics of the $\mathbf{G}$-matrix leads to the matrix equation

$$d\mathbf{G} = \left(\mathbf{M} + 2\,\mathbf{G}\,(\nabla_{\mathbf{G}}\, \bar{m} - \overline{\nabla_{\mathbf{G}}\, m})\,\mathbf{G} - \frac{v}{n}\,\mathbf{G}\right) dt + \sqrt{\frac{v}{n}}\, \boldsymbol{\Gamma} \,:\, d\mathbf{B}_{\mathbf{G}}, \tag{21}$$

where $\boldsymbol{\Gamma}$ is a fourth-order tensor describing the covariance structure for the response of $\mathbf{G}$ to drift. Defining products of fourth-order tensors $\mathbf{X}$ and $\mathbf{Y}$ by $(\mathbf{X} : \mathbf{Y})_{ijkl} = \sum_{mn} \mathbf{X}_{ijmn} \mathbf{Y}_{mnkl}$, we can write $\boldsymbol{\Gamma} = \sqrt{\boldsymbol{\Gamma}} : \sqrt{\boldsymbol{\Gamma}}$. Furthermore, we have $\sqrt{\boldsymbol{\Gamma}}_{ijkl} = \left(\sqrt{\mathbf{G}}_{ik} \sqrt{\mathbf{G}}_{jl} + \sqrt{\mathbf{G}}_{il} \sqrt{\mathbf{G}}_{jk}\right)/\sqrt{2}$, and $\sqrt{\mathbf{G}}_{ij}$ is the ijth entry of $\sqrt{\mathbf{G}}$ (as mentioned beneath Eq. (19) above, $\sqrt{\mathbf{G}}$ is any $d \times d$ matrix such that $\sqrt{\mathbf{G}}\sqrt{\mathbf{G}}^{\top} = \mathbf{G}$). Importantly, this implies $\sqrt{\mathbf{G}}_{ij}$ is *not* equal to $\sqrt{G_{ij}}$.

The product $\sqrt{\boldsymbol{\Gamma}} : d\mathbf{B}_{\mathbf{G}}$ returns a $d \times d$ matrix with ijth entry $(\sqrt{\boldsymbol{\Gamma}} : d\mathbf{B}_{\mathbf{G}})_{ij} = \sum_{kl} \sqrt{\boldsymbol{\Gamma}}_{ijkl}\, (d\mathbf{B}_{\mathbf{G}})_{kl}$. The $d \times d$ matrix-valued Brownian motion $\mathbf{B}_{\mathbf{G}}$ has independent standard Brownian motions along its diagonal entries. The off-diagonal entries are standard Brownian motions scaled by one-half (i.e., the variance of $(\mathbf{B}_{\mathbf{G}})_{ij}(t)$ is $t/2$ when $i \neq j$), and symmetric entries are equivalent so that $(\mathbf{B}_{\mathbf{G}})_{ij} = (\mathbf{B}_{\mathbf{G}})_{ji}$. The covariance structure of $\mathbf{B}_{\mathbf{G}}$ is summarized by the heuristic

$$(d\mathbf{B}_{\mathbf{G}})_{ij}\, (d\mathbf{B}_{\mathbf{G}})_{kl} = \frac{\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}}{2}\, dt. \tag{22}$$

Scaling the off-diagonals by one-half ensures variances and covariances are correctly propagated while calculating the stochastic consequences of drift for $\mathbf{G}$-matrix dynamics. Further information on symmetric normal matrices can be found in Gupta and Nagar (2018), particularly Theorem 2.5.1.

Unlike the expression for the stochastic component of $d\bar{\mathbf{z}}$, the stochastic component for $d\mathbf{G}$ does depend on the assumption of a multivariate normal trait distribution.

Using index notation, the expression (21) simplifies to a $d(d+1)/2$-dimensional system of equations summarized by

$$\begin{aligned} dG_{ij} = &\left(M_{ij} + 2 \sum_{kl} G_{ik}\, (\partial_{kl}\, \bar{m} - \overline{\partial_{kl}\, m})\, G_{lj} - \frac{v}{n}\, G_{ij}\right) dt \\ &+ \sqrt{\frac{v}{n}} \sqrt{G_{ii}\, G_{jj} + G_{ij}^2}\, dB_{G_{ij}}, \end{aligned} \tag{23}$$

for $1 \leq i \leq j \leq d$ where $\partial_{ij} = \partial/\partial G_{ij}$ and for each ij we have $B_{G_{ij}}$ is a standard Brownian motion with $B_{G_{ij}} = B_{G_{ji}}$. The covariance between $B_{G_{ij}}$ and $B_{G_{kl}}$ is encoded by the heuristic

$$dB_{G_{ij}}\, dB_{G_{kl}} = \frac{G_{ik}\, G_{jl} + G_{il}\, G_{jk}}{\sqrt{G_{ii}\, G_{jj} + G_{ij}^2}\, \sqrt{G_{kk}\, G_{ll} + G_{kl}^2}}\, dt. \tag{24}$$

Importantly, this implies that $dB_{G_{ij}}$ is *not* the ijth entry of $d\mathbf{B}_{\mathbf{G}}$.

Finally, the noise process driving abundance in general does not covary with the noise processes driving mean traits and trait covariances (i.e., $dB_n\, dB_{\bar{z}_i} = dB_n\, dB_{G_{kl}} = 0$), and, under the assumption of multivariate normality, the same holds for trait means and trait covariances (i.e., $dB_{\bar{z}_i}\, dB_{G_{kl}} = 0$). This agrees with the off-diagonal entries in 7b of

Barton (1989), but my results for the diagonal entries disagree. See Supplementary Material, Section 5.4 for more details.

The vector-matrix expression of *BG* (Eqs. (19) and (21)) is particularly well-suited for numerical exploration of models because the covariance structure of the noise processes associated with random genetic drift are written explicitly in terms of sums involving the entries of the matrix square root $\sqrt{\mathbf{G}}$. Hence, this version of the framework is easily interfaced with a common algorithm to numerically integrate systems of stochastic differential equations, the Euler-Maruyama method (Bayram et al., 2018). The numerical benefits of the *BG* version of the framework are highlighted in section 3 below to study **G**-matrix evolution.

The expression of *BG* in index notation (Eqs. (20) and (23)), while arguably more complex in appearance, are useful for deriving analytical results, and especially when applying Itô's formula to derive the dynamics of a quantity depending on $\bar{\mathbf{z}}$ and/or $\mathbf{G}$ (such as the dynamics of growth rate, $d\bar{m}$, for instance). However, there are limitations with this application of *BG*, particularly for dealing with sums of stochastic differentials such as $U = a\,d\,B_{G_{ij}} + b\,d\,B_{G_{kl}}$. If the Brownian motions $B_{G_{ij}}$ and $B_{G_{kl}}$ were independent, then we can write $U = \sqrt{a^2+b^2}\,dB$ with B being another standard Brownian motion. Because $B_{G_{ij}}$ and $B_{G_{kl}}$ are not independent, which is captured by Eq. (24) above, it is not obvious how to properly express U in terms of a, b and a single standard Brownian motion B.

To overcome these challenges, and also to simplify typical calculations, it is useful to rewrite the stochastic components of Eqs. (20) and (23) in terms of an underlying stochastic process $\mathcal{M}$. Because $\mathcal{M}$ satisfies a martingale property (roughly, this means $\mathbb{E}[\,\mathcal{M}(t+s)\,|\,\mathcal{M}(t)\,] = \mathcal{M}(t)$ for all $s > 0$), I refer to this version of the framework as the Martingale Gradient version (or *MG* for short). In the following subsection, I present this version of the framework in a way that minimizes technical pre-requisites, while maintaining the same scope as Eqs. (20) and (23) above. Additionally, heuristics are offered in Table 1 for making calculations, and these are demonstrated by deriving a stochastic differential equation for the correlation between two traits driven purely by random genetic drift.

2.2.2. The martingale gradient version (MG)

In Supplementary Material, Section 4, I show that this framework is justified based on a martingale process $\mathcal{M}$. Brownian motion is a special case of a martingale process, and stochastic equations are often expressed in terms of a Brownian stochastic differential dB. However, in this framework, stochastic equations can also be expressed in terms of the stochastic differential with respect to $\mathcal{M}$, denoted by $d\mathcal{M}$. This establishes a powerful approach for deriving fundamental insights into evolutionary processes. Further mathematical details are provided in Supplementary Material, Sections 4 and 5, but here I focus on pragmatic aspects regarding calculations that involve $d\mathcal{M}$.

I begin by artificially defining symbols related to $d\mathcal{M}$ in terms of the Brownian motions that appear in Eqs. (18), (19), and (21), and use these definitions to express the framework in terms of $d\mathcal{M}$. This is done purely for the sake of motivating the material that follows. After this, I introduce some properties of $d\mathcal{M}$ and show how these can be used to recover the Brownian motions initially used in the artificial definitions mentioned above. I then provide general heuristics summarized in Table 1 for working with $d\mathcal{M}$, and illustrate these heuristics by deriving the response of trait correlations to random genetic drift.

For now, define the symbols $d\mathcal{M}(1), d\mathcal{M}(g_i - \bar{g}_i)$ and $d\mathcal{M}\big((g_i - \bar{g}_i)(g_j - \bar{g}_j) - G_{ij}\big)$ as follows:

$$d\mathcal{M}(1) := \sqrt{v\,n}\,dB_n, \tag{25a}$$

$$d\mathcal{M}(g_i - \bar{g}_i) := \sqrt{v\,n\,G_{ii}}\,dB_{\bar{z}_i}, \tag{25b}$$

$$d\mathcal{M}\big((g_i - \bar{g}_i)(g_j - \bar{g}_j) - G_{ij}\big) := \sqrt{v\,n}\,\sqrt{G_{ii}\,G_{jj} + G_{ij}^2}\,dB_{G_{ij}}. \tag{25c}$$

Using this notation, the *MG* version of the framework can be written as

$$dn = \bar{m}\,n\,dt + d\mathcal{M}(1), \tag{26a}$$

$$d\bar{z}_i = \sum_{j=1}^{d} G_{ij}\,(\partial_j\,\bar{m} + \overline{\partial_j\,m})\,dt + \frac{1}{n}\,d\mathcal{M}(g_i - \bar{g}_i), \tag{26b}$$

$$dG_{ij} = \left(M_{ij} + 2\sum_{kl} G_{ik}\,(\partial_{kl}\,\bar{m} - \overline{\partial_{kl}\,m})\,G_{lj} - \frac{v}{n}G_{ij}\right)dt + \frac{1}{n}\,d\mathcal{M}\big((g_i - \bar{g}_i)(g_j - \bar{g}_j) - G_{ij}\big), \tag{26c}$$

where recall that $\partial_i = \partial/\partial\bar{z}_i$ and $\partial_{ij} = \partial/\partial G_{ij}$.

The covariance structure of the system is maintained through heuristics (summarized in Table 1) for computing products of the above stochastic differentials $d\mathcal{M}(x)$. To understand the heuristics, I introduce some useful notation for keeping track of averages across the distribution of traits in the population. Specifically, for functions $x(\mathbf{g})$ and $y(\mathbf{g})$, I define the symbols $\|\,x\,\|$ and $\langle\,x,\,y\,\rangle$ as follows:

$$\|\,x\,\| = \sqrt{v\,n}\,\sqrt{\overline{x^2}} = \sqrt{v\,n}\,\sqrt{\int_{\mathbb{R}^d} x^2(\mathbf{g})\,\tilde{p}(\mathbf{g})\,d\mathbf{g}}, \tag{27a}$$

$$\langle\,x,\,y\,\rangle = v\,n\,\overline{x\,y} = v\,n\int_{\mathbb{R}^d} x(\mathbf{g})\,y(\mathbf{g})\,\tilde{p}(\mathbf{g})\,d\mathbf{g}, \tag{27b}$$

where $\tilde{p}(\mathbf{g}) = \gamma(\mathbf{g})/n$ is the distribution of additive genetic values $\mathbf{g}$ in the population (assumed to be multivariate normal). To provide a few examples, one can calculate $\|1\| = \sqrt{v\,n}\,\sqrt{\int 1^2\,\tilde{p}(\mathbf{g})\,d\mathbf{g}} = \sqrt{v\,n}$ along with

$$\|g_i - \bar{g}_i\| = \sqrt{v\,n}\,\sqrt{\int_{\mathbb{R}^d} (g_i - \bar{g}_i)^2\,\tilde{p}(\mathbf{g})\,d\mathbf{g}} = \sqrt{v\,n\,G_{ii}}, \tag{28a}$$

$$\langle\,g_i - \bar{g}_i,\,g_j - \bar{g}_j\,\rangle = v\,n\int_{\mathbb{R}^d} (g_i - \bar{g}_i)\,(g_j - \bar{g}_j)\,\tilde{p}(\mathbf{g})\,d\mathbf{g} = v\,n\,G_{ij}. \tag{28b}$$

The martingale process $\mathcal{M}$ mentioned above can be thought of as a mapping that associates functions with stochastic processes. This idea is formalized in Section 4 of the Supplement. In general, the process that $\mathcal{M}$ associates with x, denoted $\mathcal{M}(x)$, will not be a Brownian motion. However, Supplementary Material, Section 4.2 provides justification for what I call the *scaling property*, which states that

$$d\mathcal{M}(\hat{x}) := \frac{d\mathcal{M}(x)}{\|\,x\,\|} \tag{29}$$

is the stochastic differential of a standard Brownian motion.

Using this heuristic, we can calculate

$$d\mathcal{M}(1) = \|1\|\,d\mathcal{M}(\hat{1}) = \sqrt{v\,n}\,d\mathcal{M}(\hat{1}), \tag{30a}$$

$$d\mathcal{M}(g_i - \bar{g}_i) = \|g_i - \bar{g}_i\|\,d\mathcal{M}(\widehat{g_i - \bar{g}_i}) = \sqrt{v\,n\,G_{ii}}\,d\mathcal{M}(\widehat{g_i - \bar{g}_i}). \tag{30b}$$

Comparing with Eqs. (18) and (20), these calculations highlight the fact that $dB_n = d\mathcal{M}(\hat{1})$ and $dB_{\bar{z}_i} = d\mathcal{M}(\widehat{g_i - \bar{g}_i})$. A similar equality holds for $dB_{G_{ij}}$, but for the sake of brevity this expression is omitted (see Supplementary Material, Section 5.4 for details).

Informally speaking, this scaling property allows us to "factor out" the standard deviation from the noise process driving the dynamics of a univariate function of the population (such as n, $\bar{z}_i$, or G_{ij}), and to replace that noise process with the product of the resulting standard deviation with a standard Brownian motion. This is the crucial step for obtaining Eqs. (18), (20), and (23) from the underlying martingale process $\mathcal{M}$, as detailed in Supplementary Material, Section 5 where the heuristics summarized in Table 1 are applied to derive *MG* and *BG*. The calculations shown in Eq. (30) are heuristic because they lead to accurate results without a rigorous understanding of their justification.

Just as $B_{\bar{z}_i}$ and $B_{\bar{z}_j}$ have a covariance encoded by the heuristic $dB_{\bar{z}_i}\,dB_{\bar{z}_j} = \big(G_{ij}/\sqrt{G_{ii}\,G_{jj}}\big)\,dt$, the processes $\mathcal{M}(x)$ and $\mathcal{M}(y)$ will also covary following a more general heuristic. More precisely, Supplementary Material, Section 4.2 justifies the *multiplicative property* which states that, for functions x, y, we have

$$d\mathcal{M}(x)\,d\mathcal{M}(y) = \langle\,x,\,y\,\rangle\,dt, \tag{31}$$

Table 1
Heuristics summarizing the scaling, multiplicative, and additive properties of the stochastic differentials $d\mathcal{M}(x)$.

Property	Heuristic
Scaling	$d\mathcal{M}(x) = \|\|x\|\| \, d\mathcal{M}(\hat{x})$, $\mathcal{M}(\hat{x})$ a BM
Multiplicative	$d\mathcal{M}(x)\, d\mathcal{M}(y) = \langle x, y \rangle \, dt$
Additive	$d\mathcal{M}(ax + by) = a\, d\mathcal{M}(x) + b\, d\mathcal{M}(y)$

where $\langle x, x \rangle = \| x \|^2$. In addition, we have $dt^2 = 0$ and $dt\, d\mathcal{M}(x) = d\mathcal{M}(x)\, dt = 0$ for any function x, which are exact equalities in standard approaches to stochastic differential equations (see Øksendal, 2013; Evans, 2012; Klebaner, 1998). In this context, phenotypic moments μ_1, μ_2 (such as $\bar{z}_i$ or G_{ij}) behave as constants such that $\langle \mu_1 x, \mu_2 y \rangle = \mu_1 \mu_2 \langle x, y \rangle$ (but see Supplementary Material, Section 4.2 for justification).

Finally, we also have the *additive property*: $d\mathcal{M}(a x + b y) = a\, d\mathcal{M}(x) + b\, d\mathcal{M}(y)$, for functions x, y and constants a, b (more generally, a, b can also be phenotypic moments such as $\bar{z}_i$ and G_{ij}, see Section 4.2 of the Supplement). This property is particularly useful for computing the correct stochastic term for an equation resulting from the application of Itô's formula, as illustrated in Section 3 below.

The heuristics associated with the *MG* version of the framework are summarized in Table 1. Example usage of these heuristics are provided through the derivation of this framework in Supplementary Material, Section 5. With these heuristics, it is straightforward to work with Eq. (26) to rigorously obtain the dynamics of quantities related to the population. For instance, the heuristics summarized in Table 1 may be used to formally derive a stochastic differential equation for the evolution of genetic correlations in response to drift. Indeed, I do this now.

3. Genetic correlations and G-matrix dynamics

In the following subsection I illustrate how to apply the above framework to obtain new models of evolutionary phenomena. Specifically, I apply the heuristics introduced in the *MG* version of the framework to derive the dynamics of genetic correlations responding only to random genetic drift. Using this derivation, I then provide biological insights into the consequences of drift for genetic correlations in asexually reproducing populations. In Section 3.2, I then discuss the significance of these results in the context of empirical research on **G**-matrix dynamics.

3.1. Consequences of drift for trait correlations

The additive genetic correlation between traits z_i and z_j is given by $\rho_{ij} = G_{ij} / \sqrt{G_{ii}\, G_{jj}}$. For this subsection, I focus on the single correlation between z_i and z_j, and simply write this as ρ. To focus on the effects of random genetic drift, and for the sake of simplicity, I ignore mutation, selection, and abundance dynamics in this analysis. For the sake of illustration, I outline only the steps needed to demonstrate how the framework is applied and to obtain the biological result. The full derivation is provided in Supplementary Material, Section 6.1.

Because we have the stochastic equations for dG_{ij}, dG_{ii}, and dG_{jj} (Eq. (26c)), and because ρ can be thought of as a function $\rho(G_{ij}, G_{ii}, G_{jj}) = G_{ij} / \sqrt{G_{ii}\, G_{jj}}$, we can apply Itô's formula, which, in this context, states the following stochastic analog of the chain-rule: $d\rho = K + L$, with $K = (\partial_{ij}\rho)\, dG_{ij} + (\partial_{ii}\rho)\, dG_{ii} + (\partial_{jj}\rho)\, dG_{jj}$ and

$$\begin{aligned} L = \tfrac{1}{2}\Big[&(\partial^2_{ij}\rho)\,(dG_{ij})^2 + (\partial^2_{ii}\rho)\,(dG_{ii})^2 + (\partial^2_{jj}\rho)\,(dG_{jj})^2 \Big] \\ &+ (\partial_{ij}\,\partial_{ii}\rho)\,(dG_{ij})\,(dG_{ii}) + (\partial_{ij}\,\partial_{jj}\rho)\,(dG_{ij})\,(dG_{jj}) \\ &+ (\partial_{ii}\,\partial_{jj}\rho)\,(dG_{ii})\,(dG_{jj}), \end{aligned} \tag{32}$$

where $\partial_{ij}\rho$ is the partial derivative of ρ with respect to G_{ij}.

With this formula in hand, all that is left is to carry out calculations based on the heuristics from the *MG* version of the framework. First off, we can directly compute the *deterministic* component of the sum K and find that this cancels to zero. This part of the calculation does not require the heuristics for $d\mathcal{M}$ introduced above. However, by using Eq. (26c) in the expression for K, and by leveraging the additive property of $d\mathcal{M}$, the *stochastic* component of K can be rewritten as

$$\begin{aligned} &\tfrac{1}{n}(\partial_{ij}\rho)\, d\mathcal{M}\big((g_i - \bar{g}_i)(g_j - \bar{g}_j) - G_{ij}\big) \\ &\quad + \tfrac{1}{n}(\partial_{ii}\rho)\, d\mathcal{M}\big((g_i - \bar{g}_i)(g_i - \bar{g}_i) - G_{ii}\big) \\ &\quad + \tfrac{1}{n}(\partial_{jj}\rho)\, d\mathcal{M}\big((g_j - \bar{g}_j)(g_j - \bar{g}_j) - G_{jj}\big) \\ &= \tfrac{1}{n}\, d\mathcal{M}\Big((\partial_{ij}\rho)\big((g_i - \bar{g}_i)(g_j - \bar{g}_j) - G_{ij}\big) \\ &\qquad + (\partial_{ii}\rho)\big((g_i - \bar{g}_i)(g_i - \bar{g}_i) - G_{ii}\big) + (\partial_{jj}\rho)\big((g_j - \bar{g}_j)(g_j - \bar{g}_j) - G_{jj}\big) \Big). \end{aligned} \tag{33}$$

Writing the argument to $d\mathcal{M}$ on the right-hand-side of Eq. (33) as H_{ij}, the scaling property shows the stochastic component of K can be written as

$$\tfrac{1}{n}\| H_{ij} \| d\mathcal{M}(\hat{H}_{ij}). \tag{34}$$

Calculation of $\| H_{ij} \|$ only requires computation of the first order derivatives $\partial_{ij}\rho$, $\partial_{ii}\rho$, $\partial_{jj}\rho$ and the fact from multivariate normal distributions that

$$\begin{aligned} &\int_{\mathbb{R}^d} (g_i - \bar{g}_i)(g_j - \bar{g}_j)(g_k - \bar{g}_k)(g_l - \bar{g}_l)\tilde{p}(g) dg \\ &\quad = G_{ij}G_{kl} + G_{ik}G_{jl} + G_{il}G_{jk}. \end{aligned} \tag{35}$$

Working through these calculations provides

$$K = \sqrt{\frac{v}{n}}(1 - \rho^2) dB_\rho, \tag{36}$$

where $B_\rho := \mathcal{M}(\hat{H}_{ij})$ is a scalar-valued standard Brownian motion.

To compute L, I first rewrite the product $(dG_{ij})\,(dG_{kl})$ using the notation introduced in the *MG* version of the framework above (i.e., using Eq. (26)), which provides

$$\begin{aligned} (dG_{ij})\,(dG_{kl}) = \tfrac{1}{n^2}\, &d\mathcal{M}\big[(g_i - \bar{g}_i)(g_j - \bar{g}_j) - G_{ij} \big] \\ &\times d\mathcal{M}\big[(g_k - \bar{g}_k)(g_l - \bar{g}_l) - G_{kl} \big]. \end{aligned} \tag{37}$$

Applying the multiplicative property for $d\mathcal{M}$, the product simplifies to

$$\begin{aligned} (dG_{ij})\,(dG_{kl}) = \tfrac{1}{n^2}\big[&\big\langle (g_i - \bar{g}_i)(g_j - \bar{g}_j), (g_k - \bar{g}_k)(g_l - \bar{g}_l) \big\rangle \\ &- v\, n\, G_{ij}\, G_{kl} \big]\, dt, \end{aligned} \tag{38}$$

where I made use of $\langle (g_i - \bar{g}_i)(g_j - \bar{g}_j), G_{kl} \rangle = v\, n\, G_{ij}\, G_{kl}$. Again applying property (35) from above, I obtain the further simplification $(dG_{ij})\,(dG_{kl}) = \frac{v}{n}(G_{ik}G_{jl} + G_{il}G_{jk})$. Interfacing this result with derivatives of $\rho(G_{ij}, G_{ii}, G_{jj})$, I arrive at

$$L = -\frac{1}{2}\frac{v}{n}\,\rho\,(1 - \rho^2)\, dt. \tag{39}$$

Hence, by applying the *MG* version of the framework, I find that trait correlations driven entirely by drift follow the ordinary stochastic differential equation

$$d\rho = -\frac{1}{2}\frac{v}{n}\rho(1 - \rho^2)dt + \sqrt{\frac{v}{n}}(1 - \rho^2) dB_\rho. \tag{40}$$

Eq. (40) shows that the effect of drift is mediated by the ratio of the variance of lifetime reproductive output v (typically set to $v = 1$ in classical quantitative genetic models such as Lande, 1976) to the effective population size n. Additionally, one can check that (because mutation and selection are absent) $\rho = \pm 1$ are stationary points. Furthermore, leveraging the fact that Eq. (40) defines a one-dimensional diffusion, we can in principle solve for its stationary distribution $\hat{q}(\rho)$ (Karlin and Taylor, 1981; Etheridge, 2010), which is done in Supplementary Material, Section 6.2. However, in attempt to do so we arrive at the non-integrable function:

$$\hat{q}(\rho) \propto \frac{1}{(1 - \rho^2)^{3/2}}. \tag{41}$$

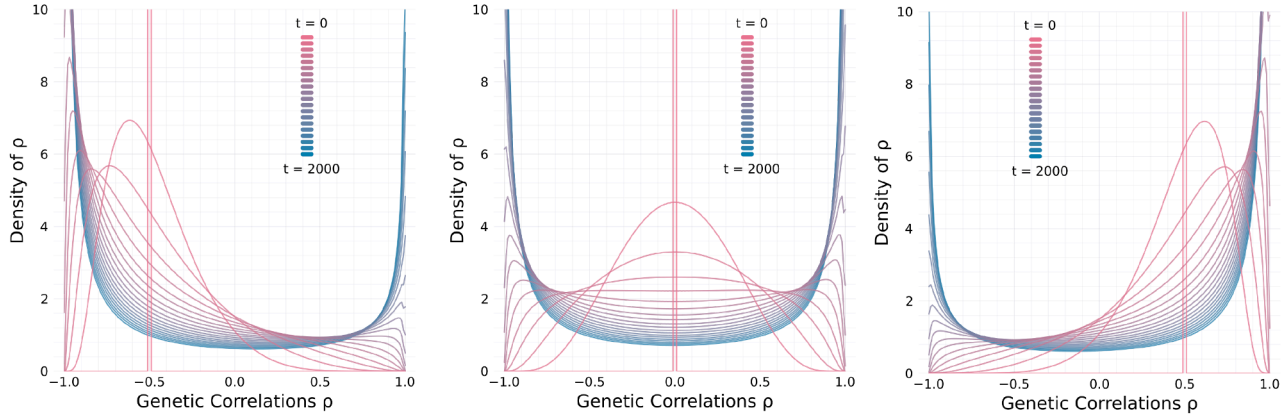


Fig. 1. The distribution of genetic correlations evolving under drift alone converges to a non-integrable stationary solution. Shown here are dynamics for initial distributions taking approximate point masses at $\rho_0 = -0.5$ (left panel), $\rho_0 = 0.0$ (middle panel), and $\rho_0 = 0.5$ (right panel). Solutions at earlier times are colored green, and later times are purple which run until $t = 2000$. The rate of drift is set to $\upsilon/n = 0.001$.

The lack of a formal stationary distribution creates a challenge for understanding the limiting behavior of genetic correlations evolving solely in response to drift. This can be partially overcome by again leveraging the theory of one-dimensional diffusions to demonstrate that the boundaries ± 1 are both attracting and unattainable (Karlin and Taylor, 1981). That is, genetic correlations tend towards their extremes, but never fix at ± 1. This is visualized in Figure 1 by plotting the distribution $q(\rho)$ of genetic correlations as numerical solutions to the forwards Kolmogorov (i.e., Fokker-Planck) equation associated with (40). Biologically, $|\rho|$ increases because drift randomly samples finite numbers of individuals, thereby causing transient correlations among the additive genetic values these individuals carry. Further biological implications are discussed in the following section.

Another way to view this result is by applying Itô's formula to $u = \tanh^{-1}(\rho)$ (detailed calculations given in Supplementary Material, Section 6.3). Doing so returns

$$du = \frac{1}{2}\frac{\upsilon}{n}\tanh(u)dt + \sqrt{\frac{\upsilon}{n}}\,d\,B_u. \tag{42}$$

Because $\rho = \tanh(u)$ is a monotone increasing function of u, statements about u map directly to ρ. In particular, $u = \rho = 0$ is unstable because $\tanh(u)$ is positive for positive u, and negative for negative u. Additionally, if u is much greater than 1, then $du \approx \upsilon\, dt/2n + \sqrt{\upsilon/n}\, d\,B_u$ and, writing this point as u_0, we have $\mathbb{E}[u_t] \approx u_0 + \upsilon\, t/2n$. The analogous approximation holds when u is much less than -1. Justification for these claims is made in Supplementary Material, Section 6.3. This agrees with the boundary classification result above that demonstrates drift has an overall tendency to drive trait correlations towards ± 1. Numerical estimates for sample paths of the solution to (40), illustrated by Fig. 2, support this conclusion.

To confirm the heuristics return correct expressions, I also simulated the evolution of the **G**-matrix driven solely by random genetic drift for a 2-dimensional trait by applying the Euler-Maruyama method (Bayram et al., 2018) to the *BG* version of the framework. I then back-calculated the genetic correlation ρ based on the simulated time-series of **G**. Fig. 1 in Supplementary Material, Section 1 demonstrates this approach agrees with Eq. (40). Further information on the numerical implementation is given in Supplementary Material, Section 1, and associated code is available at the Zenodo repository https://doi.org/10.5281/zenodo.19087096.

3.2. Evolution of The G-matrix in response to drift

The above result demonstrates that drift increases correlations between traits, at least for asexually reproducing populations where recombination is absent. This perspective offers an important refinement

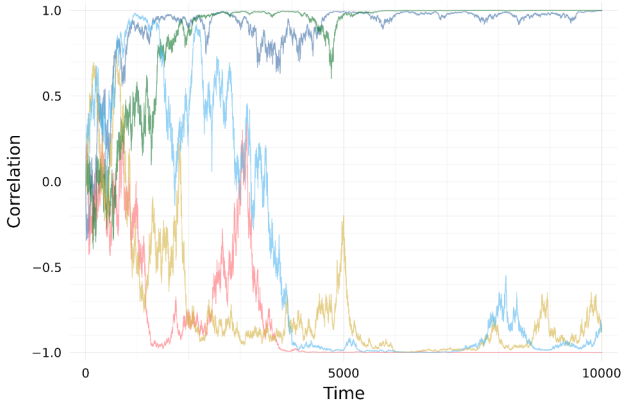


Fig. 2. Drift drives trait correlations towards ± 1. Shown here are five replicates illustrating the path-behavior of trait correlations following Eq. (40). Each replicate is initiated with $\rho_0 = 0$ and the rate of drift is set to $\upsilon/n = 0.001$.

of the conventional wisdom that **G**-matrices responding to drift should merely contract proportionally (Lande, 1979; Phillips and McGuigan, 2006; Dugand et al., 2021; Mallard et al., 2023a). In particular, it is classically thought that the orientation of the **G**-matrix should not change due to drift on average, and hence any changes in orientation may be a sign of selection (Roff, 2000; Steppan et al., 2002; Cano et al., 2004; Chapuis et al., 2008; Aguirre et al., 2013).

This idea can be obtained from Eq. (21) if the stochastic component is neglected. Then, the deterministic response due to drift is described by $\dot{\mathbf{G}} = -\upsilon\,\mathbf{G}/n$, which has the solution $\mathbf{G}_t = \mathbf{G}_0 e^{-\upsilon t/n}$, and is a continuous-time equivalent of the result found on page 409 in a paper by Lande (1979). Numerical results displayed in Fig. 2 of Supplementary Material, Section 1 indicate agreement with this classical scaling result in expectation. However, when stochastic fluctuations dominate the dynamics over the timescale of interest (so that individual sample paths diverge strongly despite a smooth deterministic trend) the average response to drift may provide little information for within-population dynamics.

Taking an experimental approach, Phillips et al. (2001) established isolated populations of *Drosophila melanogaster* from a common base population and found significant variation across replicates for the response of **G** to drift. This occurs in spite of the fact that *Drosophila melanogaster* is a sexually reproducing organism, and recombination should act to break up correlations caused by linkage (discussed further below) and thereby reduce variation across replicates. Fig. 3 emphasizes these variable outcomes theoretically by superimposing expected trait correlations over a collection of individual outcomes. The expected

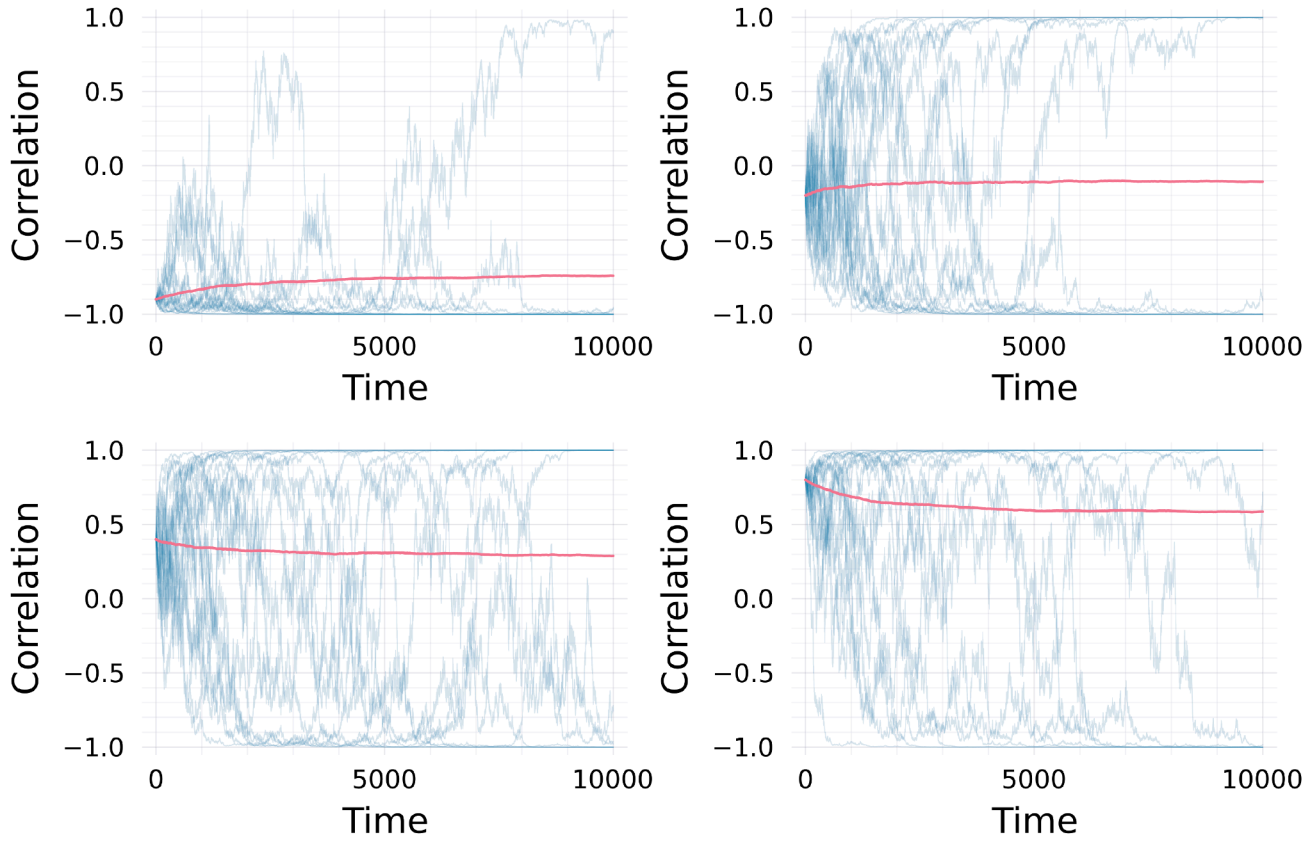


Fig. 3. Averages over replicated time-series of trait correlation do not predict within population trait correlations. Averages were computed using 3000 replicates with initial correlations $\rho_0 = -0.9$ (top-left), $\rho_0 = -0.2$ (top-right), $\rho_0 = 0.4$ (bottom-left), and $\rho_0 = 0.8$ (bottom-right). For the sake of clarity, just 20 random replicates are shown behind the average trends.

correlations are approximated by averaging over replicates, but it took a very large number of replicates (>1000) to obtain a satisfactory deterministic trend. Because population size enters only through the ratio υ/n, increasing n rescales the time axis but does not reduce pathwise variability or improve the predictive power of the average trend. The averages shift towards zero, in the sense of a slow directional trend that is not resolved to convergence over the finite time window shown, but we can see individual replicates are not predicted by this trend. Hence, to gain a more accurate picture of **G**-matrix evolution, there is a need to understand the path behavior of individual outcomes.

One approach to gain insight for the trait correlation of a given replicate, as opposed to the average, is to analyze the proportion of time that correlations spend near ± 1 over the total time the process is running. Fig. 4 plots the proportion of time that the magnitude of the trait correlation ρ spends above the threshold $1-\varepsilon$, i.e. with $|\rho| > 1-\varepsilon$, averaged over 10 replicate trajectories with initial value $\rho_0 = 0$ and $\upsilon/n = 0.001$, for $\varepsilon \in \{0.20, 0.10, 0.05, 0.02\}$. Smaller ε corresponds to increasingly strict neighborhoods of ± 1. This result shows that trait correlations typically aggregate near ± 1 in drifting, isolated, asexual populations with no mutational input. Hence, interpreting this as a signature of drift, we can say that if trait correlations in a population exhibit significant variation away from ± 1, then either the population has been drifting for only a short time relative to its effective population size, or other processes not captured by this model must be at play, such as ongoing mutation or recombination, which are explicitly excluded here. I therefore discuss mechanisms maintaining genetic correlations before concluding this section.

Pleiotropic loci and genetic linkage are two mechanisms that maintain genetic correlations (Lande, 1980). In this framework, pleiotropic mutations would arise from non-zero off-diagonals of the mutation matrix **M**. In contrast, drift facilitates chance correlations of additive genetic values between traits across individuals. Hence, this second mode of genetic correlation accumulation is directly analogous to the build-up of linkage by drift (Hill and Robertson, 1966; Ohta and Kimura, 1969; Lucek and Willi, 2021). For sexually reproducing populations, it is expected that genetic correlations should be maintained by pleiotropic loci as recombination breaks up linkage (Lande, 1980; Jones et al., 2003; Phillips and McGuigan, 2006). However, the results in this section show

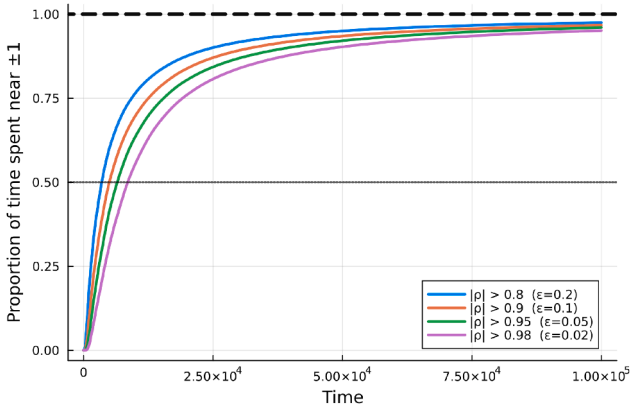


Fig. 4. Trait correlations (ρ) within populations rapidly evolve towards ± 1 under random genetic drift. Shown is the proportion of time that $|\rho|$ spends above a threshold $1-\varepsilon$, averaged over 10 replicate trajectories, with $\varepsilon \in \{0.20, 0.10, 0.05, 0.02\}$. Smaller ε corresponds to increasingly strict neighborhoods of ± 1. Simulations start from $\rho_0 = 0$ with $\upsilon/n = 0.001$.

that drift alone drives genetic correlations toward extreme values. This raises the natural question of how this tendency would be counteracted by recombination, and whether a drift-recombination balance analogous to that studied in classical population genetics (e.g., Ohta and Kimura, 1969) would emerge.

The drift-only example considered here is intended as an illustrative application of the framework rather than a quantitative fit to existing experiments. In a clonal, mutation-free setting, the model predicts that the stochastic dynamics of genetic correlations depend on time only through the rescaled variable $\tau = (v/n)t$. Equivalently, for populations with different effective sizes n, the distributions of correlation trajectories become comparable after rescaling time by n/v, so that increasing population size slows the dynamics without altering their qualitative path behavior. This prediction can be tested empirically using pathwise statistics such as the distribution of first-passage times $\tau_c = \inf\{t : |\rho(t)| > c\}$ to a high-correlation threshold c, or the long-run fraction of time that correlations satisfy $|\rho| > c$; both quantities are predicted to scale with n/v. Such tests would, in principle, be feasible in replicated clonal systems with controlled demography and minimal mutational input, including bottlenecked microbial populations (e.g., *Escherichia coli* or asexually propagated *Saccharomyces cerevisiae*) or clonal macrophytes such as duckweed (*Lemna* or *Spirodela*) maintained in controlled microcosms. Extending the framework to incorporate sexual reproduction and recombination would be a natural next step for making quantitative predictions about drift-recombination balance. Nevertheless, the present results highlight that focusing solely on average, deterministic responses can obscure substantial pathwise structure, suggesting that conventional perspectives on **G**-matrix evolution merit closer examination.

4. Discussion

The classical quantitative genetic approach has stressed the importance of considering the genetic architecture of traits for understanding evolutionary processes, and has placed particular emphasis on the use of multi-locus models (Bulmer, 1972; Chakraborty and Nei, 1982; Slatkin, 1987; Turelli and Barton, 1994; Barton and Turelli, 2004; Barton et al., 2017; Walsh and Lynch, 2018). The advantage of this approach is its clear connection with explicit genetic details. However, its primary challenge is the manipulation of complex expressions that emerge at such level of detail. To overcome this, and establish a flexible tool for modelling the integrated ecological and evolutionary dynamics of populations carrying multivariate traits, I struck a balance between incorporating genetic detail and analytical tractability. As a consequence, this framework has potential for wide-spread use across topics in evolution, ecology, and population biology.

A central feature of this framework is that a wide range of classical models can be obtained by choosing an appropriate fitness function m. For example, stochastic Lotka-Volterra dynamics for a community of species is obtained from this framework by applying the growth rate $m_i = r_i + \sum_j \alpha_{ij} n_j$ for species i, with n_j being the abundance of species j and α_{ij} the interaction coefficient. The long time behavior of the resulting stochastic Lotka-Volterra process for two species has been studied by Cattiaux and Méléard (2009). In addition, by making the interaction coefficients α_{ij} dependent on multivariate trait values $\mathbf{z}_i$, $\mathbf{z}_j$, models integrating coevolution with abundance feedbacks can be obtained similar to those studied by Gokhale et al. (2013), Cortez and Weitz (2014), Patel et al. (2018). As a special case, continuous time analogs of multivariate coevolution models are obtained by focusing on two species with fixed (or infinite) abundances, and assuming $\alpha_{ij}(\mathbf{z}_i, \mathbf{z}_j)$ depends on the Euclidean distance between $\mathbf{z}_i$ and $\mathbf{z}_j$ (Gilman et al., 2012; Débarre et al., 2014). Klausmeier et al. (2020) studied models of evolutionary rescue with univariate traits using growth rates of the forms $m(z,t) = r - \psi(\theta(t) - z)^2/2$ and $m(z,t) = r + r_0 e^{-\psi(\theta(t)-z)^2/2}$, where $\theta(t)$ is a dynamic phenotypic optimum and ψ is the strength of stabilizing selection. Applying multivariate generalizations of these growth rates to the above framework leads to extensions of an evolutionary rescue model involving demographic stochasticity studied by Xu et al. (2023). Additionally, Jones et al. (2012) also studied a model of phenotypic adaptation to a dynamic optimum, but in the context of **G**-matrix evolution. This framework can then be used as a bridge between research topics such as evolutionary rescue and **G**-matrix evolution. This list provides a small set of examples for how this framework can be used to derive new models across a broad range of topics in ecology, evolution, and population biology.

Further work is needed to extend this framework in several directions. For instance, it is possible to incorporate sexual reproduction and recombination by assuming each trait value is determined by the sum of additive genetic effects contributed by two gametes, rather than by a single genetic value. In this case, each gamete contributes an additive genetic effect g_i, and the trait is encoded as $g_i + g_i'$, where g_i and g_i' arise from convex combinations of the parental gametic values, similar to the approach taken by Lande (1980). How to formalize this extension using measure-valued processes is not obvious. In another direction, environmental stochasticity (i.e., stochastic growth rates) can be formally incorporated in a measure-valued context following the work of Mytnik (1996). Extending the framework in this direction may yield important additional insights complementing the work of Lande (2007, 2008), who studied the consequences of environmental stochasticity for long-run population growth rates. An extension in this direction may also be useful for studies investigating the consequences of environmental stochasticity on **G**-matrix evolution (e.g., Engen and Sæther, 2024).

I also point to two possible extensions that concern the role of development. The first accounts for the evolution of the mutational variance-covariance matrix **M** within the classical quantitative genetic framework. Allowing **M** to evolve enables analytical and simulation models for the evolution of evolvability, as studied by Jones et al. (2007), including numerical integration of the resulting stochastic differential equations (e.g. via Euler-Maruyama), rather than relying exclusively on individual-based models, and may yield more precise quantitative predictions amenable to experimental study (Mallard et al., 2023a). A second, conceptually distinct extension would be to replace the linear, regression-based genotype-phenotype map and residual model used here with nonlinear or mechanistic developmental maps, as in evo-devo frameworks (e.g., Milocco and Salazar-Ciudad, 2022; González-Forero, 2022, 2024a,b), in which genetic and phenotypic covariation emerge from explicit developmental dynamics rather than being parameterized through a mutational covariance matrix.

The application of this framework to study **G**-matrix evolution uncovered a more nuanced picture for the role of drift than what conventional wisdom suggests (Phillips and McGuigan, 2006). In particular, although drift indeed proportionally reduces **G**-matrices when averaged over many replicated populations, it also drives trait correlations towards their extremes within populations. This agrees with the result that the expected eigenvalues of **G** differ from the eigenvalues of the expectation of **G** under drift (Griswold et al., 2007). In addition, the impact of drift on genetic correlations can be understood as a reduction in the effective dimensionality of **G**-matrices (Hine and Blows, 2006), reflecting an extension of the principle that drift erodes heritable variation. That is, because drift drives pairwise correlations to ± 1, the distribution of multivariate traits in a population that has been evolving solely under drift may be well approximated after projecting onto a lower-dimensional trait space. Conceptually, this is similar to dimensionality reduction techniques such as principal components analysis (Kirkpatrick and Meyer, 2004). Importantly, this effect of drift on trait correlations calls into question comparative quantitative genetic methods utilizing **G**-matrices which routinely treat differences in **G**-orientation as a signal of selection in diverged populations (Roff, 2000; Steppan et al., 2002; Phillips and McGuigan, 2006; Cano et al., 2004; McGuigan, 2006; Chapuis et al., 2008; Aguirre et al., 2013; Dugand et al., 2021; Mallard et al., 2023a). These findings have broad implications for evolutionary biology, highlighting the need for revised theoretical perspectives and further empirical validation.

The study of genetic correlations and **G**-matrix dynamics using this framework can be extended by studying Eq. (21) with mutation and selection. In the special case of one-dimensional traits, stabilizing selection, and no abundance dynamics, the theory of one-dimensional diffusions (Etheridge, 2010) can be used to show that the stationary distribution of additive genetic variance follows a generalized inverse Gaussian distribution (Jorgensen, 2012). This suggests the stationary distribution of **G** may follow a matrix-variate generalization, such as the matrix generalized inverse Gaussian (MGIG) distribution (Fazayeli and Banerjee, 2016). Proposed distributions may be checked by evaluating the forwards Kolmogorov equation associated with (21) at equilibrium assuming a solution that follows the density of the proposal. Identification of the stationary distribution for (21) may then be used to study the distribution of genetic correlations maintained by interactions between mutation, selection, and random genetic drift.

In summary, this work introduces a versatile framework for modeling the stochastic eco-evolutionary dynamics of multivariate traits, providing a unifying approach that integrates mutation, selection, demographic stochasticity, and drift. By balancing mathematical rigor with accessibility, this framework enables the derivation of new models across a broad spectrum of population biology, making it a valuable tool for both theoretical and applied researchers. With its broad applicability, the framework presented here offers a foundation for future studies investigating the dynamics of populations in both theoretical and empirical contexts.

CRediT authorship contribution statement

Bob Week: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

None.

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Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.jtbi.2026.112428.

References

Aguirre, J.D., Hine, E., McGuigan, K., Blows, M.W., 2013. Comparing G: multivariate analysis of genetic variation in multiple populations. Heredity 112 (1), 21–29. https://doi.org/10.1038/hdy.2013.12

Akjouj, I., Barbier, M., Clenet, M., Hachem, W., Maïda, M., Massol, F., Najim, J., Tran, V.C., 2024. Complex systems in ecology: a guided tour with large lotka–volterra models and random matrices. Proc. R. Soc. A Math. Phys. Eng. Sci. 480 (2285). https://doi.org/10.1098/rspa.2023.0284

Arnold, S.J., Bürger, R., Hohenlohe, P.A., Ajie, B.C., Jones, A.G., 2008. Understanding the evolution and stability of the g-matrix. Evolution 62 (10), 2451–2461. https://doi.org/10.1111/j.1558-5646.2008.00472.x

Barton, N., 1989. The divergence of a polygenic system subject to stabilizing selection, mutation and drift. Genetical Res. 54 (1), 59–78. https://doi.org/10.1017/s0016672300028378

Barton, N.H., Etheridge, A.M., Véber, A., 2017. The infinitesimal model: definition, derivation, and implications. Theor. Popul. Biol. 118, 50–73. https://doi.org/10.1016/j.tpb.2017.06.001

Barton, N.H., Turelli, M., 1987. Adaptive landscapes, genetic distance and the evolution of quantitative characters. Genetical Res. 49 (2), 157–173. https://doi.org/10.1017/s0016672300026951

Barton, N.H., Turelli, M., 1989. Evolutionary quantitative genetics: how little do we know? Ann. Rev. Genet. 23 (1), 337–370. https://doi.org/10.1146/annurev.ge.23.120189.002005

Barton, N.H., Turelli, M., 1991. Natural and sexual selection on many loci. Genetics 127 (1), 229–255. https://doi.org/10.1093/genetics/127.1.229

Barton, N.H., Turelli, M., 2004. Effects of genetic drift on variance components under a general model of epistasis. Evolution 58 (10), 2111–2132. https://doi.org/10.1111/j.0014-3820.2004.tb01591.x

Bayram, M., Partal, T., Orucova Buyukoz, G., 2018. Numerical methods for simulation of stochastic differential equations. Adv. Differ. Equations 2018 (1). https://doi.org/10.1186/s13662-018-1466-5

Bhat, A.S., Guttal, V., 2025. Eco-evolutionary dynamics for finite populations and the noise-induced reversal of selection. Am. Nat. 205 (1), 1–19. https://doi.org/10.1086/733196

Bulmer, M., 1971. The effect of selection on genetic variability. Am. Nat. 105 (943), 201–211.

Bulmer, M.G., 1972. The genetic variability of polygenic characters under optimizing selection, mutation and drift. Genetical Res. 19 (1), 17–25. https://doi.org/10.1017/s0016672300014221

Bürger, R., 2000. The Mathematical Theory of Selection, Recombination, and Mutation. John Wiley & Sons.

Bürger, R., 1986. On the maintenance of genetic variation: global analysis of kimura's continuum-of-alleles model. J. Math. Biol. 24 (3), 341–351. https://doi.org/10.1007/bf00275642

Cano, J.M., Laurila, A., Palo, J., Merilä, J., 2004. Population differentiation in G matrix structure due to natural selection in rana temporaria. Evolution 58 (9), 2013–2020. https://doi.org/10.1111/j.0014-3820.2004.tb00486.x

Cattiaux, P., Méléard, S., 2009. Competitive or weak cooperative stochastic lotka–volterra systems conditioned on non-extinction. J. Math. Biol. 60 (6), 797–829. https://doi.org/10.1007/s00285-009-0285-4

Chakraborty, R., Nei, M., 1982. Genetic differentiation of quantitative characters between populations or species: I mutation and random genetic drift. Genetical Res. 39 (3), 303–314. https://doi.org/10.1017/s0016672300020978

Chapuis, E., Martin, G., Goudet, J., 2008. Effects of selection and drift on G matrix evolution in a heterogeneous environment: a multivariate QST–FST test with the freshwater snail galba truncatula. Genetics 180 (4), 2151–2161. https://doi.org/10.1534/genetics.108.092452

Constable, G. W.A., Rogers, T., McKane, A.J., Tarnita, C.E., 2016. Demographic noise can reverse the direction of deterministic selection. Proc. Natl. Acad. Sci. 113 (32). https://doi.org/10.1073/pnas.1603693113

Cortez, M.H., Weitz, J.S., 2014. Coevolution can reverse predator–prey cycles. Proc. Natl. Acad. Sci. 111 (20), 7486–7491. https://doi.org/10.1073/pnas.1317693111

Crow, J.F., Kimura, M., 1970. An Introduction to Population Genetics Theory. Harper & Row, New York.

Dawson, D., 1993. Measure-Valued Markov Processes. Springer Berlin Heidelberg. p. 1–260. https://doi.org/10.1007/bfb0084190

Dieckmann, U., Law, R., 1996. The dynamical theory of coevolution: a derivation from stochastic ecological processes. J. Math. Biol. 34 (5–6), 579–612. https://doi.org/10.1007/bf02409751

Dugand, R.J., Aguirre, J.D., Hine, E., Blows, M.W., McGuigan, K., 2021. The contribution of mutation and selection to multivariate quantitative genetic variance in an outbred population of drosophila serrata. Proc. Natl. Acad. Sci. 118 (31). https://doi.org/10.1073/pnas.2026217118

Débarre, F., Nuismer, S.L., Doebeli, M., 2014. Multidimensional (co)evolutionary stability. Am. Nat. 184 (2), 158–171. https://doi.org/10.1086/677137

Débarre, F., Otto, S.P., 2016. Evolutionary dynamics of a quantitative trait in a finite asexual population. Theor. Popul. Biol. 108, 75–88. https://doi.org/10.1016/j.tpb.2015.12.002

Débarre, F., Yeaman, S., Guillaume, F., 2015. Evolution of quantitative traits under a migration-selection balance: when does skew matter? Am. Nat. 186 (S1), S37–S47. https://doi.org/10.1086/681717

Engen, S., Sæther, B.-E., 2024. Evolutionary and ecological processes determining the properties of the G matrix. Am. Nat. 204 (5), 433–452. https://doi.org/10.1086/732159

Etheridge, A., 2000. An Introduction to Superprocesses. University lecture series, American Mathematical Society. https://books.google.de/books?id=zUHyBwAAQBAJ.

Etheridge, A., 2010. One Dimensional Diffusions. Springer Berlin Heidelberg. p. 33–51. https://doi.org/10.1007/978-3-642-16632-7_3

Etheridge, A.M., Kurtz, T.G., Letter, I., Ralph, P.L., Tsui Ho Lung, T., 2024. Looking forwards and backwards: dynamics and genealogies of locally regulated populations. Electron. J. Probab. 29 (none). https://doi.org/10.1214/24-ejp1075

Ethier, S.N., Kurtz, T.G., 1986. Markov Processes: Characterization and Convergence. John Wiley & Sons.

Evans, L.C., 2012. An Introduction to Stochastic Differential Equations. Vol. 82. American Mathematical Soc.

Ewens, W.J., 2004. Mathematical Population Genetics. Springer New York. https://doi.org/10.1007/978-0-387-21822-9

Falconer, D.S., Mackay, T., 1996. Introduction to Quantitative Genetics. Pearson.
Fazayeli, F., Banerjee, A., 2016. The Matrix Generalized Inverse Gaussian Distribution: Properties and Applications. Springer International Publishing. p. 648–664. https://doi.org/10.1007/978-3-319-46128-1_41
Fogarty, L., Wade, M.J., 2022. Niche construction in quantitative traits: heritability and response to selection. Proc. R. Soc. B Biol. Sci. 289 (1976). https://doi.org/10.1098/rspb.2022.0401
Gillespie, J.H., 1974. Natural selection for within-generation variance in offspring number. Genetics 76 (3), 601–606. https://doi.org/10.1093/genetics/76.3.601
Gillespie, J.H., 1975. Natural selection for within-generation variance in offspring number II. discrete haploid models. Genetics 81 (2), 403–413. https://doi.org/10.1093/genetics/81.2.403
Gillespie, J.H., 1977. Natural selection for variances in offspring numbers: a new evolutionary principle. Am. Nat. 111 (981), 1010–1014. https://doi.org/10.1086/283230
Gilman, R.T., Nuismer, S.L., Jhwueng, D.-C., 2012. Coevolution in multidimensional trait space favours escape from parasites and pathogens. Nature 483 (7389), 328–330. https://doi.org/10.1038/nature10853
Gilpin, W., Feldman, M.W., 2019. Cryptic selection forces and dynamic heritability in generalized phenotypic evolution. Theor. Popul. Biol. 125, 20–29. https://doi.org/10.1016/j.tpb.2018.11.002
Gokhale, C.S., Papkou, A., Traulsen, A., Schulenburg, H., 2013. Lotka–Volterra dynamics kills the red queen: population size fluctuations and associated stochasticity dramatically change host-parasite coevolution. BMC Evol. Biol. 13, 1–10.
González-Forero, M., 2022. How development affects evolution. Evolution 77 (2), 562–579. https://doi.org/10.1093/evolut/qpac003
González-Forero, M., 2024a. Evolutionary–developmental (evo-devo) dynamics of hominin brain size. Nat. Hum. Behav. 8 (7), 1321–1333. https://doi.org/10.1038/s41562-024-01887-8
González-Forero, M., 2024b. A mathematical framework for evo-devo dynamics. Theor. Popul. Biol. 155, 24–50. https://doi.org/10.1016/j.tpb.2023.11.003
Griswold, C.K., Logsdon, B., Gomulkiewicz, R., 2007. Neutral evolution of multiple quantitative characters: a genealogical approach. Genetics 176 (1), 455–466. https://doi.org/10.1534/genetics.106.069658
Guerand, J., Hillairet, M., Mirrahimi, S., 2023. A moment-based approach for the analysis of the infinitesimal model in the regime of small variance. https://arxiv.org/abs/2309.09567. https://doi.org/10.48550/arxiv.2309.09567
Gupta, A.K., Nagar, D.K., 2018. Matrix variate distributions. https://doi.org/10.1201/9780203749289
Hendry, A.P., 2017. Eco-Evolutionary Dynamics. Princeton university press.
Hill, W.G., Robertson, A., 1966. The effect of linkage on limits to artificial selection. Genetical Res. 8 (3), 269–294. https://doi.org/10.1017/s0016672300010156
Hine, E., Blows, M.W., 2006. Determining the effective dimensionality of the genetic variance–covariance matrix. Genetics 173 (2), 1135–1144. https://doi.org/10.1534/genetics.105.054627
Hofbauer, J., Sigmund, K., 1998. Evolutionary Games and Population Dynamics. Cambridge University Press. https://doi.org/10.1017/cbo9781139173179
Jones, A.G., Arnold, S.J., Bürger, R., 2003. Stability of the G-matrix in a population experiencing pleiotropic mutation, stabilizing selection, and genetic drift. Evolution 57 (8), 1747–1760. https://doi.org/10.1111/j.0014-3820.2003.tb00583.x
Jones, A.G., Arnold, S.J., Bürger, R., 2007. The mutation matrix and the evolution of evolvability. Evolution 61 (4), 727–745. https://doi.org/10.1111/j.1558-5646.2007.00071.x
Jones, A.G., Bürger, R., Arnold, S.J., Hohenlohe, P.A., Uyeda, J.C., 2012. The effects of stochastic and episodic movement of the optimum on the evolution of the G–matrix and the response of the trait mean to selection. J. Evol. Biol. 25 (11), 2210–2231. https://doi.org/10.1111/j.1420-9101.2012.02598.x
Jorgensen, B., 2012. Statistical Properties of the Generalized Inverse Gaussian Distribution. Vol. 9. Springer Science & Business Media.
Karlin, S., Taylor, H.E., 1981. A Second Course in Stochastic Processes. Elsevier.
Kessy, A., Lewin, A., Strimmer, K., 2018. Optimal whitening and decorrelation. Am. Stat. 72 (4), 309–314. https://doi.org/10.1080/00031305.2016.1277159
Kimura, M., 1964. Diffusion models in population genetics. J. Appl. Probab. 1 (2), 177–232. https://doi.org/10.2307/3211856
Kimura, M., 1965. A stochastic model concerning the maintenance of genetic variability in quantitative characters. Proc. Natl. Acad. Sci. 54 (3), 731–736. https://doi.org/10.1073/pnas.54.3.731
Kimura, M., 1968. Genetic variability maintained in a finite population due to mutational production of neutral and nearly neutral isoalleles. Genetical Res. 11 (3), 247–270. https://doi.org/10.1017/s0016672300011459
Kimura, M., Crow, J.F., 1978. Effect of overall phenotypic selection on genetic change at individual loci. Proc. Natl. Acad. Sci. 75 (12), 6168–6171. https://doi.org/10.1073/pnas.75.12.6168
Kirkpatrick, M., Meyer, K., 2004. Direct estimation of genetic principal components. Genetics 168 (4), 2295–2306. https://doi.org/10.1534/genetics.104.029181
Klausmeier, C.A., Osmond, M.M., Kremer, C.T., Litchman, E., 2020. Ecological limits to evolutionary rescue. Philos. Trans. R. Soc. B Biol. Sci. 375 (1814), 20190453. https://doi.org/10.1098/rstb.2019.0453
Klebaner, F.C., 1998. Introduction to Stochastic Calculus with Applications. World Scientific Publishing Company.
Kuosmanen, T., Särkkä, S., Mustonen, V., 2022. Turnover shapes evolution of birth and death rates https://doi.org/10.1101/2022.07.11.499527.
Lande, R., 1975. The maintenance of genetic variability by mutation in a polygenic character with linked loci. Genetical Res. 26 (3), 221–235. https://doi.org/10.1017/s0016672300016037
Lande, R., 1976. Natural selection and random genetic drift in phenotypic evolution. Evolution 30 (2), 314–334. https://doi.org/10.1111/j.1558-5646.1976.tb00911.x
Lande, R., 1979. Quantitative genetic analysis of multivariate evolution, applied to brain:body size allometry. Evolution 33 (1Part2), 402–416. https://doi.org/10.1111/j.1558-5646.1979.tb04694.x
Lande, R., 1980. The genetic covariance between characters maintained by pleiotropic mutations. Genetics 94 (1), 203–215. https://doi.org/10.1093/genetics/94.1.203
Lande, R., 1982. A quantitative genetic theory of life history evolution. Ecology 63 (3), 607–615. https://doi.org/10.2307/1936778
Lande, R., 2007. Expected relative fitness and the adaptive topography of fluctuating selection. Evolution 61 (8), 1835–1846. https://doi.org/10.1111/j.1558-5646.2007.00170.x
Lande, R., 2008. Adaptive topography of fluctuating selection in a mendelian population. J. Evol. Biol. 21 (4), 1096–1105. https://doi.org/10.1111/j.1420-9101.2008.01533.x
Lande, R., 2014. Evolution of phenotypic plasticity and environmental tolerance of a labile quantitative character in a fluctuating environment. J. Evol. Biol. 27 (5), 866–875. https://doi.org/10.1111/jeb.12360
Lande, R., Arnold, S.J., 1983. The measurement of selection on correlated characters. Evolution 37 (6), 1210–1226. https://doi.org/10.1111/j.1558-5646.1983.tb00236.x
Latter, B.D.H., 1970. Selection in finite populations with multiple alleles. ii. centripetal selection, mutation, and isoallelic variation. Genetics 66 (1), 165–186. https://doi.org/10.1093/genetics/66.1.165
Lucek, K., Willi, Y., 2021. Drivers of linkage disequilibrium across a species' geographic range. PLOS Genet. 17 (3), e1009477. https://doi.org/10.1371/journal.pgen.1009477
Lynch, M., Hill, W.G., 1986. Phenotypic evolution by neutral mutation. Evolution 40 (5), 915–935. https://doi.org/10.1111/j.1558-5646.1986.tb00561.x
Lynch, M., Walsh, B., 1998. Genetics and Analysis of Quantitative Traits. Vol. 1. Sinauer Sunderland, MA.
Mallard, F., Afonso, B., TeotÃ³nio, H., 2023a. Selection and the direction of phenotypic evolution. eLife 12. https://doi.org/10.7554/elife.80993
Mallard, F., Noble, L., Guzella, T., Afonso, B., Baer, C.F., TeotÃ³nio, H., 2023b. Phenotypic stasis with genetic divergence. Peer Community J. 3. https://doi.org/10.24072/pcjournal.349
McGuigan, K., 2006. Studying phenotypic evolution using multivariate quantitative genetics. Mol. Ecol. 15 (4), 883–896. https://doi.org/10.1111/j.1365-294x.2006.02809.x
McGuigan, K., Chenoweth, S., Blows, M., 2005. Phenotypic divergence along lines of genetic variance. Am. Nat. 165 (1), 32–43. https://doi.org/10.1086/426600
Milocco, L., Salazar-Ciudad, I., 2022. Evolution of the G matrix under nonlinear genotype-phenotype maps. Am. Nat. 199 (3), 420–435. https://doi.org/10.1086/717814
Mullon, C., Lehmann, L., 2019. An evolutionary quantitative genetics model for phenotypic (co)variances under limited dispersal, with an application to socially synergistic traits. Evolution 73 (9), 1695–1728. https://doi.org/10.1111/evo.13803
Mytnik, L., 1996. Superprocesses in random environments. Ann. Probab. 24 (4). https://doi.org/10.1214/aop/1041903212
Nuismer, S.L., Week, B., Aizen, M.A., 2018. Coevolution slows the disassembly of mutualistic networks. Am. Nat. 192 (4), 490–502. https://doi.org/10.1086/699218
Ohta, T., Kimura, M., 1969. Linkage disequilibrium due to random genetic drift. Genetical Res. 13 (1), 47–55. https://doi.org/10.1017/s001667230000272x
Øksendal, B., 2013. Stochastic Differential Equations: An Introduction with Applications. Springer Science & Business Media.
Parsons, T.L., Quince, C., Plotkin, J.B., 2010. Some consequences of demographic stochasticity in population genetics. Genetics 185 (4), 1345–1354. https://doi.org/10.1534/genetics.110.115030
Patel, S., Cortez, M.H., Schreiber, S.J., 2018. Partitioning the effects of eco-evolutionary feedbacks on community stability. Am. Nat. 191 (3), 381–394. https://doi.org/10.1086/695834
Perkins, E., 2002. Dawson–watanabe superprocesses and measure-valued diffusions. in: École d'Été de Probabilités de Saint-Flour XXIX–1999, Lecture Notes in Mathematics, vol. 1781, pp. 127–318. Springer.
Phillips, P.C., Arnold, S.J., 1989. Visualizing multivariate selection. Evolution 43 (6), 1209–1222. https://doi.org/10.1111/j.1558-5646.1989.tb02569.x
Phillips, P.C., McGuigan, K.L., 2006. Evolution of genetic variance-covariance structure. In: Fox, C.W., Wolf, J.B. (Eds.), Evolutionary Genetics: Concepts and Case Studies. Oxford University Press, Oxford, England. chapter 20.
Phillips, P.C., Whitlock, M.C., Fowler, K., 2001. Inbreeding changes the shape of the genetic covariance matrix in drosophila melanogaster. Genetics 158 (3), 1137–1145. https://doi.org/10.1093/genetics/158.3.1137
Price, G.R., 1970. Selection and covariance. Nature 227 (5257), 520–521. https://doi.org/10.1038/227520a0
Price, G.R., 1972. Extension of covariance selection mathematics. Ann. Hum. Genet. 35 (4), 485–490. https://doi.org/10.1111/j.1469-1809.1972.tb00946.x
Rackauckas, C., Nie, Q., 2017. Differentialequations.jl–a performant and feature-rich ecosystem for solving differential equations in Julia. J. Open Res. Software 5 (1), 15.
Reznick, D., 2015. Hard and soft selection revisited: how evolution by natural selection works in the real world. J. Heredity 107 (1), 3–14. https://doi.org/10.1093/jhered/esv076
Roff, D., 2000. The evolution of the G matrix: selection or drift? Heredity 84 (2), 135–142. https://doi.org/10.1046/j.1365-2540.2000.00695.x
Rogers, L.C.G., Williams, D., 2000. Diffusions, Markov Processes and Martingales: itô Calculus. Cambridge University Press. https://doi.org/10.1017/cbo9780511805141
Slatkin, M., 1987. Heritable variation and heterozygosity under a balance between mutations and stabilizing selection. Genetical Res. 50 (1), 53–62. https://doi.org/10.1017/s0016672300023338
Steppan, S.J., Phillips, P.C., Houle, D., 2002. Comparative quantitative genetics: evolution of the G matrix. Trends Ecol. Evol. 17 (7), 320–327. https://doi.org/10.1016/s0169-5347(02)02505-3
Stroock, D.W., Varadhan, S.R.S., 1997. Multidimensional Diffusion Processes. Springer Berlin Heidelberg. https://doi.org/10.1007/3-540-28999-2

Traulsen, A., Claussen, J.C., Hauert, C., 2005. Coevolutionary dynamics: from finite to infinite populations. Phys. Rev. Lett. 95 (23). https://doi.org/10.1103/physrevlett.95.238701

Turelli, M., 1984. Heritable genetic variation via mutation-selection balance: Lerch's zeta meets the abdominal bristle. Theor. Popul. Biol. 25 (2), 138–193. https://doi.org/10.1016/0040-5809(84)90017-0

Turelli, M., 1986. Gaussian Versus Non-Gaussian Genetic Analyses of Polygenic Mutation-Selection Balance. Elsevier. p. 607–628. https://doi.org/10.1016/b978-0-12-398760-0.50030-4

Turelli, M., 1988. Phenotypic evolution, constant covariances, and the maintenance of additive variance. Evolution 42 (6), 1342–1347. https://doi.org/10.1111/j.1558-5646.1988.tb04193.x

Turelli, M., 2017. Commentary: fisher's infinitesimal model: a story for the ages. Theor. Popul. Biol. 118, 46–49. https://doi.org/10.1016/j.tpb.2017.09.003

Turelli, M., Barton, N.H., 1990. Dynamics of polygenic characters under selection. Theor. Popul. Biol. 38 (1), 1–57. https://doi.org/10.1016/0040-5809(90)90002-d

Turelli, M., Barton, N.H., 1994. Genetic and statistical analyses of strong selection on polygenic traits: what, me normal? Genetics 138 (3), 913–941. https://doi.org/10.1093/genetics/138.3.913

Turner, P.E., Chao, L., 1999. Prisoner's dilemma in an RNA virus. Nature 398 (6726), 441–443. https://doi.org/10.1038/18913

Volpert, V., 2014. Elliptic Partial Differential Equations: Volume 2: Reaction-Diffusion Equations. Springer Basel. https://doi.org/10.1007/978-3-0348-0813-2

Walsh, B., Lynch, M., 2018. Evolution and Selection of Quantitative Traits. Oxford University Press.

Week, B., Nuismer, S.L., 2021. Coevolutionary arms races and the conditions for the maintenance of mutualism. Am. Nat. 198 (2), 195–205. https://doi.org/10.1086/714274

Week, B., Nuismer, S.L., Harmon, L.J., Krone, S.M., 2021. A white noise approach to evolutionary ecology. J. Theor. Biol. 521, 110660. https://doi.org/10.1016/j.jtbi.2021.110660

Whitlock, M.C., 1995. Variance-induced peak shifts. Evolution 49 (2), 252–259. https://doi.org/10.1111/j.1558-5646.1995.tb02237.x

Whitlock, M.C., Phillips, P.C., Fowler, K., 2002. Persistence of changes in the genetic covariance matrix after a bottleneck. Evolution 56 (10), 1968–1975. https://doi.org/10.1111/j.0014-3820.2002.tb00122.x

Wickman, J., Koffel, T., Klausmeier, C.A., 2023. A theoretical framework for trait-based eco-evolutionary dynamics: population structure, intraspecific variation, and community assembly. Am. Nat. 201 (4), 501–522. https://doi.org/10.1086/723406

Xu, K., Vision, T.J., Servedio, M.R., 2023. Evolutionary rescue under demographic and environmental stochasticity. J. Evol. Biol. 36 (10), 1525–1538. https://doi.org/10.1111/jeb.14224

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Original Article

Quantitative genetics of microbiome-mediated traits

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Abstract

Multicellular organisms host a rich assemblage of associated microorganisms, collectively known as their "microbiomes." Microbiomes have the capacity to influence their hosts' fitnesses, but the conditions under which such influences contribute to evolution are not clear. This is due in part to a lack of a comprehensive theoretical framework for describing the combined effects of host and associated microbes on phenotypic variation. Here, we address this gap by extending the foundations of quantitative genetic theory to include host-associated microbes, as well as alleles of hosts, as factors that explain quantitative host trait variation. We introduce a way to partition host-associated microbiomes into components relevant for predicting a microbiome-mediated response to selection. We then apply our general framework to a simulation model of microbiome inheritance to illustrate principles for predicting host trait dynamics, and to generalize classical narrow- and broad-sense heritabilities to account for microbial effects. We demonstrate that microbiome-mediated responses to host-level selection can arise from various transmission modes, not solely vertical, but that the contribution of non-vertical modes can depend strongly on host life history. Our work lays a foundation for integrating microbiome-mediated host variation and adaptation into our understanding of natural variation.

Keywords: host–microbiome, heritability, genetics, transmission, genomics

Introduction

Overview

Nearly every lineage of multicellular organisms has a complex assemblage of associated microorganisms. This "microbiome" can contribute important physiological and developmental functions to their hosts (Burns et al., 2017; Goodrich et al., 2016; McFall-Ngai, 2007; Spor et al., 2011). Indeed, evidence that microbiomes play a vital role in shaping host fitness across a broad range of taxa is rapidly emerging (Argaw-Denboba et al., 2024; Gould et al., 2018; Houwenhuyse et al., 2021; O'Brien et al., 2023; Petersen et al., 2023; Rosshart et al., 2017). Although recent work has greatly expanded our understanding of the dynamics and function of host-associated microbiomes (Bruijning et al., 2021; Rosenberg & Zilber-Rosenberg, 2018; Roughgarden, 2023; Sandoval-Motta et al., 2017a; Theis, 2018), it is still unclear the extent to which microbiomes contribute to host evolution, and how such contributions depend on the mode of microbial transmission (Russell, 2019; Wendling & Wegner, 2015; Zilber-Rosenberg & Rosenberg, 2008). This knowledge gap is due in part to the lack of a comprehensive theoretical framework for modeling how host–microbiome system-level traits emerge from the interactions of host genes, microbiome, and environment (Martiny et al., 2015; Roughgarden et al., 2017; Theis, 2018; van Vliet & Doebeli, 2019; Zilber-Rosenberg & Rosenberg, 2008).

Recently, there have been calls to develop such a theoretical framework by incorporating microbiomes into quantitative genetics (Awany & Chimusa, 2020; Benson et al., 2010; Henry et al., 2021; Mueller & Linksvayer, 2022; Wang et al., 2018), a rich mathematical theory that models the inheritance and evolution of complex traits (Lynch & Walsh, 1998; Mackay et al., 2009; Rice, 2004; Walsh & Lynch, 2018). While microbiomes have sometimes been treated as traits influenced by host genetics (Camarinha-Silva et al., 2017; Knights et al., 2014; Opstal & Bordenstein, 2015), they have only recently been considered as inherited contributors to variation of emergent host–microbiome phenotypes (Org et al., 2015; Sandoval-Motta et al., 2017a). This shift has encouraged the use of microbiome-wide association studies (MWAS) by analogy to genome-wide association studies (GWAS) (Camarinha-Silva et al., 2017; Difford et al., 2018; Pérez-Enciso et al., 2021; Qin et al., 2012). However, caution is warranted: microbiomes are dynamic ecological assemblages obeying different rules of inheritance than genomes, and so uncritical application of methods designed for GWAS may produce misleading results. Microbiomes are especially challenging to incorporate into quantitative genetics because their inheritance is not Mendelian, and microbes need not be vertically transmitted (Bruijning et al., 2021; Roughgarden, 2023; Uller & Helanterä, 2013; van Vliet & Doebeli, 2019). Taken together, a careful extension of quantitative genetics is needed that accounts for the

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fundamental differences between host genetic and microbial sources of variation. Such an extension will allow theoretical investigations into how different modes of microbial transmission influence the evolution of host and microbial joint phenotypes, and will help clarify the degree to which these processes can affect long-term evolutionary trajectories.

Another body of theoretical work that considers host-associated microbiomes is the broad framework of inclusive inheritance (Bonduriansky & Day, 2009; Danchin et al., 2011; David & Ricard, 2019; Day & Bonduriansky, 2011; Helanterä & Uller, 2010; Uller & Helanterä, 2013). In this view, direct parent–offspring microbiome transmission can be treated as a maternal effect (Kirkpatrick & Lande, 1989), social transmission as an indirect genetic effect (Bijma et al., 2007; Moore et al., 1997), and environmental acquisition as a form of ecological inheritance (Odling-Smee et al., 2013). These perspectives have been explored in recent work (Week et al., 2025), which emphasizes that while each captures important aspects of microbiome-mediated evolution, none provides a complete account of microbiome inheritance dynamics. The unified framework introduced by Day and Bonduriansky (2011) offers a powerful theoretical foundation for integrating multiple inheritance modes into a single process-based model. However, this approach relies on explicit assumptions about transmission mechanisms, fidelity, and mutability parameters that remain poorly characterized or system-specific in many microbiome contexts, limiting its usefulness as a general foundation for studying host–microbiome evolution. This underscores the need for statistical frameworks that characterize patterns of microbiome inheritance, quantify their contribution to host trait variation, and predict microbiome-mediated responses to selection without presupposing detailed transmission rules.

Previous theoretical work focusing on host–microbiome evolution has yielded valuable insights, yet important gaps remain in our understanding of how such interactions contribute to complex host phenotypes and adaptation. For instance, Henry et al. (2021) emphasize that host trait variance can, in principle, be partitioned into components attributable to host genetics and the host-associated microbiome. However, this partitioning has not been formally defined, nor are quantitative predictions provided. At the other end of the spectrum, Zeng et al. (2017) introduce a mechanistic metacommunity model that integrates microbiome dynamics with host evolution. While this work offers valuable insights into the ecological foundations of microbiome inheritance, it focuses on host fitness rather than arbitrary traits, assumes all host variance arises from the microbiome, and considers only direct parent–offspring and environmental transmission. Roughgarden (2023) has proposed a general conceptual framework for microbiome inheritance, introducing the notions of lineal and collective inheritance. Lineal inheritance maintains correlations between parents and offspring at the individual level (e.g., via direct or intimate neighborhood transmission), whereas collective inheritance maintains correlations between generations at the population level (e.g., via social or indirect environmental transmission). However, this framework adopts a population genetic perspective and does not provide a means to partition host trait variation, and linking patterns of host–microbe ancestral overlap to the microbiome-mediated component of host trait variance is essential for predicting microbiome-mediated responses to selection on host populations. Taken together, these limitations point to the need for a framework that quantifies microbiome-mediated trait variance and links it to host–microbe ancestry relationships that underlie patterns of microbial inheritance.

In this study, we aim to complement prior work by (i) formally partitioning host trait variance into components attributable to host genetics and the microbiome, extending the classical approach of Fisher (1918); (ii) introducing retrospective, microbe-centered criteria to classify microbes based on concordance of their ancestry with host ancestry, complementing the inheritance concepts of Roughgarden (2023); and (iii) combining these tools to further decompose the microbiome component of host trait variance and to derive novel quantitative predictions for the response of microbiome-mediated traits to host-level selection. By categorizing microbes based on patterns of shared ancestry, our framework is independent of specific mechanistic assumptions on the modes of microbiome transmission. Given that patterns of microbe-–host ancestral concordance are potentially measurable, this provides a unique approach to predict microbiome-mediated host adaptation without detailed knowledge of the underlying transmission mechanism. Additionally, our framework is flexible with respect to input data, accommodating coarse indices of microbial taxon presence/absence or relative abundances to fine-scale microbial single nucleotide polymorphisms (SNPs). Our variance decomposition naturally generalizes additive genetic variance, separating contributions from host genetics, distinct microbiome components, and their covariances. Using simulations, we explore the conditions under which different components of the microbiome are expected to mediate a response to selection, and propose extensions of narrow- and broad-sense heritabilities that quantify microbiome-mediated contributions to host-level evolutionary change.

The remainder of the paper proceeds as follows. We begin by reviewing foundational principles from quantitative genetics, which ground our theoretical development. We then introduce our framework for partitioning host trait variance and classifying microbial ancestry, followed by simulations that explore the conditions under which different components of the microbiome are expected to mediate a response to selection. Finally, we extend classical heritability concepts by defining microbially informed analogues of narrow- and broad-sense heritability, and introduce associated transmissibility indices that quantify the evolutionary impact of distinct microbial components.

Quantitative genetic foundational principles

Quantitative genetics developed in the late 19th century through the development of statistical models to describe how continuous phenotypic variation depended on heritable and environmental components and to predict the phenotypic response to selection (Walsh & Lynch, 2018). Even after the rediscovery of Mendel's laws in 1904, debate continued as to how continuous variation could emerge from discrete genes (Provine, 2001). R.A. Fisher settled the issue in a foundational paper (Fisher, 1918), showing that distributions of continuous phenotypes arise when alleles at multiple loci are considered. Following this, Fisher then introduced the concepts of *average excess* and *average effect* of allele substitutions in order to associate an additive quantity with each allele that best predicts the response to se-

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Table 1. Summary of key notation.

Symbol	Meaning	Symbol	Meaning
g	Host genotype	$\Delta\bar{z}$	Response to selection in host mean trait
m	Host microbiome	β	Selection gradient
z_{gm}	Genotypic–microbic value of (g, m)	h^2	Narrow-sense genetic heritability
δ_{gm}	Average excess of z_{gm}	b^2	Narrow-sense microbial transmissibility
γ_g	Additive effect of g on δ_{gm}	t^2	Narrow-sense total transmissibility
γ_i	Additive effect of locus i	H^2	Broad-sense genetic heritability
ω_m	Additive effect of m	B^2	Broad-sense microbial transmissibility
ω_j	Additive effect of taxa j	T^2	Broad-sense total transmissibility
G_A	Additive genetic variance	M_L	Lineal microbial variance
M_A	Additive microbial variance	M_N	Non-lineal microbial variance
C_A	Additive gene–microbe covariance	M_V	Novel microbial variance
M_A^ψ	Selective microbial variance	C_A^ψ	Selective gene–microbe covariance

lection (Bürger, 2000; Fisher, 1941). This general approach follows the widely used theory of multivariate linear models, a perspective we use in extending the theory to account for microbial factors. Because these concepts are at the core of theoretical quantitative genetics, we summarize the basic principles here.

For a given host genotype g, write z_g as the host trait associated with g averaged over all other trait-mediating factors (including environmental factors and host microbiome composition). The quantity z_g is referred to as a *genotypic value*. Then, writing p_g for the frequency at which host genotype g occurs in the population, we have that $\bar{z} = \sum_g p_g z_g$ is the average of z_g over all host genotypes (equivalent to the population mean trait since all other factors have already been averaged over), and the quantity $\delta_g = z_g - \bar{z}$ is the *average excess* associated with host genotype g. Fisher decomposed the average excess into an additive effect α_g (defined to be the sum of additive effects associated with each allele in host genotype g) and a residual deviation ρ_g such that $\delta_g = \alpha_g + \rho_g$.

The definition of genotypic values implies the total host phenotypic variation P can be decomposed as $P = G + E$, where $G = \sum_g p_g(z_g - \bar{z})^2$ is the variance of genotypic values across the population (called the *genetic variance*). Although E is often referred to as the *environmental variance*, it is more precisely defined as $E = P - G$, the remaining variance left unexplained by genotypic values. Furthermore, defining the *additive genetic variance* $G_A = \sum_g p_g \alpha_g^2$ and residual variance $G_R = \sum_g p_g \rho_g^2$, and defining the additive effects α_g to be chosen to minimize G_R (subject to the additivity constraint); with this choice the genetic variance decomposes as $G = G_A + G_R$ (Bürger, 2000). Although (Fisher, 1918) assumed Hardy–Weinberg equilibrium, which allowed derivation of more specific results, the overall approach only depends on the identification of the genotypic values z_g.

This definition of G_A turns out to be useful for accurately predicting the response of a trait to selection. For instance, G_A is more useful than G in predicting the response to selection because parents transmit alleles, not entire genotypes. In particular, the classical *breeders equation* takes the form $\Delta\bar{z} = G_A \beta$, where $\Delta\bar{z} = \bar{z}' - \bar{z}$ is the difference between offspring mean trait $\bar{z}'$ and parent mean trait $\bar{z}$, and $\beta = \mathrm{Cov}(w, z)/P$ is the correlation coefficient of fitness (w) with phenotype (Lande, 1976).

These definitions are also used to define *heritability* of a trait as a measure of parent–offspring resemblance. In particular, the quantities $h^2 = G_A/P$ and $H^2 = G/P$ are referred to as the *narrow-* and *broad*-sense heritabilities, respectively (reviewed by Visscher et al., 2008). With this notation, we note that the breeders equation was initially introduced by Lush (1937) in the form $R = h^2 S$, where $R = \Delta\bar{z}$ is the response to selection and S is the selection differential (see Walsh & Lynch, 2018). Key notation used throughout our paper is summarized in Table 1.

In what follows we build on these foundations to incorporate the effects of host-associated microbiomes. By generalizing the above approach, we naturally arrive at a generalization of additive genetic variance, the breeders equation, and narrow-sense and broad-sense heritabilities for microbiome-mediated traits. Before we introduce our framework, we discuss some initial assumptions for incorporating microbiomes in the following section.

Perspectives on incorporating microbes into host evolution

By building on quantitative genetics theory, we are admittedly using a very host-centric framework that leads to several constraining assumptions. In particular, we focus on a single trait that is an emergent property of the host–microbe system, such as the ability of the host to acquire energy from food that first must be processed by components of the host gut microbiome. For example, termites and their associated gut microbiomes together can process high amounts of cellulose to acquire energy (Ali et al., 2019; Arora et al., 2022; Maurice & Erdei, 2018).

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We also consider contributions from both host genes and individual microbes to this trait as potentially heritable units, meaning that they can be transmitted from a donor to a recipient, and influence the value of the emergent phenotype. Traditional quantitative genetics defines the donor and recipients solely as parents and offspring through the transmission of alleles. In sexually reproducing diploid populations, the process of gametogenesis and syngamy leads to Mendelian segregation.

In contrast, microbes can be transmitted among hosts in a variety of ways. In particular, a microbial parent (donor) and microbial offspring (recipient) may not be genetically related. Additionally, unlike the transmission of alleles via sexual reproduction, microbial recipients can have more than two donors, depending on the microbial taxa considered and mode of transmission. In some cases, the donor and receiver of transmitted host genetic and microbial units will be the same, which has been described as *lineal transmission* (Roughgarden, 2023) and *co-propagation* (Mueller & Linksvayer, 2022). In these cases, narrow-sense heritability of microbiome-mediated traits may be confounded by microbiome inheritance, possibly inflating estimates of heritability based on pedigree compared to estimates based on SNPs (Sandoval-Motta et al., 2017b).

We must also take into account that microbiome composition is governed by community assembly and not Mendelian processes, and microbial abundances can vary dynamically throughout the life of the host. These have been termed the *fidelity of transmission* and *persistence fidelity* of microbes (Mueller & Linksvayer, 2022), with the idea that microbe persistence is often less than that of an allele. Based on this consideration, one possible path forward is to study models of metacommunity dynamics as the basis of microbiome inheritance and fidelity from the lens of an extended framework of quantitative genetics (Koskella et al., 2017; Leibold et al., 2004; Miller et al., 2018; Zeng et al., 2017). Here, we focus on beginning this extension of quantitative genetics. Analysis of metacommunity models of microbiome inheritance using this framework is outside the scope of this paper, but it is likely microbiome dynamics will alter indices of microbiome-mediated host trait inheritance.

Several authors have recently attempted to extend heritability to include microbial contributions, and have proposed novel terminology such as *microbiability* and *holobiontability*. However, we follow the convention established by Rothschild et al. (2018) and clarified by Mueller and Linksvayer (2022), who use H^2 and h^2 for broad- and narrow-sense *genetic heritability*, and B^2 and b^2 for what they term *microbiome heritability*. To avoid conflating genetic and microbial sources of variation, we refer to the latter as *microbial "transmissibility."* As Mueller and Linksvayer (2022) make clear, B^2 and b^2 refer to microbial contributions to the emergent host phenotype, not to be confused with the genetic heritability of microbiomes as host traits in and of themselves (Morris & Bohannan, 2024). Here, we propose a more precise usage of this terminology to refer specifically to the proportion of host trait variance mediated by the component of host microbiome that is both transmitted across host generations and in some cases mediates a response to host-level selection (see section titled *Heritabilities and transmissibilities*).

Materials and methods

Analysis of variance of microbiome-mediated traits

The partitioning of variance reviewed in the *Quantitative genetic foundational principles* section is not restricted to genetic material, but can be applied to any collection of trait-mediating factors (e.g., soil salinity, local rainfall, temperature, oxygen concentration, and pH). In particular, for a microbiome-mediated host trait, relevant factors include the host genotype g and host microbiome m. The overall approach is sufficiently general to accommodate any summary of host genotype and host microbiome, such as presence/absence of microbial taxa and microbial SNPs obtained from 16s data or shotgun metagenomes. However, for the sake of clarity, we describe our framework focusing on host allele counts and relative abundances of microbe taxa.

We consider diploid host populations with L biallelic loci. We denote by $g_i \in \{0, 1, 2\}$ the number of alleles at locus $i \in \{1, \ldots, L\}$ (the choice of allele counted is arbitrary). Turning to the host microbiome, we assume microbiomes may be summarized by the relative abundances of S microbial taxa (relative to the total microbiome abundance), with m_j being the relative abundance of microbe taxon $j \in \{1, \ldots, S\}$. Vectors of allele counts and relative microbial abundances are, respectively, written as $g = (g_1, \ldots, g_L)^\dagger$ and $m = (m_1, \ldots, m_S)^\dagger$, where † denotes matrix transposition.

We write z_{gm} as the expected trait value for hosts carrying the genotype–microbiome pair (g, m) averaged across all other trait-mediating factors, and refer to z_{gm} as the *genotypic–microbic value* of (g, m). Additionally, we write $\bar{z}$ as the mean *genotypic–microbic value* across all hosts in the population so that $\bar{z} = \sum_{gm} p_{gm} z_{gm}$, where p_{gm} is the frequency of hosts carrying the genotype–microbiome pair (g, m) and the sum is taken over all possible combinations of (g, m). By definition, $\bar{z}$ is also the mean trait of the host population.

The *average excess* of the pair (g, m) is then $\delta_{gm} = z_{gm} - \bar{z}$, and we consider the decomposition $\delta_{gm} = \alpha_{gm} + \rho_{gm}$, where α_{gm} is an additive component that decomposes into additive genetic and additive microbial effects of the pair (g, m) on δ_{gm}, and ρ_{gm} is the residual deviation. In particular, following Fisher's general approach, we assume $\alpha_{gm} = \gamma_g + \omega_m$, where γ_g is the additive genetic effect and ω_m is the additive microbial effect on the average excess δ_{gm}.

Finally, using our assumptions for summarizing host genotype and host microbiome, the definition of additive effects implies that they can be written as

$$\gamma_g = \gamma_0 + \sum_{i=1}^{L} \gamma_i g_i, \quad \omega_m = \omega_0 + \sum_{j=1}^{S} \omega_j m_j, \tag{1}$$

where γ_i is the per count additive allelic effect of locus $i \in \{1, \ldots, L\}$ on δ_{gm} and ω_j is the additive effect of the relative abundance of taxon $j \in \{1, \ldots, S\}$ on δ_{gm}. To be clear, γ_g is distinguished from γ_i because g is a vector and i is an integer. A similar distinction holds for ω_m and ω_i. Lastly, the sum $\alpha_0 = \gamma_0 + \omega_0$ acts as an intercept for the linear model, and the components γ_0 and ω_0 will be determined by our decomposition of P_A below.

Following the general scheme outlined in the *Quantitative genetic foundational principles* section, we can write the additive component of host trait variation explained by host

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genetic and host microbiome factors as $P_A = \sum_{gm} p_{gm}\alpha_{gm}^2$. The definition of P_A is formalized by solving the least-squares problem of minimizing $P_R = \sum_{gm} p_{gm}\rho_{gm}^2$ with respect to $\alpha_0, \gamma_1, \ldots, \gamma_L, \omega_1, \ldots, \omega_S$, where $\rho_{gm} = \delta_{gm} - \alpha_{gm}$. In Supplementary Section 1, we show that setting $\mathrm{Cov}(z_{gm}, g)$ to the vector with ith entry $\mathrm{Cov}(z_{gm}, g_i)$, $\mathrm{Cov}(z_{gm}, m)$ to the vector with jth entry $\mathrm{Cov}(z_{gm}, m_j)$, Γ to the matrix with ijth entry $\mathrm{Cov}(g_i, g_j)$, Ω to the matrix with ijth entry $\mathrm{Cov}(m_i, m_j)$, and Ξ to the matrix with ijth entry $\mathrm{Cov}(g_i, m_j)$, the least-squares problem reduces to solving the linear system

$$\mathrm{Cov}(z_{gm}, g) = \Gamma\,\gamma + \Xi\,\omega, \quad \mathrm{Cov}(z_{gm}, m) = \Xi^\dagger\,\gamma + \Omega\,\omega \tag{2}$$

for $\gamma = (\gamma_1, \ldots, \gamma_L)^\dagger$ and $\omega = (\omega_1, \ldots, \omega_S)^\dagger$, where † denotes matrix transposition. The solution is then given by

$$\alpha = \Sigma^{-1}\mathrm{Cov}(z_{gm}, f) \tag{3}$$

with $\alpha = (\gamma_1, \ldots, \gamma_L, \omega_1, \ldots, \omega_S)^\dagger$, $f = (g_1, \ldots, g_L, m_1, \ldots, m_S)^\dagger$, $\alpha_0 = -\sum_i \alpha_i \bar{f}_i$, $\mathrm{Cov}(z_{gm}, f)$ the vector with ith entry $\mathrm{Cov}(z_{gm}, f_i)$, and Σ the block matrix

$$\Sigma = \begin{bmatrix} \Gamma & \Xi \\ \Xi^\dagger & \Omega \end{bmatrix}. \tag{4}$$

Similar to the case for additive genetic effects reviewed in the *Quantitative genetic foundational principles* section, this solution is unique so long as Σ has an inverse. This means that if the allele counts at any pair of loci, or if the relative abundances of any pair of microbes, or if an abundance of a microbe and an allele count of a locus perfectly correlate, the solution is no longer unique.

Since we have centered the trait (the average of δ_g is zero), it turns out that with α_0, γ, and ω as above, that $\alpha_0 = -\sum_i \gamma_i\bar{g}_i - \sum_i \omega_i\bar{m}_i$, where $\bar{g}_i$ is the average number of alleles at locus i (which is twice the allele frequency due to host genotypes being diploid) and $\bar{m}_i$ is the average relative abundance of microbe taxon i. Then, it is convenient to set $\gamma_0 = -\sum_i \gamma_i\bar{g}_i$ and $\omega_0 = -\sum_i \omega_i\bar{m}_i$. With these choices for γ_0 and ω_0, the additive components of host trait variance can be expressed in terms of allele counts and relative microbe abundances as

$$G_A = \sum_{i=1}^{L} \gamma_i^2 \mathrm{Var}(g_i) + 2\sum_{i=1}^{L}\sum_{j=1}^{i-1} \gamma_i\gamma_j \mathrm{Cov}(g_i, g_j), \tag{5a}$$

$$M_A = \sum_{i=1}^{S} \omega_i^2 \mathrm{Var}(m_i) + 2\sum_{i=1}^{S}\sum_{j=1}^{i-1} \omega_i\omega_j \mathrm{Cov}(m_i, m_j), \tag{5b}$$

$$C_A = 2\sum_{i=1}^{L}\sum_{j=1}^{S} \gamma_i\omega_j \mathrm{Cov}(g_i, m_j), \tag{5c}$$

where we use the convention that an empty sum $\sum_{j=1}^{0} x_j = 0$ is zero.

More pragmatically, these expressions can be obtained from a least-squares fit of a multivariate linear model that predicts a sample of traits $\zeta = (z_1, \ldots, z_n)^\dagger$ from associated samples of host gene sequences and microbiome compositions (collected into the data matrix X, which, for its ith row, has allele counts at L loci and relative microbe abundances across S taxa associated with the ith sample) such that $\zeta = X\,\alpha + \varepsilon$, with $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_n)^\dagger$ being the error terms for each sample. Various algorithms exist for efficiently finding a least-squares solution for the additive effects α, such as pivoted QR decomposition (Strang, 2019).

This overall approach holds for genotypic–microbic values occurring as arbitrary functions of genotype–microbiome pairs; $z_{gm} = \varphi(g, m)$ for any function φ. That is, this approach accounts for any kind of interaction among genes, among microbes, and between genes and microbes on the expression of host traits. To illustrate, we apply this approach to a model of genotypic–microbic values that includes interactions between genes and microbes in the following section.

A model of gene–microbe interactions

The approach to analysis of variance above establishes a statistical model. We can then apply this model either to empirical or simulated data. As a demonstration, we examine what the statistical model tells us when applied to a mechanistic model involving pairwise interactions between host loci and microbial taxa. Considering $z_{gm} = \varphi(g, m)$ as a function of the genotype–microbiome pair (g, m), we analyze the model

$$z_{gm} = \mathring{z}_0 + \sum_{i=1}^{L} \mathring{\gamma}_i g_i + \sum_{j=1}^{S} \mathring{\omega}_j m_j + \sum_{i=1}^{L}\sum_{j=1}^{S} \mathring{\chi}_{ij} g_i m_j, \tag{6}$$

where $\mathring{z}_0, \mathring{\gamma}_i, \mathring{\omega}_j, \mathring{\chi}_{ij}$ are the mechanistic model parameters, and $\mathring{\gamma}_i, \mathring{\omega}_j$ are not necessarily equivalent to the additive effects γ_i, ω_j as we now show.

Using the approach outlined in the previous section, the additive genetic and microbial effects from model (6) depend on $\mathrm{Cov}(z_{gm}, g_k)$ and $\mathrm{Cov}(z_{gm}, m_k)$, which are given by

$$\mathrm{Cov}(z_{gm}, g_k) = \sum_{i=1}^{L} \mathring{\gamma}_i \mathrm{Cov}(g_i, g_k) + \sum_{j=1}^{S} \mathring{\omega}_j \mathrm{Cov}(m_j, g_k) + \sum_{i=1}^{L}\sum_{j=1}^{S} \mathring{\chi}_{ij} \mathrm{Cov}(g_i m_j, g_k), \tag{7a}$$

$$\mathrm{Cov}(z_{gm}, m_k) = \sum_{i=1}^{L} \mathring{\gamma}_i \mathrm{Cov}(g_i, m_k) + \sum_{j=1}^{S} \mathring{\omega}_j \mathrm{Cov}(m_j, m_k) + \sum_{i=1}^{L}\sum_{j=1}^{S} \mathring{\chi}_{ij} \mathrm{Cov}(g_i m_j, m_k). \tag{7b}$$

Writing

$$v_k = \sum_{i=1}^{L}\sum_{j=1}^{S} \mathring{\chi}_{ij} \mathrm{Cov}(g_i m_j, g_k), \quad k = 1, \ldots, L, \tag{8a}$$

$$v_k = \sum_{i=1}^{L}\sum_{j=1}^{S} \mathring{\chi}_{ij} \mathrm{Cov}(g_i m_j, m_{k-L}), \quad k = L+1, \ldots, L+S, \tag{8b}$$

the additive genetic and microbial effects are given by $\alpha = \mathring{\alpha} + \Sigma^{-1}v$, where $\alpha = (\gamma_1, \ldots, \gamma_L, \omega_1, \ldots, \omega_S)$ are the additive genetic and microbial effects, Σ is defined at the end of the previous section, and $\mathring{\alpha} = (\mathring{\gamma}_1, \ldots, \mathring{\gamma}_L, \mathring{\omega}_1, \ldots, \mathring{\omega}_S)$ are mechanistic model parameters.

In the absence of gene–microbe interactions, so that $\mathring{\chi}_{ij} = 0$ for all $i = 1, \ldots, L$ and $j = 1, \ldots, S$, we have $v = 0$ and the

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additive effects formally defined above become equivalent to the additive effects of the model; $\gamma_i = \mathring{\gamma}_i$, $\omega_j = \mathring{\omega}_j$. However, if $\mathring{\chi}_{ij} \neq 0$ for any host locus i and any microbial taxon j, the formal additive effects and model additive effects are in general no longer equivalent ($\gamma_i \neq \mathring{\gamma}_i$, $\omega_j \neq \mathring{\omega}_j$) because the formal additive effects will also include terms due to gene–microbe interactions.

To unpack this more, consider the example where the effect of a microbe only occurs in the presence of an allele at a haploid biallelic locus such that neither the allele nor the microbe has an effect on the host trait in the absence of the other. In this case, the above model simplifies to $z_{gm} = \mathring{z}_0 + \mathring{\chi} gm$, where $g = 0, 1$ determines the presence of the allele, $m = 0, 1$ determines the presence of the microbe, and $\mathring{\chi}$ quantifies the effect of their interaction. Writing p_{gm} as the frequency of hosts carrying the pair $(g = 1, m = 1)$, p_g the frequency of hosts with the allele (so the marginal frequency of $g = 1$), and p_m the frequency of hosts with the microbe (so the marginal frequency of $m = 1$), then the additive effect of the allele is $\gamma \propto (p_m - p_{gm})(1 - p_m)$, and the additive effect of the microbe is $\omega \propto (p_g - p_{gm})(1 - p_g)$, where $\propto$ means *proportional to* and the constant of proportionality for both quantities is $\mathring{\chi}\, p_{gm}/(p_g p_m(1 - p_g)(1 - p_m) - (p_{gm} - p_g p_m)^2)$. Hence, even in the absence of any underlying additive effects in the mechanistic model, the formally defined additive effects of the statistical model are non-zero. Additionally, each of these effects depends on the frequency of the other factor.

The observation that non-additive mechanistic effects can contribute to additive components of trait variance directly extends the recognition that epistasis contributes to additive genetic variance (Barton & Turelli, 2004; Cheverud & Routman, 1996). In Supplementary Section 2, we provide expressions for additive genetic and additive microbial effects, given a model that includes all possible pairwise interactions between genes and microbes explained by products of allele counts and microbial abundances.

Microbiome-mediated response to host-level selection

Because the transmission of microbes is not necessarily direct from parent to offspring, nor does it occur with high fidelity relative to genetic transmission (e.g., microbiome composition exhibits significant variation between related individuals, see Tavalire et al., 2021; Tierney et al., 2019), we cannot transfer many of the important simplifying assumptions of classical quantitative genetics regarding inheritance to the study of microbiome-mediated traits. These simplifying assumptions have also been central to the power of the classical quantitative genetic approach, such as the use of pedigree analysis in the animal model (Wilson et al., 2009). Additionally, the degree to which additive host trait variation explained by microbes mediates an inter-generational response to selection on host individuals is not obvious. To overcome these issues, and identify components of host microbiomes that contribute host population-level selection responses, we propose a classification scheme to account for different patterns of ancestral concordance between microbes and the hosts that carry them. We then apply a set of simplifying assumptions to clarify our initial analysis, and consider consequences for a microbially mediated heritable response to selection on host individuals. This analysis motivates definitions of microbial heritabilities and host trait transmissibilities that we introduce below in the *Heritabilities and transmissibilities* section.

Classifying microbes by concordance with host ancestry

As a first step in addressing the challenge described above, we classify host-associated microbes according to the ancestral relationship between their lineages and those of the hosts they inhabit. Specifically, we consider the degree to which microbial lineages are associated with host lineages. A lineage is a chain of (genetic) parent–offspring relationships, and so the distinction regards the degree to which transmission of microbes matches transmission of genetic material. These definitions make no assumptions about transmission mode (e.g., social, environmental, and parent–offspring). This leads to the following partition:

(1) **Lineal ancestry:** Microbes are classified as *lineal* if their lineages have historically remained associated with the same host lineage across multiple generations (Figure 1, left panel). In other words, if the ancestor of a given microbe within an individual host at some point back in time can be found in a host who is a direct (genealogical) ancestor of that individual. These microbes are most likely to have co-diverged, co-adapted, or co-transmitted with their host lineage and are therefore most likely to contribute to selection responses at the host population level.

(2) **Non-lineal ancestry:** Microbes are classified as *non-lineal* if their lineages have been associated with host lineages but do not consistently follow host lineages (Figure 1, center panel). For instance, these microbes may frequently disperse horizontally between related hosts such as siblings, or between less related conspecific hosts, or may move back and forth between hosts and the environment. Their ancestry is partially concordant with host ancestry, and their contribution to host adaptation may vary depending on transmission routes and patterns of host association.

(3) **Novel ancestry:** Microbes are classified as *novel* if their ancestral lineages have not spent any time with host ancestors (Figure 1, right panel). These microbes originate from sources external to the host population (e.g., biotic or abiotic environmental reservoirs) and represent recently acquired microbial lineages with no history of host association. By definition, novel microbes are not inherited through host lineages and are unlikely to have experienced selection aligned with host reproductive success. All novel microbes are by definition environmentally acquired, but not all environmentally acquired microbes are novel.

These definitions, formulated at the scale of microbial individuals, are illustrated in Figure 1. The goal of this scheme is to partition host microbiomes into components that are (1) unlikely to contribute to host population–level selection responses (novel microbes), (2) most likely to do so (lineal microbes), or (3) of uncertain contribution (non-lineal microbes). For some host–microbiome systems, it may be necessary to add additional categories, for instance, between non-lineal and novel for microbes that are nearly novel, but with ancestral lineages that have spent significant time in the host population. This partitioning of host microbiomes

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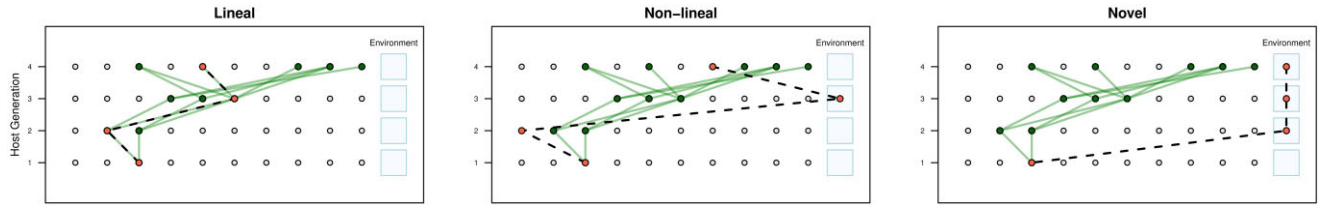


Figure 1. Illustration of our partitioning of host microbiomes by ancestral concordance. Each panel shows a different pattern of host–microbe ancestral concordance. Rows represent host generations (most recent at the bottom), with squares indicating the environment. Circular nodes are host individuals; green lines and circles trace the focal host's ancestry, while dashed lines and tomato-colored circles trace the focal microbe's lineage. **Left:** A lineal microbe whose ancestry closely tracks the host lineage: inheritance of microbes (dotted line) always follows genetic ancestry (green lines). **Middle:** A non-lineal microbe with ancestry mostly among hosts but not following genetic ancestry, and occasionally entering the environment. **Right:** A novel microbe whose ancestry lies entirely outside the host population.

is conceptually similar to the notions of lineal and collective inheritance introduced by Roughgarden (2023). Indeed, a lineal microbe in our framework would be considered lineally inherited in Roughgarden's framework. The primary distinction is that Roughgarden's concepts focus on inheritance processes, but our partitioning focuses on categorizing microbes by their tendency to be inherited (possibly lineally, collectively, or not at all). For instance, the class of non-lineal microbes spans a broad spectrum of associations with the host; many may be transient and therefore have limited capacity to contribute to collective inheritance.

To leverage this approach for understanding how microbiomes mediate responses to selection, we next quantify the degree to which signals of novel, lineal, and non-lineal ancestry are reflected in each microbial taxon within a host individual. These values can be averaged across host individuals to obtain a summary measure of ancestral concordance for each microbial taxon. Assessing host–microbe ancestral concordance at the level of microbial taxa is especially useful in our framework, as it aligns with the scale at which we quantify microbial effects on host traits.

Partitioning host trait variance by ancestral concordance

Although many microbial taxa likely exhibit several patterns of ancestral concordance, for this initial inquiry we consider a simplified scenario where each taxon consists of individuals that are either entirely novel, lineal, or non-lineal. This simplifies the further partitioning host trait variance according to the partitioning of microbiomes proposed above, and this simplicity allows us to highlight the consequences for host population-level selection responses. More realistic models that account for the complexity of microbial transmission could be studied in future work. In particular, assuming the S microbial taxa in the host microbiome can be subdivided into S_L lineal taxa, S_N non-lineal taxa, and S_V novel taxa (so that $S = S_L + S_N + S_V$), we can rewrite the additive microbial effect on the average excess $\delta_{gm} = z_{gm} - \bar{z}$ as $\omega_m = \ell_m + n_m + \upsilon_m$, where

$$\ell_m = \ell_0 + \sum_{i=1}^{S_L} \ell_i m_{L,i}, \quad n_m = n_0 + \sum_{i=1}^{S_N} n_i m_{N,i},$$

$$\upsilon_m = \upsilon_0 + \sum_{i=1}^{S_V} \upsilon_i m_{V,i}, \tag{9}$$

where the additional subscripts indicate the pattern of ancestral concordance (*L*ineal, *N*on-lineal, No*V*el) each group of taxa has with its host. Taking this partitioning a step further, we can write the additive microbial variance as

$$M_A = M_L + M_N + M_E + C_{LN} + C_{LE} + C_{NE}, \tag{10}$$

where $M_L = \sum_{g,m} p_{gm} \ell_m^2$ is the component of additive variance due to lineal microbes (which we refer to as the *additive lineal variance*), $M_N = \sum_{g,m} p_{gm} n_m^2$ is the component due to non-lineal microbes (the *additive non-lineal variance*), $M_V = \sum_{g,m} p_{gm} \upsilon_m^2$ is due to novel microbes (the *additive novel variance*), $C_{LN} = 2\sum_{g,m} p_{gm} \ell_m n_m$ is due to non-random associations between lineal and non-lineal microbes (the *additive lineal–non-lineal covariance*), $CLV = 2\sum_{g,m} p_{gm} \ell_m \upsilon_m$ is due to non-random associations between the abundances of lineal and novel microbes (the *additive lineal–novel covariance*), and $C_{NV} = 2\sum_{g,m} p_{gm} n_m \upsilon_m$ is due to non-random associations between the abundances of non-lineal and novel microbes (the *additive non-lineal–novel covariance*). For later use, we also introduce the *additive gene–lineal covariance* $C_L = 2\sum_{g,m} p_{gm} \gamma_g \ell_m$ as the component of *additive gene–microbe covariance* due to lineal microbes, the *additive gene-non–lineal covariance* $C_N = 2\sum_{g,m} p_{gm} \gamma_g n_m$ as the component due to non-lineal microbes, and the *additive gene–novel covariance* $C_V = 2\sum_{g,m} p_{gm} \gamma_g \upsilon_m$ as the component due to novel microbes.

Results

Interactions between host-level selection and ancestral concordance

By definition, novel microbes cannot contribute to a microbiome-mediated response of the host trait to selection on hosts. In contrast, lineal microbes will likely make the most reliable contributions to a microbiome-mediated response to host-level selection. The degree to which non-lineal microbes contribute to a microbiome-mediated response depends on the stage in host life history that selection occurs relative to the stage in host life history that non-lineal microbes are transmitted. For instance, if non-lineal microbes are first shed into the environment before the host parental population experiences host-level selection (so selection does not shape the set of non-lineal sources), then non-lineal microbes will not contribute to a microbiome-mediated response. On the other hand, if non-lineal sources include only the microbiomes of host parents that, for example, survived an episode of viability-based selection, then non-lineal microbes may contribute significantly to a microbiome-mediated response.

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Based on the above reasoning, to obtain an accurate model for predicting a microbiome-mediated response to selection, novel microbes must first be identified and culled from the set of factors considered, and their abundances should be averaged over when calculating z_{gm}. Additionally, it must be determined whether or not non-lineal microbes are sourced from selected parents or from the broader population of unselected parents, because this is likely associated with whether or not they contribute to a response to selection. Having identified the microbial taxa that contribute to a response to selection, we write M_A^{ψ} (C_A^{ψ}) for the additive microbial variance (gene–microbe covariance) calculated using only the abundances of those taxa, and refer to this quantity as the *selective* additive microbial variance (gene–microbe covariance). Building on this logic, we anticipate three basic conclusions.

First, if we summarize the abundances of lineal microbes using presence/absence, and assume host offspring inherit these binary values for each taxon independently, then the inheritance of these microbes is structurally equivalent to the inheritance of genetic alleles at freely recombining loci. Hence, in this case, it is clear that lineal microbes can contribute to a response to selection. This clearly still holds when presence/absence is replaced with higher-resolution summaries of microbial abundances such as approximated relative abundances. In particular, we expect the selective additive microbial variance M_A^{ψ} to always include the additive lineal variance M_L, and the selective additive gene–microbe covariance C_A^{ψ} to always include the additive gene-lineal covariance C_L (defined at the end of the *Partitioning host trait variance by ancestral concordance* section).

Second, if non-lineal microbes are sourced from the broader population of "unselected" parents (i.e., if the source of non-lineal microbes is independent of fitness), then we anticipate non-lineal microbes to not contribute to a response to selection. In this case, the selective additive microbial variance would still simplify to the additive lineal variance, $M_A^{\psi} = M_L$, and the selective additive gene–microbe covariance would also still simplify as $C_A^{\psi} = C_L$.

However, as a third conclusion, if non-lineal microbes are sourced from selected parents, or more generally, values of the selected trait are correlated with probability of being a non-lineal source of microbes, we anticipate that their abundances will also contribute to a response to selection. In this case, selective additive microbial variation would become $M_A^{\psi} = M_L + M_N + C_{LN}$ and selective additive gene–microbe covariance becomes $C_A^{\psi} = C_L + C_N$. To test our expectations described above, we implemented a simulation model based on the general setup described at the beginning of the *Analysis of variance of microbiome-mediated traits* section, which we now describe.

Simulation description

Our simulation model assumes that host traits follow an additive model equivalent to that analyzed in the *A model of gene–microbe interactions* section, except without any gene–microbe interactions (so $\mathring{\chi}_{ij} = 0$ for each i, j). As a function of phenotype, host fitness is set to $W(z) = \exp(sz)$, which leads to directional selection for larger host traits when $s > 0$ (we set $s = 10^{-3}$). We simulate 100 freely recombining host genetic loci. The abundances of lineal microbes in host offspring are Poisson distributed around the mid-parents of those taxa. For non-lineal microbes, we consider two models: 1) each host offspring chooses a single host individual uniformly from the broader unselected parental population as its non-lineal source and 2) each host offspring chooses a single host individual from the selected parental population with probability proportional to its fitness. Once the non-lineal donor is chosen, the host offspring inherits non-lineal abundances that are Poisson distributed around the abundances of non-lineal taxa in the donor. Finally, novel microbe abundances are drawn independently and identically from a Poisson distribution for each taxon in each host in each host generation. We assume 100 microbial taxa in each microbiome component, with each taxon having an average abundance of 50 in each host parent. This model of microbial inheritance is illustrated in Figure 2.

Results are obtained by first drawing normally distributed additive effects, and then observed and predicted responses to selection are averaged over randomly drawn genotype–microbiome pairs for host parents, repeated selection experiments, and repeated formation of offspring from selected parents. Figure 3 illustrates mean trait dynamics under our model when different combinations of factors mediate traits, and when non-lineal microbes are sourced from pre-selected or post-selected parents. In addition, time series of correlations between host allele counts and microbe abundances are shown in Figure 4. Parameter values used for simulating data presented in Figures 3 and 4 are provided in Supplementary Table 1. Parameter values used for simulating data presented in Figure 5 are provided in Supplementary Table 2. Code to reproduce these results (written in Julia) is provided at the Zenodo repository: https://zenodo.org/records/16793112.

Simulation results

Our simulation results (summarized in Figure 5) demonstrate that when a host trait is significantly mediated by heritable microbes (such as lineal and possibly non-lineal microbes), then the classical breeders equation, which only accounts for genetic factors, severely underestimates the response to selection. In particular, for microbiome-mediated traits, we find that the observed response to selection exceeds predictions based solely on host allele counts: $\Delta\bar{z} \geq G_A\,\beta$.

Conversely, our model shows that including all trait-mediating microbes in the additive variance substantially overestimates the selection response. This result occurs because many microbes will be novel and thus not transmitted across host generations (which explains why the lines associated with GLNV and GLN coincide for both panels of Figure 3). More precisely, including trait-mediating microbial taxa that are not transmitted across host generations (especially novel microbes, but possibly also non-lineal microbes) inflates the additive microbial variance, which leads to observations that are exceeded by predictions: $\Delta\bar{z} \leq (G_A + M_A + C_A)\,\beta$.

In general, results from our simulations agree with our expectations described above. That is, by including *only* the transmissible microbes that contribute to an intergenerational response to selection on host individuals, we obtain the heuristic $\Delta\bar{z} = (G_A + M_A^{\psi} + C_A^{\psi})\,\beta$.

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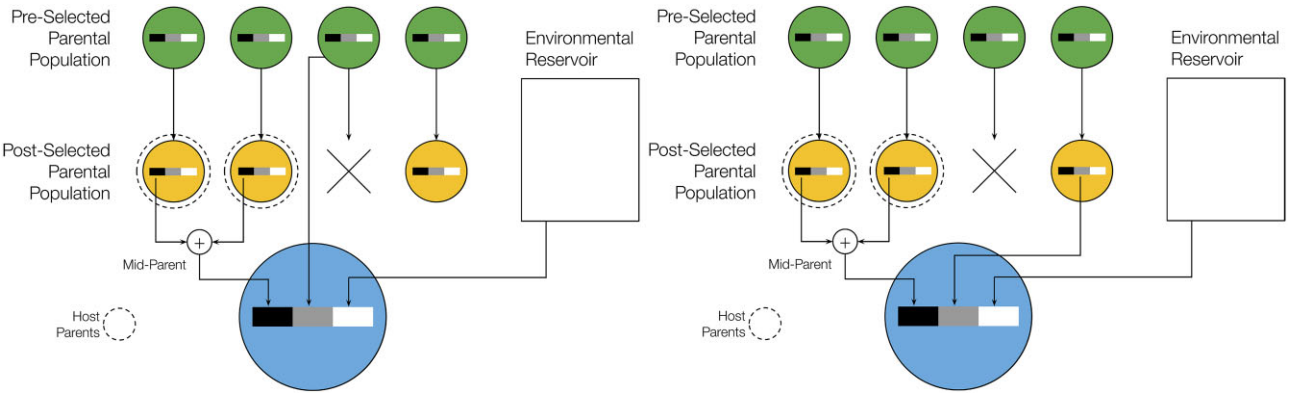


Figure 2. Graphical representation for the two models of microbiome inheritance used in our simulations. In the left panel, non-lineal microbes for a given offspring are acquired from a randomly chosen individual in the pre-selected parental population. In the right panel, non-lineal microbes are acquired from a randomly chosen individual in the post-selected population. Circles represent hosts and rectangles within hosts the lineal (black), non-lineal (gray), and novel (white) microbiome components. The green circles at the top of each panel represent the parental population before an episode of viability selection, and the mustard circles represent the parental population after an episode of viability selection (with vertical straight arrows indicating the trajectories of host individuals). The vertical rectangle on the right represents an environmental reservoir that novel microbes are sampled from. Hosts highlighted with dashed concentric circles are the parents of the focal offspring (the blue circle at the bottom). The plus symbol joining the lineal components of the parental microbiomes indicates taking an average. Arrows pointing toward offspring brackets represent microbiome transmission.

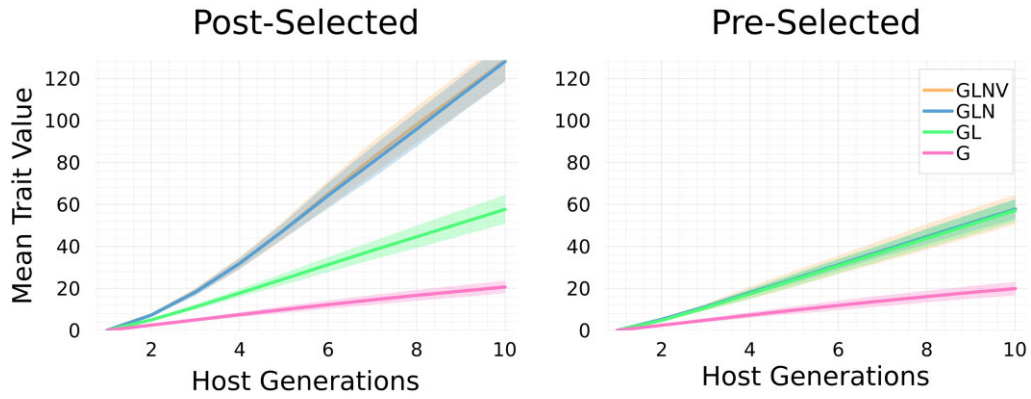


Figure 3. Time-series plots of average simulated host trait dynamics. Different colors correspond to different combinations of factors. Pink corresponds to only host genes mediating a response to selection (G). Green corresponds to only host genes and lineal microbes (GL). Blue corresponds to host genes, lineal, and non-lineal microbes (GLN). Mustard corresponds to host genes, lineal, non-lineal, and novel microbes (GLNV) (mostly hidden by blue lines/areas). For each combination of factors, simulations were repeated 20 times. Solid lines are averages across these repeated runs, and shaded regions correspond to standard deviations across repeated runs. The panels are divided by non-lineal microbes acquired from post-selected parents (left) and pre-selected parents (right).

The dichotomy of whether non-lineal microbes contribute to a selection response is artificial by design. More realistically, because different microbe individuals of the same taxon in a given host microbiome may have different patterns of ancestral associations with their hosts, microbial taxa cannot be neatly categorized by ancestral concordance as done here. Additionally, non-lineal microbe individuals may vary in the degree to which they contribute to a selection response because they may be transmitted either before or after the parental population experiences selection depending on the life histories of the hosts and microbes. Such within-taxon variation could be captured in future work by modeling individual microbes' ancestral concordance as drawn from a taxon-specific distribution. Lastly, it is also possible that microbes can be transmitted beyond consecutive generations, but under our simulation model we expect the response to selection to be further attenuated in such cases.

Heritabilities and transmissibilities

Results from the previous sections provide insights into formulating a definition of microbial transmissibility. In particular, the indices we introduce here assume only microbes acquired from sources that contribute to a response to host-level selection (such as the lineal and possibly non-lineal microbes, see the *Interactions between host-level selection and ancestral concordance* section) are included as trait-mediating factors while calculating components of additive trait variation. Then, just as the *narrow-sense genetic heritability*, defined as $h^2 = G_A/P$, is useful for predicting the response to selection and can be measured using parent–offspring correlations, we define the *narrow-sense microbial transmissibility* as $b^2 = M_A^{\psi}/P$, which is also useful for predicting the response to selection. However, unlike h^2, only the lineal component of b^2 will be measurable from parent–offspring correlations. If non-lineal microbes contribute to a response to selection, then their contribution to b^2 may be

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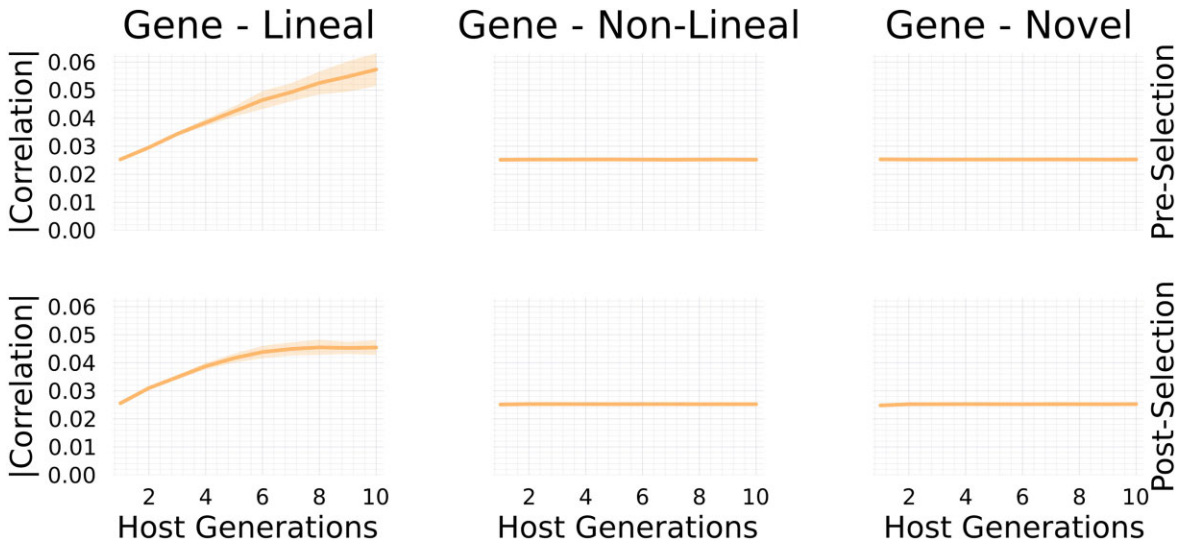


Figure 4. Time-series plots of absolute values of correlations between host allele counts and microbe abundances. Correlations between allele counts and abundances of lineal microbes (left plots) increase faster when non-lineal microbes are sourced from the pre-selected host parental population in comparison to when non-lineal microbes are sourced from the post-selected host parental population. In contrast, correlations with abundances of non-lineal microbes and novel microbes remain at neutral levels (right plots).

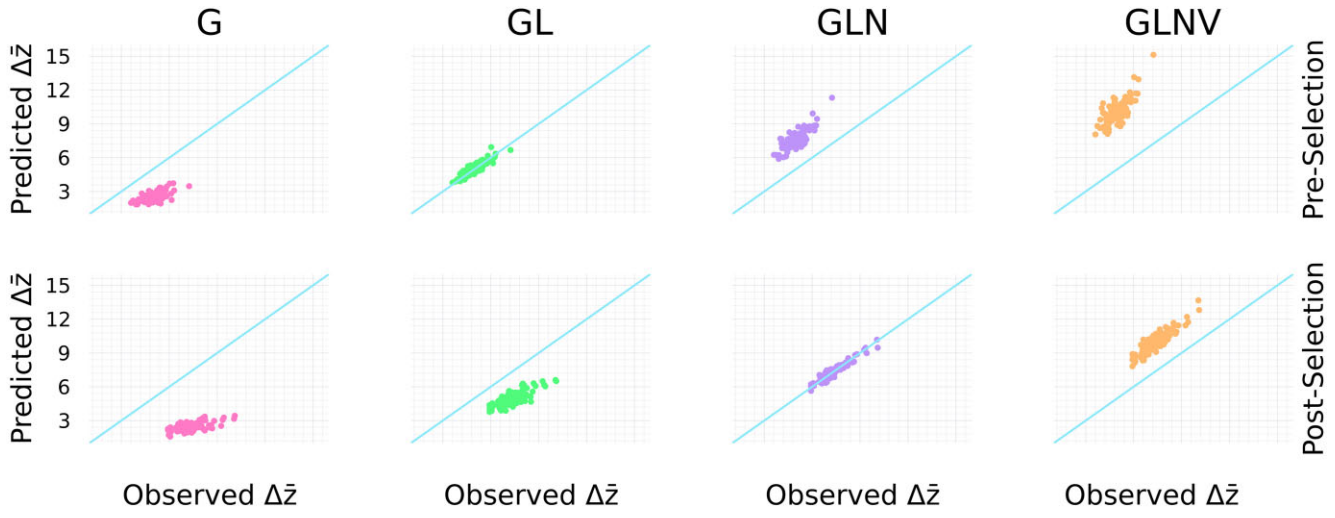


Figure 5. Comparison of change in host mean trait over a single consecutive generation ($\Delta\bar{z}$) observed from simulations (x-axis) to that predicted by including different components of the host microbiome (y-axis). Each dot corresponds to an average over repeated runs for a given set of randomly drawn additive effects. Colors and column labels follow the same pairing described in the caption for Figure 3. Diagonal lines have unit slope and zero intercept. This figure demonstrates that $\Delta\bar{z}$ is *overestimated* when *including* either novel microbes (GLNV in top and bottom rows) or non-lineal microbes acquired from pre-selected parents (GLN in the top row), is *underestimated* when *excluding* either all microbes (G in top and bottom rows) or non-lineal microbes acquired from post-selected parents (GL in the bottom row). Predictions match observations on average when including host genes and lineal microbes, but excluding non-lineal microbes acquired from pre-selected parents and novel microbes (GL in the top row), and when including only host genes, lineal microbes, and non-lineal microbes acquired from post-selected parents (GLN in the bottom row).

quantified by first identifying the non-lineal donor–recipient pairs between the parent and offspring populations, and then measure correlations between donors and recipients. Of course, this approach to estimating b^2 depends on our simplifying assumptions on the partitioning of host microbiome into lineal, non-lineal, and novel components. Because each taxon may consist of a mixture of lineal, non-lineal, and novel microbes, estimates of b^2 will likely require a more sophisticated correlational analysis.

On their own, the narrow-sense heritability and narrow-sense microbial transmissibility are only useful for predicting a response to selection when the host trait is either entirely genetically mediated or entirely microbially mediated. To obtain accurate predictions for the response to selection when host traits are partially genetically mediated and partially microbially mediated, we introduce notions of *total transmissibility*. These quantities are meant to capture all components of host trait variation explained by factors that facilitate a host response to selection (which we refer to as *selective factors* from hereon). Writing P_A^{ψ} as the component of host trait variation explained only by additive effects of selective factors, we define the *narrow-sense total transmis-*

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sibility as $t^2 = P_A^{\psi}/P$. By focusing on additive effects, we can use the same general approach outlined in the *Analysis of variance of microbiome-mediated traits* section to quantify P_A^{ψ}. Here, because we assume the only selective factors are host genes and microbes, the narrow-sense total transmissibility becomes

$$t^2 = \frac{G_A + M_A^{\psi} + C_A^{\psi}}{P}. \tag{11}$$

Hence, in general $t^2 \neq h^2 + b^2$. Instead, writing $k_{hb} = C_A^{\psi}/P$, we can heuristically think of expanding the square $(h+b)^2$ to arrive at $t^2 = h^2 + k_{hb} + b^2$, where k_{hb} takes the place of the undefined symbol $2hb$. Because k_{hb} is a covariance, it can be positive or negative. So ignoring k_{hb} while calculating t^2 can positively or negatively bias predictions for a microbiome-mediated response to selection. Our work then suggests the form of the breeders equation introduced by Lush (1937), $R = h^2 S$ (where $R = \Delta\bar{z}$ is the response to selection and S is the selection differential), naturally generalizes to $R = t^2 S = (h^2 + k_{hb} + b^2)S$. This reduces to the original expression in the absence of microbiome-mediated effects.

In our framework, broad-sense genetic heritability retains the same definition from classical quantitative genetics. In particular, the genotypic value z_g is the genotypic–microbic value z_{gm} averaged over host microbiomes. Then the *broad-sense genetic heritability* is defined as the proportion of host trait variance explained by genotypic variation: $H^2 = \mathrm{Var}(z_g)/P$. In analogy, the *microbic value* z_m is the genotypic–microbic value z_{gm} averaged over host genotypes. To clarify that we are focusing specifically on host trait variation explained by microbes that contribute to a selection response, we propagate the superscript $^{\psi}$. In particular, z_m^{ψ} is z_{gm} averaged over host genotypes and components of host microbiomes that do not contain selective factors. This suggests that the *broad-sense microbial transmissibility* should be defined as the proportion of host trait variation explained by selective microbic variation: $B^2 = \mathrm{Var}(z_m^{\psi})/P$. Setting $G_R = \mathrm{Var}(z_g) - G_A$ and $M_R^{\psi} = \mathrm{Var}(z_m^{\psi}) - M_A^{\psi}$, the residual variances left unexplained by additive genetic and selective additive microbial effects, respectively, the broad-sense heritabilities can then be expressed as

$$H^2 = \frac{G_A + G_R}{P}, \quad B^2 = \frac{M_A^{\psi} + M_R^{\psi}}{P}. \tag{12}$$

In general, the sum of these broad-sense heritabilities does not capture all of host trait variation explained by host genotype–microbiome pairs (g, m) because they do not account for covariances between allele counts and microbe abundances (quantified by C_A^{ψ}) or for residual variation left unexplained by genotypic values and microbic values (quantified by $P_R^{\psi} - G_R - M_R^{\psi}$, where $P_R^{\psi} = P - P_A^{\psi}$). More generally, if the host trait is mediated by the selective factors $f = (f_1, \ldots, f_N)$, and z_f is the average trait value among hosts carrying factors f, then we define the *broad-sense total transmissibility* as the proportion of phenotypic variance explained by selective factors: $T^2 = \mathrm{Var}(z_f)/P$. Because we assume the only selective factors are host genes and microbes, we have $z_f = z_{gm}$, the genotypic–microbic value. Then, defining the symbol $K_{HB} = (C_A^{\psi} + P_R^{\psi} - G_R - M_R^{\psi})/P$, we obtain an expression for the broad-sense transmissibility as

$$T^2 = H^2 + K_{HB} + B^2. \tag{13}$$

Again, we can intuitively think about calculating the expression for T^2 by expanding the square $(H+B)^2$, with K_{HB} taking the place of the undefined symbol $2HB$. Because $G_R + M_R^{\psi} \leq P_R^{\psi}$, the component of K_{HB} due to differences in residual variation will always be non-negative. However, because C_A^{ψ} may be negative, K_{HB} will be negative when $C_A^{\psi} < G_R + M_R^{\psi} - P_R^{\psi}$. As a result, excluding K_{HB} from calculating T^2 can positively or negatively bias estimates for the proportion of host trait variation explained by genetic and selective microbial factors.

The utility of the definition of narrow-sense total transmissibility introduced in this section for predicting a response to selection relies on a set of assumptions that simplify the process of microbial inheritance. Future work is needed to test their utility, given the complexity of within-host dynamics (Gerber, 2014), the microbiome assembly process (Costello et al., 2012), and host–host transmission (Sarkar et al., 2024).

Discussion

There is growing evidence that microbes can be integral to the form, function, and fitness of larger organisms (Bordenstein & Theis, 2015). Microbes associated with plant and animal hosts have been shown to contribute to fundamental organismal processes, including development (McFall-Ngai et al., 2013), nutrition (David et al., 2013), pathogen protection (Koch & Schmid-Hempel, 2011), and even behavior (Cryan & Dinan, 2012). Microbes have been suggested to provide their plant and animal hosts with the capacity for rapid evolution (Bisschop et al., 2022; Rosenberg & Zilber-Rosenberg, 2016) and perhaps even to contribute to the evolutionary rescue of larger organisms from rapid environmental change (Lennon et al., 2019; Pillai et al., 2016).

This increasing interest in host–microbe interactions has led to a crucial need for a deeper understanding of how host-associated microbes influence the phenotypes of their hosts, and ultimately their evolution. But developing this understanding has been hampered by the lack of a comprehensive theoretical framework for considering the combined influences of host genetics, host-associated microbes and the surrounding environment on traits that emerge from such host–microbe systems, and corresponding statistical frameworks to disentangle these influences. Quantitative genetics is uniquely positioned to provide this foundation because it is focused on understanding the inheritance and evolution of the complex traits that result from multiple interacting sources of variation. While quantitative genetics was initially developed with genetic factors as the sole heritable units, it has since been extended to include factors subject to non-genetic inheritance, such as epigenetic alleles and cultural memes (Cavalli-Sforza & Feldman, 1981; David & Ricard, 2019; Day & Bonduriansky, 2011). Here, we extend quantitative genetics by formally incorporating microbiomes as potentially heritable units that are similar to, but distinct from, host genetic factors.

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Importantly, microbes are not merely their host's *second genome*; there are fundamental differences between microbes in a microbiome and genes in a genome. These differences make applying quantitative genetic approaches directly to host microbiomes problematic. As a consequence, there is a need to expand theory of quantitative genetics to incorporate unique aspects of microbiome biology, as we have attempted to do here. We have built on previous efforts to establish quantitative genetic frameworks for microbiome-mediated traits (Henry et al., 2021) by generalizing the foundations of quantitative genetic theory. Our approach provides a formal definition for the component of phenotypic variation explained by additive effects of host genetic and microbial factors (P_A), and shows how to partition this additive variance into additive genetic variance (G_A) and the newly defined *additive microbial variance* (M_A, the component of host trait variance explained by additive effects of microbe factors) and *additive gene–microbe covariance* (C_A, which quantifies additive effects due to covariances between allele counts and microbe abundances across host individuals) such that $P_A = G_A + M_A + C_A$.

Furthermore, to make accurate predictions for microbiome-mediated responses to selection on hosts, we found it necessary to introduce additional partitioning of host trait variation explained by host microbiomes. In particular, we suggest host microbiomes may be decomposed into three components: the *lineal* component, which contains microbes passed along host lineages, the *novel* component, which contains microbes acquired from the environment that have no previous associations with the host species, and the *non-lineal* component, which contains all remaining microbes. We anticipate that whether or not microbes facilitate a response to selection depends on the microbiome component to which they belong. When microbial taxa that contribute to a selection response have been identified, we suggest quantifying M_A^{ψ} as the component of host trait variance explained *only* by the additive effects of those taxa. Using a simulation model, we find support for a generalized breeders equation taking the form $\Delta\bar{z} = (G_A + M_A^{\psi} + C_A^{\psi})\,\beta$. Building on this, we suggest definitions that generalize narrow-sense and broad-sense heritabilities to account for microbes that contribute to a selection response.

From this initial expansion, we can make a number of conclusions that are worth further theoretical and experimental exploration. For instance, Figures 3 and 4 together demonstrate that not all evolutionarily important microbes significantly correlate with host genes. This finding therefore negates a requirement of lineal transmission. In particular, Figure 3 shows that non-lineal microbes sourced from post-selected parents significantly contribute to a sustained response to selection over several host generations. In addition, Figure 4 shows that the within-host abundances of these same microbes do not significantly correlate with host allele counts (where significant here means to have greater magnitudes of correlations than novel microbe abundances). Another conclusion is in the converse direction: that not all microbes with abundances that significantly correlate with host allele counts contribute significantly to host trait variation. This result underscores the possibility that microbes transmitted across host generations do not necessarily mediate a response to host-level selection. For example, such correlations may occur when the host population is structured and exhibits random genetic drift. Furthermore, it is worth considering the degree to which processes analogous to classical random genetic drift emerge from microbiome-mediated host trait dynamics.

Implications for association studies

Both of the conclusions in the preceding paragraph—that not all evolutionary important microbes correlate with host genes and not all microbes that correlate with host genes contribute to host trait variation—have important implications for how we study host–microbe interactions and their impact on host phenotype. The approaches commonly used for identifying the genes or microbes underlying a particular phenotype (e.g., GWAS or MWAS) require an understanding of how statistical associations form among phenotypes, genes, and microbes (Camarinha-Silva et al., 2017; Difford et al., 2018; Pérez-Enciso et al., 2021; Qin et al., 2012). Because these associations are not necessarily causal, and because causal pathways may not be detected via statistical associations, there is a need for a more comprehensive theoretical foundation to guide approaches such as GWAS and MWAS. Importantly, microbiomes are dynamic ecological assemblages obeying different rules of inheritance than genomes, and so uncritical application of methods designed for GWAS may produce misleading results. The framework we introduce addresses this issue by clarifying when analogies between genes and microbes hold and when they break down, helping to replace false equivalences with a more accurate foundation for interpreting host–microbe trait associations. Furthermore, this framework offers a conceptual guide for interpreting MWAS results and for identifying microbial associations that are more likely to reflect biologically meaningful or evolutionarily relevant effects. Rather than assuming causality from variance explained, our framework points to ways of prioritizing microbial taxa based on patterns of shared ancestry with host lineages and association with host-level selection response, providing a clearer path for future empirical and statistical work.

In addition to extending the theoretical foundations of quantitative genetics, a key challenge moving forward will be to develop inference methods capable of estimating the host and microbial parameters defined in our framework. An interesting approach for future work will be to expand genomic regression approaches to include microbiomes for inferring parameters such as T^2 and b^2 (Xavier et al., 2019). As with classical quantitative genetics, inference will require statistical models that account for relatedness among hosts and microbial lineages, and that distinguish causal from merely associated components. Importantly, as shown in recent work, failure to account for differences between causal loci and observed markers in genomic regression can lead to misleading heritability estimates (de los Campos et al., 2015). Analogous pitfalls are likely to occur in microbiome-based studies, underscoring the need for explicit statistical theory to accompany any expanded quantitative genetic framework.

Implications for agriculture and wild populations

In livestock research, several studies have estimated the contribution of the gut microbiota to feed efficiency traits (Déru et al., 2022; Martinez-Boggio et al., 2024; Zhang et al., 2024) and in crops to stress tolerance and yield-

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related traits (Dwivedi et al., 2025; Mueller & Linksvayer, 2022), often by modeling microbial variation as an independent random effect alongside host genetic effects in linear mixed models. Unlike these empirical approaches, which frequently reduce dimensionality without biological interpretation (e.g., via principal components or operational taxonomic unit filtering thresholds), our framework provides biologically grounded decompositions based on patterns of host–microbe ancestral concordance. These decompositions may be estimated from lineage-traced or metagenomic data, and in doing so, provide a principled means of attributing trait variation to distinct classes of microbes. This allows our framework to complement and systematize empirical models used in breeding applications, while also offering a unified quantitative structure for comparing microbiome-mediated effects to other extended inheritance mechanisms. By explicitly modeling classes of microbial effects in parallel to classical host genetic effects, our approach can help disentangle microbiome contributions from confounded influences such as maternal effects, indirect genetic effects, and ecological inheritance, thereby clarifying their relative roles in shaping host phenotypes.

Beyond its relevance to agriculture, the framework we present here may also offer insights into how wild populations respond to natural selection. Long-term studies in wild systems have frequently documented cases where evolutionary change falls short of predictions based on observed heritability and directional selection (Kruuk et al., 2002; Pemberton, 2010; Pujol et al., 2018). Microbiome inheritance and its influence on host traits may contribute to this mismatch, but the mechanisms involved are likely complex and remain poorly understood. Different modes of microbial transmission and ecological dynamics could interact with host genetics in ways that violate standard quantitative genetic assumptions. Future work is needed to clarify how microbially mediated traits shape evolutionary predictions in natural populations. By formally integrating host-associated microbiomes into the foundations of quantitative genetic theory, our framework offers a potentially fruitful direction for addressing this challenge.

Limitations and future directions

The framework presented throughout this paper has both general components that are independent of model assumptions and specific components that are obtained by applying simplifying assumptions. In particular, both the approach to analysis of phenotypic variance in the *Analysis of variance of microbiome-mediated traits* section and our approach to partition host microbiomes by ancestral concordance in the *Classifying microbes by concordance with host ancestry* section are independent of model assumptions. However, our results on the response of a microbiome-mediated trait to selection (illustrated by Figure 5 and equations presented in the the*simulation results* section) were obtained following a series of simplifying assumptions. These include independence of microbiome components (see the *Partitioning host trait variance by ancestral concordance* section) and assumptions regarding the inheritance of microbiomes (summarized in Figure 2). In effect, our simulation model ignores within-host microbiome dynamics, the assembly process, and host–host microbe transmission, which has allowed us to focus on the effects of a novel inheritance mechanism on the dynamics of a microbiome-mediated quantitative character, and to identify the conditions under which distinct classes of microbes contribute to selection responses at the host level. Future work is needed to study models in which these assumptions are relaxed, and to apply our general framework of phenotypic variance analysis to both simulated and empirical data. More specifically, given the ecological nature of microbiomes, a fruitful avenue forward would be to interface our extension of quantitative genetics with metacommunity models of microbiome dynamics (Koskella et al., 2017; Leibold et al., 2004; Miller et al., 2018; Zeng et al., 2017). Such work can shed light on microbial ecological mechanisms that explain microbiome inheritance.

Conclusion

Microbes can be integral to the functioning of their animal and plant hosts, yet it is not well understood how microbiomes contribute to host fitness and evolution. This knowledge gap is due at least in part to the lack of a comprehensive theoretical framework for modeling how host–microbiome system-level traits emerge from the interactions of host genes, microbiome, and environment. We provide one such framework, by expanding theoretical quantitative genetics to include unique aspects of host-associated microbiomes, including multiple forms of Mendelian and non-Mendelian inheritance. This expansion leads to a formalization of several important concepts, including microbial heritabilities and transmissibilities, as well as fundamental quantities such as additive gene–microbe covariances. In addition, our framework provides an approach to partitioning quantitative trait variation into host-genetic and microbial components, allowing the theoretical exploration of how the joint contribution of host and microbiome to trait variation influences the evolution of host–microbiome systems. We consider our theoretical expansion as a first step toward a comprehensive incorporation of host-associated microbes into quantitative genetics.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

Code for reproducing Figures 1 and 3–5 has been archived on Zenodo, available at: https://zenodo.org/records/16793112.

Author contributions

Conceptualization: B.J.M.B., P.L.R., W.A.C., H.F.T., and B.W.; methodology and formal analysis: B.W. and P.L.R.; software, validation, investigation, data curation, and visualization: B.W.; resources: B.J.M.B.; writing – original draft: H.F.T., B.J.M.B., P.L.R., and W.A.C.; writing – review and editing: B.W., B.J.M.B., P.L.R., and W.A.C.; supervision and funding acquisition: B.J.M.B., P.L.R., and W.A.C.; project administration: B.J.M.B. and B.W.; and senior authorship: B.J.M.B. and W.A.C. are co-senior authors.

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Conflict of interest

The authors declare no conflict of interest.

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References

Ali, H. R. K., Hemeda, N. F., & Abdelaliem, Y. F. (2019). Symbiotic cellulolytic bacteria from the gut of the subterranean termite psammotermes hypostoma desneux and their role in cellulose digestion. *AMB Express*, *9*(1), 111. https://doi.org/10.1186/s13568-019-0830-5

Argaw-Denboba, A., Schmidt, T. S. B., Di Giacomo, M., Ranjan, B., Devendran, S., Mastrorilli, E., Lloyd, C. T., Pugliese, D., Paribeni, V., Dabin, J., Pisaniello, A., Espinola, S., Crevenna, A., Ghosh, S., Humphreys, N., Boruc, O., Sarkies, P., Zimmermann, M., Bork, P., & Hackett, J. A. (2024). Paternal microbiome perturbations impact offspring fitness. *Nature*, *629*, 652–659. https://doi.org/10.1038/s41586-024-07336-w

Arora, J., Kinjo, Y., Šobotník, J., Buček, A., Clitheroe, C., Stiblik, P., Roisin, Y., Žifčáková, L., Park, Y. C., Kim, K. Y., Sillam-Dussès, D., Hervé, V., Lo, N., Tokuda, G., Brune, A., & Bourguignon, T. (2022). The functional evolution of termite gut microbiota. *Microbiome*, *10*(1), 78. https://doi.org/10.1186/s40168-022-01258-3

Awany, D., & Chimusa, E. R. (2020). Heritability jointly explained by host genotype and microbiome: Will improve traits prediction? *Briefings in Bioinformatics*, *22*(3), bbaa175. https://doi.org/10.1093/bib/bbaa175

Barton, N., & Turelli, M. (2004). Effects of genetic drift on variance components under a general model of epistasis. *Evolution*, *58*, 2111–2132. https://doi.org/10.1111/j.0014-3820.2004.tb01591.x

Benson, A. K., Kelly, S. A., Legge, R., Ma, F., Low, S. J., Kim, J., Zhang, M., Oh, P. L., Nehrenberg, D., Hua, K., Kachman, S. D., Moriyama, E. N., Walter, J., Peterson, D. A., & Pomp, D. (2010). Individuality in gut microbiota composition is a complex polygenic trait shaped by multiple environmental and host genetic factors. *Proceedings of the National Academy of Sciences of the United States of America*, *107*, 18933–18938. https://doi.org/10.1073/pnas.1007028107

Bijma, P., Muir, W. M., & Van Arendonk, J. A. M. (2007). Multilevel selection 1: Quantitative genetics of inheritance and response to selection. *Genetics*, *175*, 277–288. https://doi.org/10.1534/genetics.106.062711

Bisschop, K., Kortenbosch, H. H., van Eldijk, T. J. B., Mallon, C. A., Salles, J. F., Bonte, D., & Etienne, R. S. (2022). Microbiome heritability and its role in adaptation of hosts to novel resources. *Frontiers in Microbiology*, *13*, 703183. https://doi.org/10.3389/fmicb.2022.703183

Bonduriansky, R., & Day, T. (2009). Nongenetic inheritance and its evolutionary implications. *Annual Review of Ecology, Evolution, and Systematics*, *40*, 103–125. https://doi.org/10.1146/annurev.ecolsys.39.110707.173441

Bordenstein, S. R., & Theis, K. R. (2015). Host biology in light of the microbiome: Ten principles of holobionts and hologenomes. *PLOS Biology*, *13*, e1002226. https://doi.org/10.1371/journal.pbio.1002226

Bruijning, M., Henry, L. P., Forsberg, S. K. G., Metcalf, C. J. E., & Ayroles, J. F. (2021). Natural selection for imprecise vertical transmission in host–microbiota systems. *Nature Ecology and Evolution*, *6*, 77–87. https://doi.org/10.1038/s41559-021-01593-y

Bürger, R. (2000). *The mathematical theory of selection, recombination, and mutation*. John Wiley and Sons.

Burns, A. R., Miller, E., Agarwal, M., Rolig, A. S., Milligan-Myhre, K., Seredick, S., Guillemin, K., & Bohannan, B. J. M. (2017). Interhost dispersal alters microbiome assembly and can overwhelm host innate immunity in an experimental zebrafish model. *Proceedings of the National Academy of Sciences of the United States of America*, *114*, 11181–11186. https://doi.org/10.1073/pnas.1702511114

Camarinha-Silva, A., Maushammer, M., Wellmann, R., Vital, M., Preuss, S., & Bennewitz, J. (2017). Host genome influence on gut microbial composition and microbial prediction of complex traits in pigs. *Genetics*, *206*, 1637–1644. https://doi.org/10.1534/genetics.117.200782

Cavalli-Sforza, L. L., & Feldman, M. W. (1981). *Cultural transmission and evolution: A quantitative approach, no. 16*. Princeton University Press.

Cheverud, J. M., & Routman, E. J. (1996). Epistasis as a source of increased additive genetic variance at population bottlenecks. *Evolution*, *50*, 1042–1051. https://doi.org/10.1111/j.1558-5646.1996.tb02345.x

Costello, E. K., Stagaman, K., Dethlefsen, L., Bohannan, B. J. M., & Relman, D. A. (2012). The application of ecological theory toward an understanding of the human microbiome. *Science*, *336*, 1255–1262. https://doi.org/10.1126/science.1224203

Cryan, J. F., & Dinan, T. G. (2012). Mind-altering microorganisms: The impact of the gut microbiota on brain and behaviour. *Nature Reviews Neuroscience*, *13*, 701–712. https://doi.org/10.1038/nrn3346

Danchin, E., Charmantier, A., Champagne, F. A., Mesoudi, A., Pujol, B., & Blanchet, S. (2011). Beyond DNA: Integrating inclusive inheritance into an extended theory of evolution. *Nature Reviews Genetics*, *12*, 475–486. https://doi.org/10.1038/nrg3028

David, I., & Ricard, A. (2019). A unified model for inclusive inheritance in livestock species. *Genetics*, *212*, 1075–1099. https://doi.org/10.1534/genetics.119.302375

David, L. A., Maurice, C. F., Carmody, R. N., Gootenberg, D. B., Button, J. E., Wolfe, B. E., Ling, A. V., Devlin, A. S., Varma, Y., Fischbach, M. A., Biddinger, S. B., Dutton, R. J., & Turnbaugh, P. J. (2013). Diet rapidly and reproducibly alters the human gut microbiome. *Nature*, *505*, 559–563. https://doi.org/10.1038/nature12820

Day, T., & Bonduriansky, R. (2011). A unified approach to the evolutionary consequences of genetic and nongenetic inheritance. *The American Naturalist*, *178*, E18–E36. https://doi.org/10.1086/660911

de los Campos, G., Sorensen, D., & Gianola, D. (2015). Genomic heritability: What is it? *PLOS Genetics*, *11*, e1005048. https://doi.org/10.1371/journal.pgen.1005048

Déru, V., Tiezzi, F., Carillier-Jacquin, C., Blanchet, B., Cauquil, L., Zemb, O., Bouquet, A., Maltecca, C., & Gilbert, H. (2022). Gut microbiota and host genetics contribute to the phenotypic variation of digestive and feed efficiency traits in growing pigs fed a conventional and a high fiber diet. *Genetics Selection Evolution*, *54*(1), 55. https://doi.org/10.1186/s12711-022-00742-6

Difford, G. F., Plichta, D. R., Løvendahl, P., Lassen, J., Noel, S. J., Højberg, O., Wright, A. D. G., Zhu, Z., Kristensen, L., Nielsen, H. B. r., Guldbrandtsen, B., & Sahana, G. (2018). Host genetics and the rumen microbiome jointly associate with methane emissions in

Downloaded from https://academic.oup.com/evolut/article/79/11/2487/8244151 by UB Kiel user on 13 December 2025

dairy cows. *PLOS Genetics*, *14*, e1007580. https://doi.org/10.1371/journal.pgen.1007580

Dwivedi, S. L., Vetukuri, R. R., Kelbessa, B. G., Gepts, P., Heslop-Harrison, P., Araujo, A. S., Sharma, S., & Ortiz, R. (2025). Exploitation of rhizosphere microbiome biodiversity in plant breeding. *Trends in Plant Science*, *30*(9), 1033–1045. https://doi.org/10.1016/j.tplants.2025.04.004

Fisher, R. A. (1918). XV.—the correlation between relatives on the supposition of mendelian inheritance. *Transactions of the Royal Society of Edinburgh*, *52*, 399–433. https://doi.org/10.1017/s0080456800012163

Fisher, R. A. (1941). Average excess and average effect of a gene substitution. *Annals of Eugenics*, *11*, 53–63. https://doi.org/10.1111/j.1469-1809.1941.tb02272.x

Gerber, G. K. (2014). The dynamic microbiome. *FEBS Letters*, *588*, 4131–4139. https://doi.org/10.1016/j.febslet.2014.02.037

Goodrich, J. K., Davenport, E. R., Waters, J. L., Clark, A. G., & Ley, R. E. (2016). Cross-species comparisons of host genetic associations with the microbiome. *Science*, *352*, 532–535. https://doi.org/10.1126/science.aad9379

Gould, A. L., Zhang, V., Lamberti, L., Jones, E. W., Obadia, B., Korasidis, N., Gavryushkin, A., Carlson, J. M., Beerenwinkel, N., & Ludington, W. B. (2018). Microbiome interactions shape host fitness. *Proceedings of the National Academy of Sciences of the United States of America*, *115*(51), E11951–E11960. https://doi.org/10.1073/pnas.1809349115

Helanterä, H., & Uller, T. (2010). The price equation and extended inheritance. *Philosophy and Theory in Biology*, *2*. https://doi.org/10.3998/ptb.6959004.0002.001

Henry, L. P., Bruijning, M., Forsberg, S. K. G., & Ayroles, J. F. (2021). The microbiome extends host evolutionary potential. *Nature Communications*, *12*. 5141https://doi.org/10.1038/s41467-021-25315-x

Houwenhuyse, S., Stoks, R., Mukherjee, S., & Decaestecker, E. (2021). Locally adapted gut microbiomes mediate host stress tolerance. *The ISME Journal*, *15*, 2401–2414. https://doi.org/10.1038/s41396-021-00940-y

Kirkpatrick, M., & Lande, R. (1989). The evolution of maternal characters. *Evolution*, *43*, 485–503. https://doi.org/10.1111/j.1558-5646.1989.tb04247.x

Knights, D., Silverberg, M. S., Weersma, R. K., Gevers, D., Dijkstra, G., Huang, H., Tyler, A. D., van Sommeren, S., Imhann, F., Stempak, J. M., Huang, H., Vangay, P., Al-Ghalith, G. A., Russell, C., Sauk, J., Knight, J., Daly, M. J., Huttenhower, C., & Xavier, R. J. (2014). Complex host genetics influence the microbiome in inflammatory bowel disease. *Genome Medicine*, *6*, 107. https://doi.org/10.1186/s13073-014-0107-1

Koch, H., & Schmid-Hempel, P. (2011). Socially transmitted gut microbiota protect bumble bees against an intestinal parasite. *Proceedings of the National Academy of Sciences of the United States of America*, *108*, 19288–19292. https://doi.org/10.1073/pnas.1110474108

Koskella, B., Hall, L. J., & Metcalf, C. J. E. (2017). The microbiome beyond the horizon of ecological and evolutionary theory. *Nature Ecology and Evolution*, *1*, 1606–1615. https://doi.org/10.1038/s41559-017-0340-2

Kruuk, L. E. B., Slate, J., Pemberton, J. M., Brotherstone, S., Guinness, F., & Clutton-Brock, T. (2002). Antler size in red deer: Heritability and selection but no evolution. *Evolution*, *56*, 1683–1695. https://doi.org/10.1111/j.0014-3820.2002.tb01480.x

Lande, R. (1976). Natural selection and random genetic drift in phenotypic evolution. *Evolution*, *30*, 314–334.

Leibold, M. A., Holyoak, M., Mouquet, N., Amarasekare, P., Chase, J. M., Hoopes, M. F., Holt, R. D., Shurin, J. B., Law, R., Tilman, D., Loreau, M., & Gonzalez, A. (2004). The metacommunity concept: A framework for multi-scale community ecology. *Ecology Letters*, *7*, 601–613. https://doi.org/10.1111/j.1461-0248.2004.00608.x

Lennon, J. T., Mueller, E. A., Wisnoski, N. I., & Peralta, A. L. (2019). Microbial rescue effects: How microbiomes can save hosts from extinction. https://doi.org/10.31219/osf.io/snr49

Lush, J. (1937). *Animal breeding plans*. Collegiate Press, Incorporated.

Lynch, M., & Walsh, B. (1998). *Genetics and analysis of quantitative traits*(Vol. *1*). Sinauer.

Mackay, T. F. C., Stone, E. A., & Ayroles, J. F. (2009). The genetics of quantitative traits: Challenges and prospects. *Nature Reviews Genetics*, *10*, 565–577. https://doi.org/10.1038/nrg2612

Martinez-Boggio, G., Monteiro, H. F., Lima, F. S., Figueiredo, C. C., Bisinotto, R. S., Santos, J. E. P., Mion, B., Schenkel, F. S., Ribeiro, E. S., Weigel, K. A., Rosa, G. J. M., & Peñagaricano, F. (2024). Revealing host genome–microbiome networks underlying feed efficiency in dairy cows. *Scientific Reports*, *14*. 26060. https://doi.org/10.1038/s41598-024-77782-z

Martiny, J. B. H., Jones, S. E., Lennon, J. T., & Martiny, A. C. (2015). Microbiomes in light of traits: A phylogenetic perspective. *Science*, *350*, aac9323. https://doi.org/10.1126/science.aac9323

Maurice, N., & Erdei, L. (2018). *Termite gut microbiome*(pp. 69–99). Springer International Publishing.

McFall-Ngai, M. (2007). Care for the community. *Nature*, *445*, 153–153. https://doi.org/10.1038/445153a

McFall-Ngai, M., Hadfield, M. G., Bosch, T. C. G., Carey, H. V., Domazet-Lošo, T., Douglas, A. E., Dubilier, N., Eberl, G., Fukami, T., Gilbert, S. F., Hentschel, U., King, N., Kjelleberg, S., Knoll, A. H., Kremer, N., Mazmanian, S. K., Metcalf, J. L., Nealson, K., Pierce, N. E., & Wernegreen, J. J. (2013). Animals in a bacterial world, a new imperative for the life sciences. *Proceedings of the National Academy of Sciences of the United States of America*, *110*, 3229–3236. https://doi.org/10.1073/pnas.1218525110

Miller, E. T., Svanbäck, R., & Bohannan, B. J. (2018). Microbiomes as metacommunities: Understanding host-associated microbes through metacommunity ecology. *Trends in Ecology and Evolution*, *33*, 926–935. https://doi.org/10.1016/j.tree.2018.09.002

Moore, A. J., Brodie, E. D., & Wolf, J. B. (1997). Interacting phenotypes and the evolutionary process: I. Direct and indirect genetic effects of social interactions. *Evolution*, *51*, 1352–1362. https://doi.org/10.1111/j.1558-5646.1997.tb01458.x

Morris, A. H., & Bohannan, B. J. M. (2024). Estimates of microbiome heritability across hosts. *Nature Microbiology*, *9*, 3110–3119. https://doi.org/10.1038/s41564-024-01865-w

Mueller, U. G., & Linksvayer, T. A. (2022). Microbiome breeding: Conceptual and practical issues. *Trends in Microbiology*, *30*, 997–1011. https://doi.org/10.1016/j.tim.2022.04.003

O'Brien, A. M., Laurich, J. R., & Frederickson, M. E. (2023). Evolutionary consequences of microbiomes for hosts: Impacts on host fitness, traits, and heritability. *Evolution*, *78*, 237–252. https://doi.org/10.1093/evolut/qpad183

Odling-Smee, J., Erwin, D. H., Palkovacs, E. P., Feldman, M. W., & Laland, K. N. (2013). Niche construction theory: A practical guide for ecologists. *The Quarterly Review of Biology*, *88*, 3–28. https://doi.org/10.1086/669266

Opstal, E. J. v., & Bordenstein, S. R. (2015). Rethinking heritability of the microbiome. *Science*, *349*, 1172–1173. https://doi.org/10.1126/science.aab3958

Org, E., Parks, B. W., Joo, J. W. J., Emert, B., Schwartzman, W., Kang, E. Y., Mehrabian, M., Pan, C., Knight, R., Gunsalus, R., Drake, T. A., Eskin, E., & Lusis, A. J. (2015). Genetic and environmental control of host-gut microbiota interactions. *Genome Research*, *25*, 1558–1569. https://doi.org/10.1101/gr.194118.115

Pemberton, J. M. (2010). Evolution of quantitative traits in the wild: Mind the ecology. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *365*, 2431–2438. https://doi.org/10.1098/rstb.2010.0108

Pérez-Enciso, M., Zingaretti, L. M., Ramayo-Caldas, Y., & de los Campos, G. (2021). Opportunities and limits of combining microbiome and genome data for complex trait prediction.

Downloaded from https://academic.oup.com/evolut/article/79/11/2487/8244151 by UB Kiel user on 13 December 2025

Genetics Selection Evolution, *53*, 65. https://doi.org/10.1186/s12711-021-00658-7

Petersen, C., Hamerich, I. K., Adair, K. L., Griem-Krey, H., Torres Oliva, M., Hoeppner, M. P., Bohannan, B. J. M., & Schulenburg, H. (2023). Host and microbiome jointly contribute to environmental adaptation. *The ISME Journal*, *17*, 1953–1965. https://doi.org/10.1038/s41396-023-01507-9

Pillai, P., Gouhier, T. C., & Vollmer, S. V. (2016). Ecological rescue of host-microbial systems under environmental change. *Theoretical Ecology*, *10*, 51–63. https://doi.org/10.1007/s12080-016-0310-3

Provine, W. B. (2001). *The origins of theoretical population genetics: With a new afterword*. University of Chicago Press.

Pujol, B., Blanchet, S., Charmantier, A., Danchin, E., Facon, B., Marrot, P., Roux, F., Scotti, I., Teplitsky, C., Thomson, C. E., & Winney, I. (2018). The missing response to selection in the wild. *Trends in Ecology and Evolution*, *33*, 337–346. https://doi.org/10.1016/j.tree.2018.02.007

Qin, J., Li, Y., Cai, Z., Li, S., Zhu, J., Zhang, F., Liang, S., Zhang, W., Guan, Y., Shen, D., Peng, Y., Zhang, D., Jie, Z., Wu, W., Qin, Y., Xue, W., Li, J., Han, L., Lu, D., & Wang, J. (2012). A metagenome-wide association study of gut microbiota in type 2 diabetes. *Nature*, *490*, 55–60. https://doi.org/10.1038/nature11450

Rice, S. (2004). *Evolutionary theory: Mathematical and conceptual foundations*. Sinauer.

Rosenberg, E., & Zilber-Rosenberg, I. (2016). Microbes drive evolution of animals and plants: The hologenome concept. *mBio*, *7*, 10–1128. https://doi.org/10.1128/mbio.01395-15

Rosenberg, E., & Zilber-Rosenberg, I. (2018). The hologenome concept of evolution after 10 years. *Microbiome*, *6*, 78.https://doi.org/10.1186/s40168-018-0457-9

Rosshart, S. P., Vassallo, B. G., Angeletti, D., Hutchinson, D. S., Morgan, A. P., Takeda, K., Hickman, H. D., McCulloch, J. A., Badger, J. H., Ajami, N. J., Trinchieri, G., Pardo-Manuel de Villena, F., Yewdell, J. W., & Rehermann, B. (2017). Wild mouse gut microbiota promotes host fitness and improves disease resistance. *Cell*, *171*, 1015–1028.e13. https://doi.org/10.1016/j.cell.2017.09.016

Rothschild, D., Weissbrod, O., Barkan, E., Kurilshikov, A., Korem, T., Zeevi, D., Costea, P. I., Godneva, A., Kalka, I. N., Bar, N., Shilo, S., Lador, D., Vila, A. V., Zmora, N., Pevsner-Fischer, M., Israeli, D., Kosower, N., Malka, G., Wolf, B. C., & Segal, E. (2018). Environment dominates over host genetics in shaping human gut microbiota. *Nature*, *555*, 210–215. https://doi.org/10.1038/nature25973

Roughgarden, J. (2023). Holobiont evolution: Population theory for the hologenome. *The American Naturalist*, *201*, 763–778. https://doi.org/10.1086/723782

Roughgarden, J., Gilbert, S. F., Rosenberg, E., Zilber-Rosenberg, I., & Lloyd, E. A. (2017). Holobionts as units of selection and a model of their population dynamics and evolution. *Biological Theory*, *13*, 44–65. https://doi.org/10.1007/s13752-017-0287-1

Russell, S. L. (2019). Transmission mode is associated with environment type and taxa across bacteria-eukaryote symbioses: A systematic review and meta-analysis. *FEMS Microbiology Letters*, *366*, fnz013. https://doi.org/10.1093/femsle/fnz013

Sandoval-Motta, S., Aldana, M., & Frank, A. (2017a). Evolving ecosystems: Inheritance and selection in the light of the microbiome. *Archives of Medical Research*, *48*, 780–789. https://doi.org/10.1016/j.arcmed.2018.01.002

Sandoval-Motta, S., Aldana, M., Martínez-Romero, E., & Frank, A. (2017b). The human microbiome and the missing heritability problem. *Frontiers in Genetics*, *8*, 80. https://doi.org/10.3389/fgene.2017.00080

Sarkar, A., McInroy, C. J., Harty, S., Raulo, A., Ibata, N. G., Valles-Colomer, M., Johnson, K. V. A., Brito, I. L., Henrich, J., Archie, E. A., Barreiro, L. B., Gazzaniga, F. S., Finlay, B. B., Koonin, E. V., Carmody, R. N., & Moeller, A. H. (2024). Microbial transmission in the social microbiome and host health and disease. *Cell*, *187*, 17–43. https://doi.org/10.1016/j.cell.2023.12.014

Spor, A., Koren, O., & Ley, R. (2011). Unravelling the effects of the environment and host genotype on the gut microbiome. *Nature Reviews Microbiology*, *9*, 279–290. https://doi.org/10.1038/nrmicro2540

Strang, G. (2019). *Linear algebra and learning from data*(Vol. *4)*. Wellesley-Cambridge Press.

Tavalire, H. F., Christie, D. M., Leve, L. D., Ting, N., Cresko, W. A., & Bohannan, B. J. M. (2021). Shared environment and genetics shape the gut microbiome after infant adoption. *mBio*, *12*, 10–1128. https://doi.org/10.1128/mbio.00548-21

Theis, K. R. (2018). Hologenomics: Systems-level host biology. *mSystems*, *3*, 10–1128. https://doi.org/10.1128/msystems.00164-17

Tierney, B. T., Yang, Z., Luber, J. M., Beaudin, M., Wibowo, M. C., Baek, C., Mehlenbacher, E., Patel, C. J., & Kostic, A. D. (2019). The landscape of genetic content in the gut and oral human microbiome. *Cell Host and Microbe*, *26*, 283–295. https://doi.org/10.1016/j.chom.2019.07.008

Uller, T., & Helanterä, H. (2013). Non-genetic inheritance in evolutionary theory: A primer. *Non-Genetic Inheritance*, *1*, 27–32. https://doi.org/10.2478/ngi-2013-0003

van Vliet, S., & Doebeli, M. (2019). The role of multilevel selection in host microbiome evolution. *Proceedings of the National Academy of Sciences of the United States of America*, *116*, 20591–20597. https://doi.org/10.1073/pnas.1909790116

Visscher, P. M., Hill, W. G., & Wray, N. R. (2008). Heritability in the genomics era—concepts and misconceptions. *Nature Reviews Genetics*, *9*, 255–266. https://doi.org/10.1038/nrg2322

Walsh, B., & Lynch, M. (2018). *Evolution and selection of quantitative traits*. Oxford University Press.

Wang, J., Chen, L., Zhao, N., Xu, X., Xu, Y., & Zhu, B. (2018). Of genes and microbes: Solving the intricacies in host genomes. *Protein and Cell*, *9*, 446–461. https://doi.org/10.1007/s13238-018-0532-9

Week, B., Russel, S., Schulenburg, H., Bohannan, B., & Bruijning, M. (2025). Applying evolutionary theory to understand host–microbiome evolution. *Nature Ecology & Evolution*, *Sep 8*, 1–12. https://doi.org/10.1038/s41559-025-02846-w

Wendling, C. C., & Wegner, K. M. (2015). Adaptation to enemy shifts: Rapid resistance evolution to local vibrio spp. in invasive pacific oysters. *Proceedings of the Royal Society B: Biological Sciences*, *282*, 20142244. https://doi.org/10.1098/rspb.2014.2244

Wilson, A. J., Réale, D., Clements, M. N., Morrissey, M. M., Postma, E., Walling, C. A., Kruuk, L. E. B., & Nussey, D. H. (2009). An ecologist's guide to the animal model. *Journal of Animal Ecology*, *79*, 13–26. https://doi.org/10.1111/j.1365-2656.2009.01639.x

Xavier, A., Muir, W. M., & Rainey, K. M. (2019). bwgr: Bayesian whole-genome regression. *Bioinformatics*, *36*, 1957–1959. https://doi.org/10.1093/bioinformatics/btz794

Zeng, Q., Wu, S., Sukumaran, J., & Rodrigo, A. (2017). Models of microbiome evolution incorporating host and microbial selection. *Microbiome*, *5*. 127https://doi.org/10.1186/s40168-017-0343-x

Zhang, W., Lan, F., Zhou, Q., Gu, S., Li, X., Wen, C., Yang, N., & Sun, C. (2024). Host genetics and gut microbiota synergistically regulate feed utilization in egg-type chickens. *Journal of Animal Science and Biotechnology*, *15*, 123. https://doi.org/10.1186/s40104-024-01076-7

Zilber-Rosenberg, I., & Rosenberg, E. (2008). Role of microorganisms in the evolution of animals and plants: The hologenome theory of evolution. *FEMS Microbiology Reviews*, *32*, 723–735. https://doi.org/10.1111/j.1574-6976.2008.00123.x

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Host-Parasite Coevolution in Continuous Space Leads to Variation in Local Adaptation across Spatial Scales

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ABSTRACT: Previous host-parasite coevolutionary theory has focused on understanding the determinants of local adaptation using spatially discrete models. However, these studies fall short of describing patterns of host-parasite local adaptation across spatial scales. In contrast, empirical work demonstrates that patterns of adaptation depend on the scale at which they are measured. Here, we propose a mathematical model of host-parasite coevolution in continuous space that naturally leads to a scale-dependent definition of local adaptation. In agreement with empirical research, we find that patterns of adaptation vary across spatial scales. In some cases, not only the magnitude of local adaptation but also the identity of the locally adapted species will depend on the spatial scale at which measurements are taken. Building on our results, we suggest a way to consistently measure parasite local adaptation when continuous space is the driver of cross-scale variation. We also describe a way to test whether continuous space is driving cross-scale variation. Taken together, our results provide a new perspective that can be used to understand empirical observations previously unexplained by theoretical expectations and deepens our understanding of the mechanics of host-parasite local adaptation.

Keywords: host-parasite coevolution, local adaptation, continuous space, spatial scale.

Introduction

Interactions between hosts and parasites have shaped patterns of diversity across all scales of biological organization. For example, coevolution with parasites can alter epidemiological dynamics (Best et al. 2010; Débarre et al. 2012; Lion and Gandon 2015), promote the evolution of sexual reproduction (Otto and Nuismer 2004; Lively 2010), yield novel mutualisms (Yamamura 1993), and influence patterns of speciation across a broad range of taxa (Agrawal and Zhang 2021). In each of these examples, the geography of the interspecific interactions plays a critical role in determining ecological and evolutionary outcomes. Previous theoretical studies of host-parasite coevolution have incorporated geography using models in which dispersal occurs between spatially discrete locations (Gandon et al. 1996; Boots and Sasaki 1999; Nuismer et al. 2000; Gandon 2002; Gandon and Michalakis 2002; Nuismer 2006; Ridenhour and Nuismer 2007; Gandon and Nuismer 2009; Débarre et al. 2012; Lion and Gandon 2015). However, because most species disperse in continuous space (Nathan et al. 2008) and because the consequences of spatially continuous systems may only be poorly approximated by discrete space models, there is a need for models that help us understand the drivers of host-parasite local adaptation in spatially continuous systems.

A theoretical model of coevolution in continuous space may also explain empirically observed variation in spatial patterns of host-parasite adaptation across spatial scales. Researchers often document a spatial pattern in which greater support for parasite local adaptation is found at larger spatial scales and little to no support is found at shorter spatial scales (Hanks and Denno 1994; Kaltz et al. 1999; Thrall et al. 2002; Schönrogge et al. 2006; Tack et al. 2014). We lack a mechanistic explanation for why this pattern should emerge because previous theory has not explicitly related parasite local adaptation to the geographic distances that measurements are taken at. Furthermore, these studies rely on indices of local adaptation that do not explicitly account for geographic distance. Therefore, there is a need to further integrate theoretical predictions and empirical observations of spatial patterns of coevolution. In this article, we aim to close this gap by analyzing the interaction among host-parasite coevolution, random genetic drift, and gene flow in a continuous two-dimensional habitat by introducing a novel quantitative genetic model and an index of local adaptation that explicitly accounts for the geographic distance measurements are taken at.

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We begin by introducing our model, which is a two-species generalization of Slatkin's (1978) model of quantitative trait evolution in continuous space. We then introduce our novel measure of local adaptation that explicitly accounts for geographic distance. Analysis of our model yields an interspecific spatial covariance function, called the cross-covariance function, that allows us to calculate the covariance of traits for host and parasite populations as a function of the geographic distance separating those populations. In particular, the cross-covariance function accounts for spatial autocorrelation of trait values in each species, which occurs when traits measured at different locations tend to look more similar the closer they are geographically. Combining our model of host-parasite coevolution with our continuous space index of local adaptation, we find that parasite local adaptation can be expressed in terms of the cross-covariance function. Our main finding is that parasite local adaptation depends on the spatial scale measurements are taken at relative to the spatial scales of phenotypic autocorrelation for both species. In some cases, not only the magnitude of local adaptation but also the identity of the locally adapted species can vary across spatial scales because of spatially autocorrelated traits. Building on our theoretical results, we make a suggestion for obtaining a consistent measurement of parasite local adaptation when cross-scale variation is due to spatial autocorrelation. In the discussion section, we also suggest an approach to test whether cross-scale variation is due to spatial autocorrelation and describe two directions for generalizing our model.

Methods

The Model

Our model tracks the evolution of local mean traits for a pair of species codistributed across a continuous two-dimensional geographic landscape. For each species, the genomic architecture of their traits is based on an infinitesimal approximation such that the trait of an individual can be thought of as the sum of an infinite number of allelic effects (with no epistasis or dominance), each of infinitesimal size (reviewed in Barton et al. 2017). The primary components of our model (selection, reproduction, and dispersal) can be thought of as different stages in the life cycle of an individual. We assume the life cycle begins by determining fitness in response to selective forces, including interspecific interactions, followed by the production of offspring that disperse to new locations, inherit trait values that are normally distributed around parental trait values, and repeat the cycle of life. However, instead of explicitly tracking individuals, our model focuses on the dynamics of mean traits averaged across individuals at each location in a two-dimensional geographic landscape.

In this section, we begin by outlining our approach to account for biotic and abiotic selection. We then discuss our model of dispersal and random genetic drift before combining these components into our working model. Model parameters are summarized in table 1. A more thorough and technically detailed introduction to and justification for our model is presented in sections S1 and S2 of the supplemental PDF, respectively.

Selection. We assume that fitness consequences for interactions between hosts and parasites are mediated by the difference in quantitative traits $z_H - z_P$, where z_S is the trait value of an individual in species $S = H, P$ for host and parasite, respectively. More precisely, we assume that the probability of infection increases the more similar traits are and decreases with trait mismatch. Given a successful infection, we assume that the host incurs a fitness cost and the parasite receives a fitness benefit. As a consequence, the host trait will evolve to evade the parasite trait in response to biotic selection, while the parasite trait evolves to match the host trait.

To draw biologically meaningful conclusions, we analyze our model at equilibrium. Then, to ensure our model has an equilibrium, we also account for abiotic selection that is stabilizing with strengths A_H, A_P. As our equilibrium analysis of the model is agnostic to globally averaged trait values, without loss of generality we assume that stabilizing selection pulls both traits toward zero. Additionally, our approach to computing spatial covariance functions (outlined in sec. S3 of the supplemental PDF) requires that biotic selection be weak relative to abiotic stabilizing selection for both species ($B_S \ll A_S$, $S = H, P$).

Following classical quantitative genetics, we assume that trait distributions of local populations are normal with mean $\bar{z}_S(\boldsymbol{x})$ for species S at the two-coordinate location $\boldsymbol{x} = (x_1, x_2)$. Although this assumption may be violated as dispersal can lead to skewed trait distributions (Débarre et al. 2015), this effect does not occur when dispersal distance is short relative to the spatial scale of trait variation. As will be shown in the results section, we find that dispersal distances are always short relative to spatial scales of trait variation when selection is weak.

Table 1: Model parameters

Symbol	Description
A_H	Host abiotic stabilizing selection
A_P	Parasite abiotic stabilizing selection
B_H	Host biotic selection
B_P	Parasite biotic selection
σ_H	Host dispersal distance
σ_P	Parasite dispersal distance
D_H	Rate of host (phenotypic) drift
D_P	Rate of parasite (phenotypic) drift

Drift. Random genetic drift can lead to variation in evolutionary trajectories across spatial locations. In our model, drift provides the ultimate source of geographic variation in phenotypes; this geographic variation then interacts with gene flow and selection to yield distinct spatial patterns. The classic model for the response of a quantitative character to random genetic drift is given by Lande (1976). This model states that the change in mean trait in response to drift between consecutive, nonoverlapping generations follows a normal distribution with variance equal to the ratio of additive genetic variance to effective population size. The continuous time analog of this model is trait evolution following Brownian motion, which has been widely applied as a phenomenological model in the field of phylogenetic comparative methods (Felsenstein 1973; Manceau et al. 2016). Mechanistically, this result has been formalized in continuous time by Week et al. (2021). In our continuous space setting, we extend this model so that different locations experience uncorrelated effects of drift (see sec. S1 of the supplemental PDF for mathematical details), with rate D_S for species S. We denote the spatial noise process capturing drift for species S by $\xi_S(\boldsymbol{x})$, which is normalized to have unit rate. That is, $D_S\xi_S(\boldsymbol{x})$ is the instantaneous evolutionary response to drift for species S at location $\boldsymbol{x}$.

Dispersal. We assume that displacement between parental and offspring birthplaces follows a bivariate Gaussian distribution centered on zero with displacement in each direction drawn independently. These assumptions prevent any net directionality in dispersal. For species S, we assume that displacements in the two directions are drawn with a standard deviation σ_S, which we refer to as the dispersal distance.

Gaussian dispersal leads to change in $\bar{z}_S(\boldsymbol{x})$ toward a local average, where the scale of "local" is determined by σ_S. The local mean trait will increase or decrease depending on the concavity of the spatial mean trait surface. As the concavity of a surface is quantified by a second spatial derivative, the effect of Gaussian dispersal on the instantaneous rate of change in local mean trait value is related to the second spatial derivative.

Selection, Dispersal, and Drift. We combine the evolutionary forces described in the previous sections to obtain our working model as

$$\frac{\partial \bar{z}_H}{\partial t} = \overbrace{-A_H\bar{z}_H}^{\text{abiotic selection}} - \overbrace{B_H(\bar{z}_P - \bar{z}_H)}^{\text{biotic selection}} + \overbrace{\frac{\sigma_H^2}{2}\left(\frac{\partial^2\bar{z}_H}{\partial x_1^2} + \frac{\partial^2\bar{z}_H}{\partial x_2^2}\right)}^{\text{dispersal}} + \overbrace{D_H\xi_H}^{\text{drift}}, \tag{1a}$$

$$\frac{\partial \bar{z}_P}{\partial t} = -A_P\bar{z}_P + B_P(\bar{z}_H - \bar{z}_P) + \frac{\sigma_P^2}{2}\left(\frac{\partial^2\bar{z}_P}{\partial x_1^2} + \frac{\partial^2\bar{z}_P}{\partial x_2^2}\right) + D_P\xi_P, \tag{1b}$$

where A_S is the strength of abiotic stabilizing selection, B_S is the strength of biotic selection, σ_S is the dispersal distance, and D_S is the rate of random genetic drift for species S.

A full technical description of our model can be found in section S1 of the supplemental PDF. The system of equations (1) forms a pair of linear stochastic partial differential equations (Khoshnevisan 2008; Hairer 2009). In the absence of coevolution, the well-studied equilibrium solution to this model is a Gaussian random field (Whittle 1954; Whittle 1963; Lindgren et al. 2011; Lindgren 2012). Background on Gaussian random fields is provided in section S2 of the supplemental PDF. As our model of coevolution implies a linear interaction between two fields, the methods used by Whittle (1954, 1963) to study the case of no interaction can be extended to show that the equilibrium solution of equations (1) is also a Gaussian random field. To illustrate our model, a single realization of mean trait values across space for the two species is provided in figure 1.

Equilibrium solutions to our model are completely characterized by five quantities: (1) the expected host mean trait at each location μ_H, (2) the expected parasite mean trait at each location μ_P, (3) the covariance between host mean traits sampled at any displacement $C_H(\boldsymbol{x})$, (4) the covariance between parasite mean traits sampled at any displacement $C_P(\boldsymbol{x})$, and (5) the cross-covariance between host and parasite mean traits sampled at any displacement $C_{HP}(\boldsymbol{x})$. As our primary interest is in the spatial covariances of mean traits, we ignore the expected values μ_H, μ_P. In fact, the key quantity derived from our model for drawing conclusions on host-parasite local adaptation is the interspecific cross-covariance between traits $C_{HP}(\boldsymbol{x})$. We describe our analytical approach to approximating this cross-covariance function in section S3 of the supplemental PDF. To make use of this cross-covariance function for understanding cross-scale variation in patterns of local adaptation, we introduce a new index of local adaptation in the following section that explicitly accounts for the effects of geographic scale.

Local Adaptation in Continuous Space

There are two popular definitions of local adaptation at the population level (Kawecki and Ebert 2004; Blanquart et al. 2013; but for a third definition, see Nuismer and Gandon 2008). The first, known as "home versus away," is the mean fitness of a population (where mean fitness of a population is defined as the average fitness among

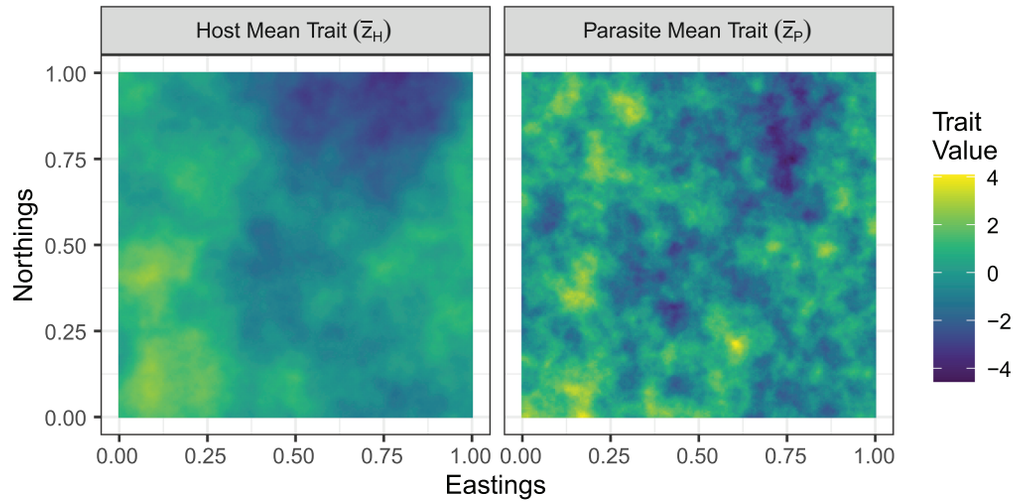


Figure 1: A single realization of our model. Shown are local mean trait values $\bar{z}_H(\boldsymbol{x})$, $\bar{z}_P(\boldsymbol{x})$, given by equations (1a) and (1b), respectively, across geographic space $\boldsymbol{x} = (x_1, x_2)$, where x_1 quantifies eastings and x_2 quantifies northings, for the host (*left*) and parasite (*right*). Local mean trait values are colored according to the scale on the right. Here we have made the dispersal distance of the host to be five times that of the parasite, with all other pairs of parameters equal.

all individuals in that population) in its local environment minus the average mean fitness of that population when transplanted to any other location. An alternative definition, known as "local versus foreign," is the mean fitness of a population in its local environment minus the average mean fitness of populations transplanted from any other location to that local environment. These definitions are particularly well suited for metapopulations, composed of a finite number K of discrete locations, because a randomly drawn foreign location (for home versus away) or population (for local versus foreign, but from here on we simply write location) occurs with probability $1/K$. However, this definition fails to account for the effects of geographic scale on measurements of local adaptation. Therefore, we introduce a definition that explicitly accounts for the geographic scale at which measurements are taken.

To obtain an index of local adaptation for species distributed continuously in space that accounts for geographic scale, we compute the population growth rate (referred to as a Malthusian growth rate in Crow and Kimura 1970) for a population in its local environment (say at location $\boldsymbol{x}$) minus the growth rate for the focal population when transplanted to a different location $\boldsymbol{y}$. This definition corresponds to a home versus away definition of local adaptation, as described above. Although classical indices of local adaptation are defined in terms of fitness as lifetime expected number of offspring, we chose population growth rate because it leads to relatively simple mathematical expressions. However, given the close correspondence between growth rate and fitness, conclusions drawn using either one should be qualitatively similar.

We denote by $m_H(z, \boldsymbol{x})$ the population growth rate of hosts with trait z encountering parasites located at $\boldsymbol{x}$. Similarly, $m_P(z, \boldsymbol{x})$ is the growth rate of parasites with trait z encountering hosts located at $\boldsymbol{x}$. Expressions for these growth rates are given by equations (S33) in section S4 of the supplemental PDF. The population growth rate for individuals of species S transplanted from location $\boldsymbol{x}$ to location $\boldsymbol{y}$, written $\bar{m}_S(\boldsymbol{x}, \boldsymbol{y})$, is then the average of $m_S(z, \boldsymbol{y})$ across trait values of individuals in species S at location $\boldsymbol{x}$.

As our model considers mean traits as random variables, the averaged population growth rates $\bar{m}_H(\boldsymbol{x}, \boldsymbol{y})$, $\bar{m}_P(\boldsymbol{x}, \boldsymbol{y})$ are also random variables. We therefore define local adaptation in terms of expectations of these growth rates. The measure of local adaptation we propose, $\ell_S(\boldsymbol{x}, \boldsymbol{y})$, allows us to calculate the expected difference between population growth rates for individuals of species S drawn from location $\boldsymbol{x}$ reared locally compared with individuals transplanted to location y. Therefore, this definition explicitly accounts for the spatial distance between locations $\boldsymbol{x}$ and $\boldsymbol{y}$. Mathematically, our definition of local adaptation is expressed as

$$\ell_S(\boldsymbol{x}, \boldsymbol{y}) = \mathbb{E}[\bar{m}_S(\boldsymbol{x}, \boldsymbol{x}) - \bar{m}_S(\boldsymbol{x}, \boldsymbol{y})]. \tag{2}$$

Following this notation, a local versus foreign definition of local adaptation (as described above) would correspond to $\ell_S(\boldsymbol{x}, \boldsymbol{y}) = \mathbb{E}[\bar{m}_S(\boldsymbol{x}, \boldsymbol{x}) - \bar{m}_S(\boldsymbol{y}, \boldsymbol{x})]$. However, under our model we find $\mathbb{E}[\bar{m}_S(\boldsymbol{y}, \boldsymbol{x})] = \mathbb{E}[\bar{m}_S(\boldsymbol{x}, \boldsymbol{y})]$, and thus the two definitions coincide. This is a consequence of solutions to our model being spatially isotropic, which is defined in section S2 of the supplemental PDF. Although

the index of local adaptation we introduce here can be applied to quantify both host and parasite local adaptation, in practice most empirical researchers focus on measuring parasite local adaptation only. For clarity and brevity, we then discuss only parasite local adaptation below, but note here that this index applies equally well to host local adaptation.

In the results section, we combine this definition of local adaptation with the population growth rates found in section S4 of the supplemental PDF to uncover patterns of local adaptation between hosts and parasites coevolving in continuous space.

Results

Spatial Covariance Functions

Intraspecific Spatial Covariance. Taking our approach outlined in section S3 of the supplemental PDF, we obtain analytic approximations for the (intraspecific) spatial covariance and (interspecific) spatial cross-covariance functions of host and parasite mean trait values at equilibrium. We find spatial covariance functions for the host and parasite respectively take the forms

$$C_H(\boldsymbol{x}) = \sqrt{2}V_H \frac{\|\boldsymbol{x}\|}{\lambda_H} K_1\left(\sqrt{2}\frac{\|\boldsymbol{x}\|}{\lambda_H}\right), \tag{3a}$$

$$C_P(\boldsymbol{x}) = \sqrt{2}V_P \frac{\|\boldsymbol{x}\|}{\lambda_P} K_1\left(\sqrt{2}\frac{\|\boldsymbol{x}\|}{\lambda_P}\right), \tag{3b}$$

where $\|\boldsymbol{x}\|$ is the geographic distance between sampled locations; V_H, V_P are the colocated variances (i.e., $V_S = C_S(0)$); λ_H, λ_P are the spatial scales of autocorrelation in each species; and K_1 is the modified Bessel function of the second kind, order 1 (Abramowitz and Stegun 1965). These spatial covariance functions belong to the class of Matérn covariance functions that have been widely employed in the fields of spatial statistics (Stein 1999; Lindgren et al. 2011) and machine learning (Rasmussen and Williams 2006). In figure 2, we illustrate the relationship between patterns of phenotypic spatial variation and associated spatial covariance functions.

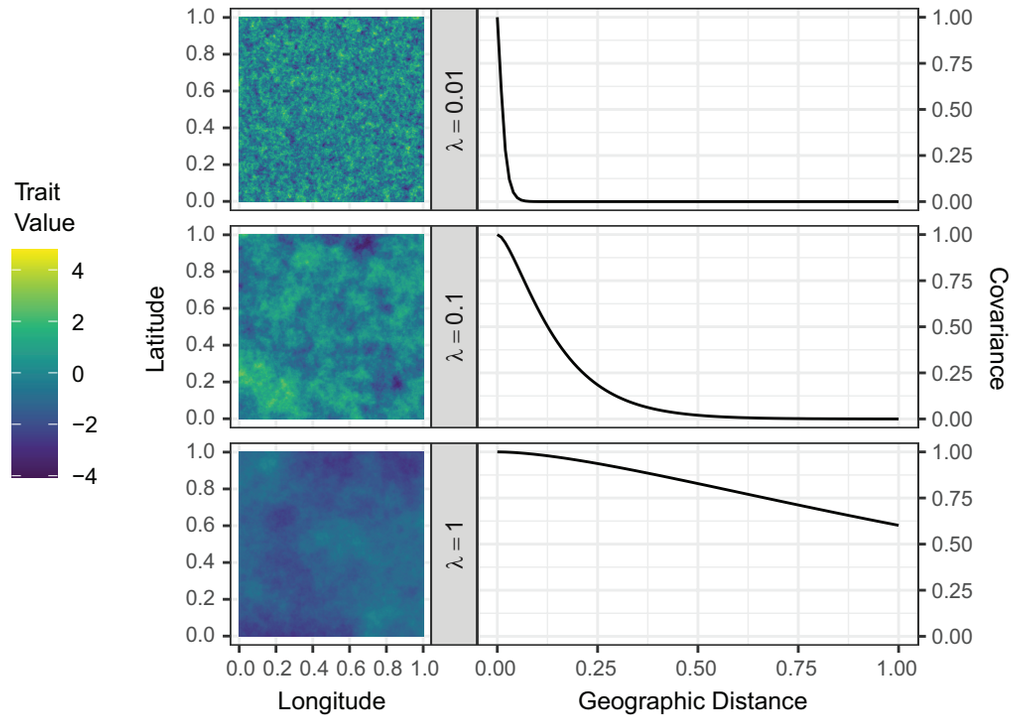


Figure 2: Mean trait surfaces $\bar{z}(\boldsymbol{x})$ (*left*), given by equations (1), and associated intraspecific covariance functions $C(\boldsymbol{x})$ (*right*), given by equations (3), across three different spatial scales of autocorrelation $\lambda = 0.01, 0.1, 1$, with colocated variance $V = 1$. This illustrates the relationship between the spatial scale of phenotypic autocorrelation and the variance of trait values across a fixed region. More precisely, increasing the spatial scale of autocorrelation λ (due to either increased dispersal distance or relaxed selection strengths; see eqq. [4]) leads to decreased trait variation in a fixed region.

Our results also demonstrate that the spatial scales of phenotypic isolation by distance in the host and parasite can be expressed in terms of model parameters respectively as

$$\lambda_H = \frac{\sigma_H}{\sqrt{A_H}}, \tag{4a}$$

$$\lambda_P = \frac{\sigma_P}{\sqrt{A_P}}. \tag{4b}$$

From these expressions, we see that these spatial scales are proportional to the dispersal distances in the respective species; the farther individuals tend to move, the larger the spatial scales one must observe to find significant phenotypic variation. We also see that increased abiotic stabilizing selection decreases these spatial scales in each species. Under the assumption of weak abiotic stabilizing selection (which would imply $A_H, A_P \ll 1$ and which is required in our justification of population growth rates; see section S4 of the supplemental PDF), our expression for the spatial scale of phenotypic variation coincides with that found by Slatkin (1978).

The colocated phenotypic variances, V_H and V_P, represent uncertainty in mean trait value at a particular location due to different possible realizations of drift. This should not be confused with the typical notion of phenotypic variance, which represents the variance of trait values among individuals in a population. As solutions to our model are spatially homogeneous random fields (i.e., they have the same statistical properties at any given spatial location), mean traits of individuals of species S sampled at locations separated by distances much greater than λ_S escape the effects of spatial autocorrelation and return essentially independent and identically distributed random variables with variances equal to the colocated variance V_S. That is, the colocated variances also provide measures of global diversity of mean traits across space. In terms of our model parameters, the colocated variances can be expressed as

$$V_H = \frac{D_H^2}{A_H \sigma_H^2}, \quad V_P = \frac{D_P^2}{A_P \sigma_P^2}. \tag{5}$$

From the expressions for V_H and V_P, we can identify relationships between global diversity of mean traits in each species and the different evolutionary processes of selection, dispersal, and drift. First, increased dispersal distance σ_S decreases V_S because, in the limit of infinite dispersal distance, our model collapses to a panmictic population of infinite size. Additionally, because abiotic selection is stabilizing around spatially homogeneous optima, increased A_S decreases V_S. Last, the variance due to drift D_S, the ultimate source of spatial differentiation in our model, increases with additive genetic variance and decreases with effective population size (Lande 1976; Week et al. 2021).

Interspecific Spatial Cross-Covariance. In contrast to the intraspecific spatial covariance functions above, the spatial cross-covariance function, which quantifies interspecific trait covariance measured at two potentially different locations, does not yield a closed-form expression. Specifically, we find that the cross-covariance function is approximated by

$$\begin{aligned} C_{HP}(\boldsymbol{x}) = \int_{\mathbb{R}^2} & \frac{B_P}{A_P} \frac{K_0(\|\boldsymbol{y}\|/\lambda_P)}{\lambda_P^2} C_H(\boldsymbol{x}-\boldsymbol{y}) \\ & - \frac{B_H}{A_H} \frac{K_0(\|\boldsymbol{y}\|/\lambda_H)}{\lambda_H^2} C_P(\boldsymbol{x}-\boldsymbol{y}) d\boldsymbol{y}. \end{aligned} \tag{6}$$

Although the expression for $C_{HP}(\boldsymbol{x})$ is quite technical, we can still draw some basic conclusions. For one, we see that interspecific cross-covariance can occur in the absence of coevolution. For example, in the case where $B_H = 0$ and $B_P > 0$, the host trait evolves only in response to abiotic selection, dispersal, and drift, but the parasite trait continues to evolve in response to the host trait. In this case, interspecific cross-covariance is maintained only by the parasite tracking the host. We can also use this expression to see that interspecific cross-covariance requires intraspecific autocovariance in at least one species (if $C_H(\boldsymbol{x}) = C_P(\boldsymbol{x}) = 0$, then $C_{HP}(\boldsymbol{x}) = 0$). Consequently, because both $C_H(\boldsymbol{x}), C_P(\boldsymbol{x}) \to 0$ in the limit of large geographic distance (i.e., $\|\boldsymbol{x}\| \to \infty$), $C_{HP}(\boldsymbol{x})$ also goes to zero in the limit of large distances. That is, interspecific cross-covariance disappears at distances large enough that intraspecific autocovariance is negligible in both species.

Patterns of Local Adaptation across Spatial Scales

Analytical Results. Combining our model with our definition of local adaptation in continuous space (see eq. [2]), we find

$$\ell_H(\boldsymbol{x}, \boldsymbol{y}) = \tilde{B}_H(C_{HP}(\boldsymbol{x}-\boldsymbol{y}) - C_{HP}(\boldsymbol{0})), \tag{7a}$$

$$\ell_P(\boldsymbol{x}, \boldsymbol{y}) = \tilde{B}_P(C_{HP}(\boldsymbol{0}) - C_{HP}(\boldsymbol{x}-\boldsymbol{y})), \tag{7b}$$

where $\tilde{B}_S$ is B_S divided by additive genetic variance in species S (see sec. S1 of the supplemental PDF). As all of the functions involved with this result depend only on spatial distance $d \geq 0$, we simplify our notation by writing $\ell_S(d) = \ell_S(\boldsymbol{x}, \boldsymbol{y})$ when $d = \|\boldsymbol{x}-\boldsymbol{y}\|$. Similarly, we write $C_H(d)$, $C_P(d)$, and $C_{HP}(d)$ for the spatial covariance functions evaluated at geographic distance d. In figure 3, we present our index of parasite local adaptation as a function of geographic distance across nine combinations of λ_H and λ_P, with all other parameter pairs made equal.

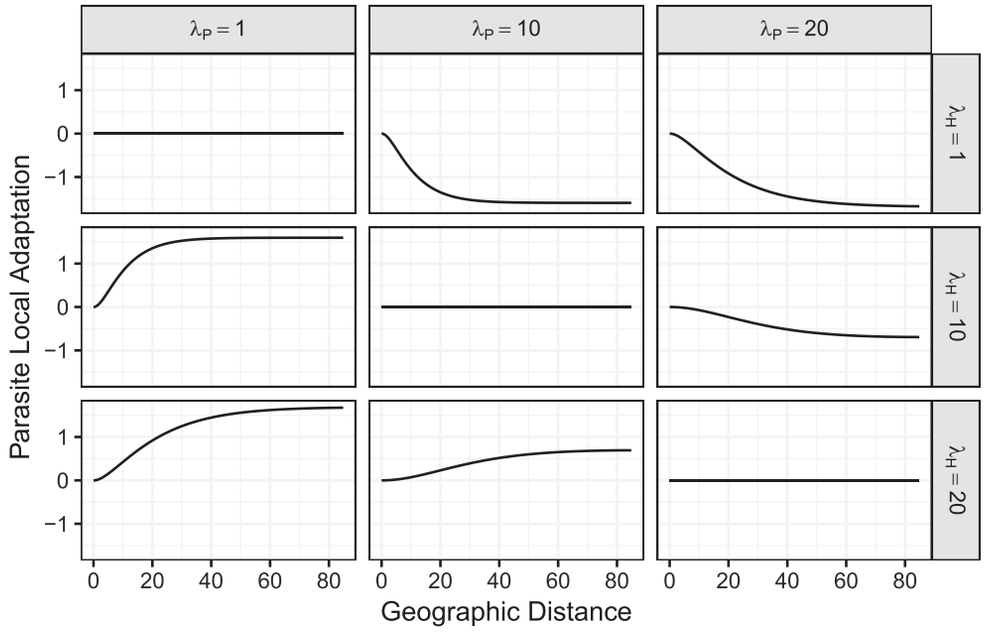


Figure 3: Parasite local adaptation ℓ_P, given by equation (7b), as a function of the geographic distance d between local parasite populations and the populations of hosts they are confronted with, scaled by biotic selection and colocated variances across nine combinations of $\lambda_H, \lambda_P = 1, 10, 20$, with $V_H = V_P = 1$ and $B_H/A_H = B_P/A_P = 1/250$. In the cases where $\lambda_H = \lambda_P$, we have $\ell_H(d) = \ell_P(d) = 0$ for all $d \geq 0$. In general, when all parameters are made equal between the two species, local adaptation does not emerge at any spatial scale.

Spatial Scale Dependency of Local Adaptation. As shown in figure 3, the magnitude of parasite local adaptation depends on the spatial scale measurements are taken at. More precisely, figure 3 suggests that local adaptation will be underestimated for study designs that define allopatric populations to be within the range of spatial autocorrelation for either species. In addition, if we allow for asymmetric background parameter values, which we might expect for realistic systems, our model predicts the potential for not only the magnitude of local adaptation to depend on the spatial scale measurements are taken, but also the identity of the locally adapted species. We present two such cases in figure 4. This result demonstrates that to obtain consistent estimates of local adaptation, measurements need to be taken at sufficiently large spatial scales.

A Consistent Estimate for Parasite Local Adaptation. The most rigorous way to estimate parasite local adaptation would be to confront parasites from a focal population with hosts sampled at populations from many different locations in a large provenance trial (Blanquart et al. 2013). However, such an experiment would be incredibly labor intensive and prone to variable results as predicted by our model. Fortunately, our model suggests an efficient way to consistently measure parasite local adaptation. Namely, when measurements are taken at sufficiently large spatial scales, $C_{HP}(d) \approx 0$ so that parasite local adaptation $\ell_P(d)$ is proportional to $C_{HP}(0)$ (see eqq. [7]). That is, parasite local adaptation can be consistently measured by computing the interspecific spatial covariance of sympatric mean traits across locations separated by distances that are large relative to both spatial scales of intraspecific autocorrelation λ_H and λ_P. Importantly, this approach, similar to previous approaches based on spatial covariance of allele frequencies (Blanquart et al. 2013), requires only information on host and parasite mean traits sampled across space (rather than the outcome of a manipulative experiment) and is thus much more feasible. Museum and herbarium collections may offer a vital resource for characterizing host/parasite phenotypic diversity across space (e.g., Rabosky et al. 2016). With a geographic map of host and parasite phenotypes in hand, a researcher could determine the spatial scale at which intraspecific phenotypic autocorrelation becomes negligible and thus the spatial scale at which interspecific spatial covariance of mean traits should be calculated to estimate local adaptation.

Discussion

Here, we have developed a model that tracks the evolution of local mean traits of coevolving species distributed in two-dimensional continuous space. This model accounts

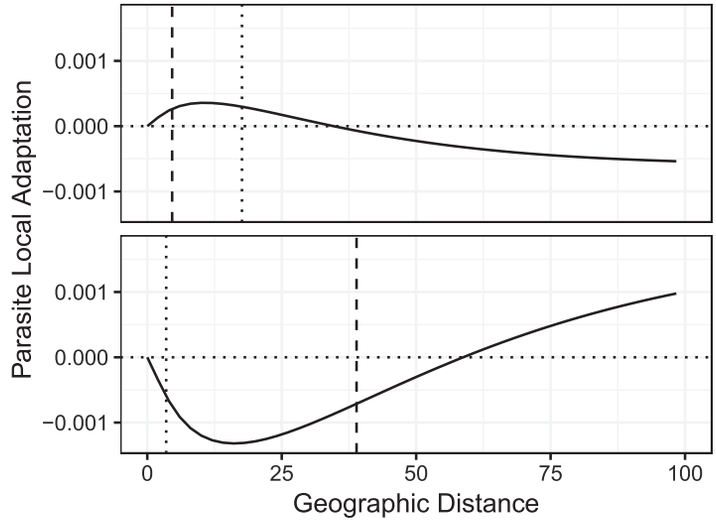


Figure 4: Parasite local adaptation ℓ_P, given by equation (7b), as a function of geographic distance d between local parasite populations and the populations of hosts they are confronted with, for two cases where the identity of the locally adapted species depends on the spatial scale at which measurements are taken. The horizontal dotted line marks a measurement of zero parasite local adaptation. Vertical lines mark spatial scales of intraspecific autocorrelation in the host (dashed vertical lines) and parasite (dotted vertical lines).

for host-parasite interactions mediated by a trait-matching/mismatching mechanism, abiotic stabilizing selection, Gaussian dispersal, and random genetic drift. Solutions to this model are Gaussian random fields that are characterized by two intraspecific spatial autocovariance functions along with an interspecific spatial cross-covariance function. We have also introduced a novel index of local adaptation that explicitly accounts for the geographic distance separating allopatric populations. Combining our novel index of local adaptation in continuous space with the cross-covariance function obtained from our model, we found that variation in measurements of parasite local adaptation across spatial scales can be explained by intraspecific spatial autocorrelation that emerges when species disperse in continuous space. More precisely, figure 3 demonstrates that measurements taken within the range of spatial autocorrelation for either species will underestimate the magnitude of parasite local adaptation or maladaptation. Building on this result, we suggest taking measurements at spatial scales for which intraspecific autocorrelation is negligible for both species to obtain consistent estimates for parasite local adaptation.

The mathematical model we propose may help shed light on several cases where estimates of parasite local adaptation may have been dependent on the spatial scales at which they were measured. For example, Tack et al. (2014) showed that maladaptation of the parasitic powdery mildew *Podosphaera plantaginis* on the invasive weed *Plantago lanceolata* was strong at scales of 6–200 km but weak at scales of less than 6 km. In another study, Thrall et al. (2002) found evidence of local adaptation of the autoecious rust pathogen *Melampsora lini* on its herbaceous perennial herb host *Linum marginale* to be more apparent at scales of 5–10 km than at scales of 200–400 m. Hanks and Denno (1994) investigated local adaptation of the phytophagous armored scale insect *Pseudaulacaspis pentagon* on individual mulberry trees of the species *Morus alba* and found stronger evidence of local adaptation for insects transplanted to distant trees ($\geq$300 m) than for insects transplanted to nearby trees ($<$5 m). Schönrogge et al. (2006) found the hoverfly *Microdon mutabilis* to be extremely locally adapted to its host ant *Formica lemani* when confronted with host individuals separated by distances of 3–20 km, moderately locally adapted at distances of 0.1–3 km, and not locally adapted at $<$100 m. Last, Kaltz et al. (1999) studied local adaptation of the insect-transmitted fungal pathogen *Microbotryum violaceum* to its host plant *Silene latifolia* and found weak parasite local maladaptation when confronted with nearby populations ($<$10 km), strong parasite maladaptation when confronted with populations at intermediate distances ($<$30 km), and variation in maladaptation for remote populations ($<$170 km) ranging from strong to weak maladaptation.

These empirical studies demonstrate the key finding of our model: that there can be variation in the magnitude of local adaptation across spatial scales in naturally occurring systems. For example, in all of the studies referenced in the preceding paragraph, the estimated strength of local adaptation is a function of the geographic distance between the

populations for which it is assessed. A second useful prediction from our model is that because of the phenotypic spatial autocorrelation within each species, measurements of local adaptation taken at larger distances will provide more consistent estimates. Therefore, in each of the cases referenced above (in which local adaptation varies as a function of the spatial scale at which it is measured) we would recommend summarizing local adaptation in the system using comparisons between the most distant population pairs. However, because our model includes stochasticity (due to random genetic drift), it also predicts that there should be some amount of variation in individual measurements of local adaptation, even at large spatial scales. Kaltz et al. (1999) highlight this variation, finding high variance in infection success of the *Microbotryum* parasite on *Silene* hosts at the largest spatial distances. These infection success measurements do appear to increase, on average, with distance, and given the second prediction above, the model would support summarizing these results using the mean measurements at the largest spatial lags.

In addition to providing a theoretical explanation for the observation of spatial variation in host-parasite local adaptation, our model predicts that the shorter-dispersing species should be locally adapted to the farther-dispersing species, assuming that all other parameters are equal (see fig. 3). Of the five studies listed above (by no means an exhaustive list), two (Hanks and Denno 1994; Tack et al. 2014) fit this prediction. In Hanks and Denno (1994) the parasite disperses a shorter distance than its host and is locally adapted, and in Tack et al. (2014) the parasite disperses a longer distance than its host and is locally maladapted. Deviations from this prediction may then imply interspecific asymmetry of model parameters other than dispersal distances. Equation (S13) in section S3 of the supplemental PDF shows that the identity of the locally adapted species depends on symmetry of each pair of model parameters (abiotic selection strengths A_H, A_P; biotic selection strengths B_H, B_P; dispersal distances σ_H, σ_P; and rates of drift D_H, D_P). Consequently, our model suggests that the identity of the locally adapted species in the other three studies (Kaltz et al. 1999; Thrall et al. 2002; Schönrogge et al. 2006), in each of which the locally adapted species disperses farther than its partner, is mediated by differential selection strengths or differential rates of drift.

Our suggested approach for obtaining consistent measurements of parasite local adaptation rests on the hypothesis that intraspecific spatial autocorrelation explains variation in measurements of coadaptation across spatial scales. This hypothesis, implied by our theoretical findings, can be tested using geographically referenced phenotypic data. More precisely, standard methods applied in the analysis of spatial data (e.g., Dale and Fortin 2014) may be used to determine the distances for which spatial autocorrelation is negligible in each species. In turn, these estimates can be compared to the geographical distances between allopatric populations defined in previous empirical studies. Our hypothesis would be rejected if variation in measurements of local adaptation occurs across several spatial scales that are larger than both of the spatial scales of autocorrelation in the host and parasite species. In this case, additional theoretical studies would be required to explain such cross-scale variation.

Additional hypotheses for the cross-scale variation in spatial patterns of adaptation can be developed by extending our model in several directions. For example, heavy-tailed dispersal kernels, which are likely common in nature (Houtan et al. 2007; Bullock et al. 2016; García and Borda-de-Água 2016; Jordano 2016), can be incorporated into our model by replacing the dispersal operator with fractional spatial derivatives (for mathematical details, see Laskin 2000; Bayın 2016). As fractional derivatives are defined in terms of spectral representations, our approach, which recovers covariance functions from spectral representations, is particularly well suited to this generalization of our model. Such a generalization of our model may then be used to predict variation in patterns of adaptation as a consequence of dispersal mode. Another direction this work can be taken is to consider spatial variation of the abiotic environment by modeling abiotic optima as additional Gaussian random fields (making for four random fields in total; one for the host trait, one for the parasite trait, one for the host abiotic optimum, and one for the parasite abiotic optimum). This second direction would allow for theoretical predictions of spatial variation in local adaptation based on spatial scales of abiotic environmental variation. Hu et al. (2013) provides further information on the mathematical details involved with both of these extensions.

Although there is already a large body of theory on geographically structured host-parasite coevolution (e.g., Gandon et al. 1996; Nuismer et al. 2000; Gandon 2002; Gandon and Michalakis 2002; Nuismer et al. 2003; Nuismer 2006; Ridenhour and Nuismer 2007; Gandon and Nuismer 2009; Débarre et al. 2012; Lion and Gandon 2015), these studies are limited in their ability to explain variation in patterns of host-parasite adaptation across spatial scales because they depend on indices of local adaptation that do not explicitly account for geographic distance. In contrast, empirical studies have emphasized that such cross-scale variation is common (Hanks and Denno 1994; Kaltz et al. 1999; Thrall et al. 2002; Schönrogge et al. 2006; Tack et al. 2014). Our model, combined with our novel index of local adaptation, is, to our knowledge, the first to provide a mechanistic explanation for measurements of parasite local adaptation that vary across spatial scales. Furthermore, it is consistent with empirical observations that local adaptation

in host-parasite systems tends to be stronger at larger spatial scales relative to shorter spatial scales.

In addition to the geographically explicit index of local adaptation we have introduced here, our mathematical model of continuous space coevolution has several novel features. In comparison to previous models of coevolution in continuous space (Nuismer et al. 2000, 2003), our model differs in three major ways. First, whereas previous continuous space models have focused on interspecific interactions mediated by allelic variation at genetic loci, our model focuses on coevolution of a pair of quantitative traits that determine the outcomes of interactions. Second, while previous work has focused on a single continuous spatial axis, our model accounts for the more biologically relevant case of two continuous spatial axes. Last, our model accounts for the stochastic dynamics of random genetic drift, which rarely yields tractable evolutionary models in two-dimensional continuous space (Felsenstein 1975; Etheridge 2004; Barton et al. 2010). It is therefore likely that investigations relaxing important assumptions of our model, such as spatially homogeneous abundance densities and weak biotic selection, will require the study of individual-based simulations.

Conclusion

In summary, our work here provides initial steps toward understanding coevolution and local adaptation in continuous space using an analytically tractable mathematical model. Our theoretical findings form testable hypotheses for previously unexplained variation observed in empirical host-parasite systems and suggest an approach to consistently measure parasite local adaptation in the wild. Taken together, this work establishes a novel theoretical approach to model spatial coevolution and lends a new perspective for understanding spatial patterns of interspecific coadaptation.

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Statement of Authorship

B.W. contributed to conceptualization, model development/analysis, visualization, coding simulation, writing—original draft, and writing—review and editing. G.B. contributed to conceptualization, supervision, writing—original draft, writing—review and editing, and funding acquisition.

Data and Code Availability

Data and code for reproducing each of the figures in this article and for deriving some key analytical results have been collected in a Zenodo repository (https://doi.org/10.5281/zenodo.8008017; Week and Bradburd 2023).

Literature Cited

Abramowitz, M., and I. A. Stegun. 1965. Handbook of mathematical functions with formulas, graphs, and mathematical tables. Dover, Mineola, NY.

Agrawal, A. A., and X. Zhang. 2021. The evolution of coevolution in the study of species interactions. Evolution 75:1594–1606.

Barton, N. H., A. M. Etheridge, and A. Véber. 2010. A new model for evolution in a spatial continuum. Electronic Journal of Probability 15:162–216.

———. 2017. The infinitesimal model: definition, derivation, and implications. Theoretical Population Biology 118:50–73.

Bayın, S. Ş. 2016. Definition of the Riesz derivative and its application to space fractional quantum mechanics. Journal of Mathematical Physics 57:123501.

Best, A., S. Webb, A. White, and M. Boots. 2010. Host resistance and coevolution in spatially structured populations. Proceedings of the Royal Society B 278:2216–2222.

Blanquart, F., O. Kaltz, S. L. Nuismer, and S. Gandon. 2013. A practical guide to measuring local adaptation. Ecology Letters 16:1195–1205.

Boots, M., and A. Sasaki. 1999. "Small worlds" and the evolution of virulence: infection occurs locally and at a distance. Proceedings of the Royal Society B 266:1933–1938.

Bullock, J. M., L. M. González, R. Tamme, L. Götzenberger, S. M. White, M. Pärtel, and D. A. P. Hooftman. 2016. A synthesis of empirical plant dispersal kernels. Journal of Ecology 105:6–19.

Crow, J. F., and M. Kimura. 1970. An introduction to population genetics theory. Harper & Row, New York.

Dale, M., and M.-J. Fortin. 2014. Spatial analysis. 2nd ed. Cambridge University Press, Cambridge.

Débarre, F., S. Lion, M. van Baalen, and S. Gandon. 2012. Evolution of host life-history traits in a spatially structured host-parasite system. American Naturalist 179:52–63.

Débarre, F., S. Yeaman, and F. Guillaume. 2015. Evolution of quantitative traits under a migration-selection balance: when does skew matter? American Naturalist 186:S37–S47.

Etheridge, A. M. 2004. Survival and extinction in a locally regulated population. Annals of Applied Probability 14:188–214.

Felsenstein, J. 1973. Maximum-likelihood estimation of evolutionary trees from continuous characters. American Journal of Human Genetics 25:471–492.

———. 1975. A pain in the torus: some difficulties with models of isolation by distance. American Naturalist 109:359–368.
Gandon, S. 2002. Local adaptation and the geometry of host-parasite coevolution. Ecology Letters 5:246–256.
Gandon, S., Y. Capowiez, Y. Dubois, Y. Michalakis, and I. Olivieri. 1996. Local adaptation and gene-for-gene coevolution in a metapopulation model. Proceedings of the Royal Society B 263:1003–1009.
Gandon, S., and Y. Michalakis. 2002. Local adaptation, evolutionary potential and host-parasite coevolution: interactions between migration, mutation, population size and generation time. Journal of Evolutionary Biology 15:451–462.
Gandon, S., and S. L. Nuismer. 2009. Interactions between genetic drift, gene flow, and selection mosaics drive parasite local adaptation. American Naturalist 173:212–224.
García, C., and L. Borda-de-Água. 2016. Extended dispersal kernels in a changing world: insights from statistics of extremes. Journal of Ecology 105:63–74.
Hairer, M. 2009. An introduction to stochastic PDEs. arXiv, https://doi.org/10.48550/arXiv.0907.4178.
Hanks, L. M., and R. F. Denno. 1994. Local adaptation in the armored scale insect *Pseudaulacaspis pentagona* (Homoptera: Diaspididae). Ecology 75:2301–2310.
Houtan, K. S. V., S. L. Pimm, J. M. Halley, R. O. Bierregaard, and T. E. Lovejoy. 2007. Dispersal of amazonian birds in continuous and fragmented forest. Ecology Letters 10:219–229.
Hu, X., D. Simpson, F. Lindgren, and H. Rue. 2013. Multivariate Gaussian random fields using systems of stochastic partial differential equations. arXiv, https://doi.org/10.48550/arXiv.1307.1379.
Jordano, P. 2016. What is long-distance dispersal? and a taxonomy of dispersal events. Journal of Ecology 105:75–84.
Kaltz, O., S. Gandon, Y. Michalakis, and J. A. Shykoff. 1999. Local maladaptation in the anther-smut fungus *Microbotryum violaceum* to its host plant *Silene latifolia*: evidence from a cross-inoculation experiment. Evolution 53:395–407.
Kawecki, T. J., and D. Ebert. 2004. Conceptual issues in local adaptation. Ecology Letters 7:1225–1241.
Khoshnevisan, D. 2008. A primer on stochastic partial differential equations. Pages 1–38 *in* Lecture notes in mathematics. Springer, Berlin.
Lande, R. 1976. Natural selection and random genetic drift in phenotypic evolution. Evolution 30:314–334.
Laskin, N. 2000. Fractional quantum mechanics. Physical Review E 62:3135–3145.
Lindgren, F., H. Rue, and J. Lindström. 2011. An explicit link between Gaussian fields and Gaussian Markov random fields: the stochastic partial differential equation approach. Journal of the Royal Statistical Society B 73:423–498.
Lion, S., and S. Gandon. 2015. Evolution of spatially structured host-parasite interactions. Journal of Evolutionary Biology 28:10–28.
Lively, C. M. 2010. Antagonistic coevolution and sex. Evolution: Education and Outreach 3:19–25.
Manceau, M., A. Lambert, and H. Morlon. 2016. A unifying comparative phylogenetic framework including traits coevolving across interacting lineages. Systematic Biology 66:551–568.
Nathan, R., W. M. Getz, E. Revilla, M. Holyoak, R. Kadmon, D. Saltz, and P. E. Smouse. 2008. A movement ecology paradigm for unifying organismal movement research. Proceedings of the National Academy of Sciences of the USA 105:19052–19059.
Nuismer, S. L. 2006. Parasite local adaptation in a geographic mosaic. Evolution 60:24–30.
Nuismer, S. L., and S. Gandon. 2008. Moving beyond common-garden and transplant designs: insight into the causes of local adaptation in species interactions. American Naturalist 171:658–668.
Nuismer, S. L., J. N. Thompson, and R. Gomulkiewicz. 2000. Coevolutionary clines across selection mosaics. Evolution 54:1102–1115.
———. 2003. Coevolution between hosts and parasites with partially overlapping geographic ranges. Journal of Evolutionary Biology 16:1337–1345.
Otto, S. P., and S. L. Nuismer. 2004. Species interactions and the evolution of sex. Science 304:1018–1020.
Rabosky, A. R. D., C. L. Cox, D. L. Rabosky, P. O. Title, I. A. Holmes, A. Feldman, and J. A. McGuire. 2016. Coral snakes predict the evolution of mimicry across new world snakes. Nature Communications 7:11484.
Rasmussen, C. E., and C. K. Williams. 2006. Gaussian processes for machine learning. MIT Press, Cambridge, MA.
Ridenhour, B. J., and S. L. Nuismer. 2007. Polygenic traits and parasite local adaptation. Evolution 61:368–376.
Schönrogge, K., M. G. Gardner, G. W. Elmes, E. K. V. Napper, D. J. Simcox, J. C. Wardlaw, J. Breen, et al. 2006. Host propagation permits extreme local adaptation in a social parasite of ants. Ecology Letters 9:1032–1040.
Slatkin, M. 1978. Spatial patterns in the distributions of polygenic characters. Journal of Theoretical Biology 70:213–228.
Stein, M. L. 1999. Interpolation of spatial data. Springer, New York.
Tack, A. J. M., F. Horns, and A.-L. Laine. 2014. The impact of spatial scale and habitat configuration on patterns of trait variation and local adaptation in a wild plant parasite. Evolution 68:176–189.
Thrall, P. H., J. J. Burdon, and J. D. Bever. 2002. Local adaptation in the *Linum marginale–Melampsora lini* host-pathogen interaction. Evolution 56:1340–1351.
Week, B., and G. Bradburd. 2023. Data from: Host-parasite coevolution in continuous space leads to variation in local adaptation across spatial scales. American Naturalist, Zenodo, https://doi.org/10.5281/zenodo.8008017.
Week, B., S. L. Nuismer, L. J. Harmon, and S. M. Krone. 2021. A white noise approach to evolutionary ecology. Journal of Theoretical Biology 521:110660.
Whittle, P. 1954. On stationary processes in the plane. Biometrika 41:434–449.
———. 1963. Stochastic-processes in several dimensions. Bulletin of the International Statistical Institute 40:974–994.
Yamamura, N. 1993. Vertical transmission and evolution of mutualism from parasitism. Theoretical Population Biology 44:95–109.

References Cited Only in the Online Enhancements

Bolker, B., and S. W. Pacala. 1997. Using moment equations to understand stochastically driven spatial pattern formation in ecological systems. Theoretical Population Biology 52:179–197.
Da Prato, G., and J. Zabczyk. 2014. Stochastic equations in infinite dimensions. Cambridge University Press, Cambridge.
Evans, L. C. 2010. Partial differential equations. Vol. 19. American Mathematical Society, Providence, RI.
Lindgren, G. 2012. Stationary stochastic processes: theory and applications. CRC Press, Boca Raton, FL.
Méléard, S., and S. Roelly. 1993. Interacting measure branching processes: some bounds for the support. Stochastics and Stochastic Reports 44:103–121.

Rue, H., and L. Held. 2005. Gaussian Markov random fields: theory and applications. CRC Press, Boca Raton, FL.

Walsh, J. B. 1986. An introduction to stochastic partial differential equations. Pages 265–439 *in* École d'Été de probabilités de saint flour XIV-1984. Springer, Berlin.

Week, B., and S. L. Nuismer. 2021. Coevolutionary arms races and the conditions for the maintenance of mutualism. American Naturalist 198:195–205.

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"New species of Brugmansia, of the family of the Rafflesiaceæ." From the review of Forbes' *A Naturalist's Wanderings in the Eastern Archipelago (The American Naturalist*, 1885, 19:975–977).